

The Championship Gap: A Data Science Investigation into Real Madrid and Barcelona's UEFA Champions League Performance 2012–2018

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Abstract

This paper serves as the second installment of the football data analysis studies, presenting a forensic data science investigation into one of football's most compelling questions: how did Real Madrid claim four UEFA Champions League titles between 2012 and 2018 while Barcelona, home to arguably the greatest player in history, won only one? Both clubs were widely regarded as the finest teams in the world during this period, yet their championship outcomes diverged significantly. Rather than relying on emotional judgment or reputation, this paper applies objective data analysis to identify the measurable performance differences that explain the gap.

The paper examines each team's performance independently against their respective UEFA Champions League opponents across eleven modules: 1, Research background and the central question. 2, Data sources, scope, and key limitations. 3, The five-pillar analytical framework overview. 4-8, Analytical findings per pillar — shot analysis, passing analysis, danger pass, valued action, and defending respectively. 9, Tactical and structural synthesis explaining the performance gap. 10, Conclusions on key contributing factors. 11, Future plan introducing an AI agent to automate the framework for 2019–2024 UEFA Champions League seasons, building on the methodological foundation established in Parts 1 and 2.

Dedication

I would like to dedicate this paper to my favorite player Luka Modrić, who is approaching the twilight of an extraordinary career that has redefined what intelligence and grace mean in football. His brilliance on the pitch inspires me to conduct this set of studies to identify meaningful quantitative characteristics that explain the intelligence, quality, tactics and strategy of the game.

Acknowledgments

All references will be included in the paper. I would like to sincerely appreciate the work and education conducted by Jan Van Haaren, [Jan Van Haaren | LinkedIn](#), whose efforts in building an evidence-based culture at Club Brugge deeply resonate with the spirit of this study. As he noted, the most difficult challenge is not the analysis itself but changing the culture — convincing the industry to trust and invest in data over intuition and tradition. This paper is a step toward that same goal. I would also like to acknowledge Anthropic's Claude, whose AI tools significantly improved the efficiency of this study — Claude Chat supported the analytical thinking and interaction throughout the research process, while Claude Code accelerated the data analysis, programming, and visualization work through its strong programming capabilities.

Module 4 Shot Analysis

4.1 Introduction

Of all the measurable elements in football, the shot is the most direct expression of attacking intent and the closest quantifiable link to the outcome of a match. A team may dominate possession, complete thousands of passes, and press relentlessly, yet none of it translates into points without the ability to generate shots and convert them into goals. Shot analysis is therefore the natural starting point for any forensic examination of team performance.

However, not all shots are equal. A shot from 35 meters with no pressure tells a very different story from a shot inside the six-yard box following a through ball. Volume alone is insufficient. This module therefore examines shot quality alongside shot quantity, using a set of metrics that together provide a more complete picture of how each team attacked and how efficiently they converted their opportunities.

Metrics Used in This Analysis

The following metrics are used consistently throughout this module for both Real Madrid and Barcelona:

Shots: Total shots attempted per match, reflecting attacking volume and pressure applied on the opponent's goal.

Goals: Actual goals scored per match by outfield players, excluding own goals.

xG (Expected Goals): A probability-based measure of shot quality estimating the likelihood of a shot resulting in a goal. Commercial xG models typically incorporate multiple factors including shot location, angle, body part, assist type, and defensive pressure. However, since the WhoScored/Opta dataset used in this study does not include commercial xG values, the author constructed a simplified xG approximation based primarily on shot distance and location. While this model does not capture the full complexity of a commercial xG system, it serves as a reasonable and consistent proxy for shot quality comparison. Since both Real Madrid and Barcelona are evaluated using the identical model, the comparative validity between the two teams is maintained throughout this analysis.

Rationality (xG/Shot): The average xG per shot, measuring the decision-making quality behind shot selection. A higher rationality score indicates a team is taking higher quality shots rather than shooting indiscriminately.

Box Shot %: The proportion of shots taken from inside the penalty box, reflecting how effectively a team penetrates the opponent's defensive structure before shooting.

Accuracy %: The proportion of shots that are on target, measuring basic shot execution.

Big Misses: Shots where the xG was high but no goal was scored, indicating missed high-quality opportunities that may have changed the outcome of a match.

Opponent Shots and Opponent xG: The same metrics applied to the opponent, providing context for how well each team suppressed the opposition's attacking threat.

Finishing Delta: Goals – xG. Whether the team scored above or below the model's expectation.

Quality Margin: xG – Opponent xG. The surplus of chance quality created over the opponent.

Field Tilt %: The proportion of total shots in a match taken by the team, reflecting territorial dominance and sustained attacking pressure.

How to Read the Shot Maps

Each match in this module is accompanied by a shot map displaying the spatial distribution of shots for both teams. Real Madrid's shots are shown in red and Barcelona's shots in blue, with the opponent's shots shown in the contrasting color. The size of each bubble represents the xG value of the shot — larger bubbles indicate higher quality chances. Filled bubbles represent goals scored, while hollow bubbles represent shots that did not result in a goal. Player initials are labeled alongside each shot for reference.

Structure of This Module

The analysis is presented chronologically, covering six UEFA Champions League seasons from 2012-2013 to 2017-2018. For each season, the analysis includes a season overview comparing both teams across all key metrics, a tier breakdown separating performance against Tier 1 (Elite, $Elo > 1850$) and Tier 2 (Standard, $Elo \leq 1850$) opponents, player-level highlights identifying the key shot contributors for each team, and where applicable, a focused examination of critical matches that were decisive for that season's outcome. The module closes with a cross-season summary comparing both teams across all six seasons to identify patterns of consistency, divergence, and evolution over time.

4.2 Season 2012-2013

The 2012-2013 UEFA Champions League season marked the beginning of the study period and established an early contrast between Real Madrid and Barcelona's attacking profiles. Both teams progressed through the Group Stage comfortably before facing more demanding opposition in the knockout rounds. Real Madrid was eliminated in the Semi-Finals by Borussia Dortmund, while Barcelona suffered a heavier defeat at the same stage against Bayern Munich — a result that would foreshadow a recurring pattern in Barcelona's European campaign.

Season Overview

Table 8 presents the season average shot metrics for both teams across all matches in the 2012-2013 UEFA Champions League campaign.

Table 8 Season Overview — Shot Analysis 2012-2013

Metric	Real Madrid	Barcelona
Avg Shots per Match	19.3	14.8
Avg Goals per Match	2.2	1.5
xG per Match	1.25	0.84
Rationality (xG/Shot)	0.067	0.061
Box Shot %	51.6%	50.0%
Accuracy %	37.4%	36.9%
Big Misses per Match	0.42	0.00
Opp Shots per Match	12.6	10.3
Opp xG per Match	0.87	0.72
Finishing Delta (Goals – xG)	0.91	0.66
Quality Margin (xG – Opp xG)	0.38	0.12
Field Tilt %	58.7%	56.8%

Real Madrid generated significantly more shots per match than Barcelona, averaging 19.3 compared to Barcelona's 14.8. This volume advantage extended to goals scored, xG generated, and Field Tilt, where Real Madrid held a territorial advantage of 58.7% compared to Barcelona's 56.8%. Both teams showed similar shot accuracy at approximately 37%, suggesting that the primary differentiator in this season was attacking volume and chance creation rather than finishing precision. Real Madrid's higher rationality score of 0.067 compared to Barcelona's 0.061 further indicates that Real Madrid were selecting slightly higher quality shooting positions on average.

The Quality Margin — the difference between a team's xG and their opponent's xG per match — reveals the net advantage in chance quality each team created. Real Madrid's Quality Margin of 0.38 exceeded Barcelona's 0.12, indicating that Real Madrid created a larger surplus of chance quality over their opponents across the season. Barcelona, despite their reputation for dominance, generated only a marginal quality advantage over the teams they faced — their opponents created almost as much xG as Barcelona did.

An apparent contradiction emerges when examining opponent shots. Barcelona conceded fewer shots per match than Real Madrid — 10.3 compared to 12.6 — suggesting tighter defensive organisation at the season-average level. Yet Barcelona's much lower Quality Margin indicates that the shots their opponents did generate were of relatively high quality. Barcelona suppressed shot volume but were less effective at controlling shot quality.

The Finishing Delta — the difference between actual goals scored and xG generated — was positive for both teams, 0.91 for Real Madrid and 0.66 for Barcelona. Because this study's xG model is distance-only and blind to defensive pressure, a positive Finishing Delta should not be read simply as clinical finishing or good fortune. It is better understood as a signature of the shooting environment: shots taken under less defensive pressure may convert above their distance-based expectation [HYPOTHESIS]. Real Madrid's larger Finishing Delta is therefore consistent with a team generating a meaningful share of its shots in lower-pressure situations, though this interpretation cannot be confirmed from shot data alone [HYPOTHESIS].

Real Madrid also recorded 0.42 big misses per match compared to Barcelona's 0.00 — meaning Real Madrid were generating high quality chances that went unconverted, while Barcelona rarely created opportunities that qualified as big misses. This reflects Real Madrid's higher volume of high-xG attempts, some of which were not converted, whereas Barcelona's shot profile contained fewer genuinely high-quality chances overall.

Tier Comparison

Table 9 presents the tier breakdown separating performance against Tier 1 (Elite, Elo > 1850) and Tier 2 (Standard, Elo ≤ 1850) opponents.

Table 9 Tier Comparison — Shot Analysis 2012-2013

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Avg Shots per Match	20.9	16.3	14.3	15.1
Avg Goals per Match	1.6	3.3	0.75	1.9
xG per Match	1.32	1.11	0.76	0.88
Rationality (xG/Shot)	0.065	0.071	0.068	0.057
Box Shot %	49.0%	56.8%	49.8%	50.2%
Accuracy %	30.0%	52.3%	25.1%	42.8%
Big Misses per Match	0.50	0.25	0.00	0.00
Opp Shots per Match	12.6	12.5	13.3	8.75
Opp xG per Match	1.05	0.52	1.15	0.51
Finishing Delta (Goals – xG)	0.30	2.14	-0.01	0.99
Quality Margin (xG – Opp xG)	0.28	0.60	-0.39	0.38
Field Tilt %	60.4%	55.2%	49.3%	60.6%

The tier breakdown reveals a significant and important divergence between the two clubs. Against Tier 1 opponents, Real Madrid maintained a Field Tilt of 60.4%, demonstrating continued territorial dominance even against elite opposition. Barcelona's Field Tilt against Tier 1 opponents dropped sharply to 49.2% — falling below the 50% threshold, meaning elite opponents actually achieved greater territorial control against Barcelona than Barcelona achieved against them. This is a notable finding. A team widely regarded as one of the finest in football history was being outshot territorially by elite European opposition.

Real Madrid's goal output dropped from 3.3 per match against Tier 2 opponents to 1.6 against Tier 1, reflecting the expected difficulty of converting against stronger defences. Barcelona experienced a more severe drop, from 1.9 goals per match against Tier 2 to just 0.75 against Tier 1. The accuracy figures tell a similar story — Real Madrid maintained

30.0% accuracy against elite opponents while Barcelona dropped to 25.1%, the lowest figure across both teams and both tiers in this season.

The Quality Margin figures expose the most critical difference at Tier 1 level. Real Madrid maintained a positive Quality Margin against elite opponents, continuing to generate more chance quality than their opponents even in the most demanding matches. Barcelona's Quality Margin against Tier 1 opponents was negative — their elite opponents generated more xG than Barcelona did. Against the best teams in Europe, Barcelona were being out-created, not merely held level. This is the single most important shot-analysis finding of the season: a team widely regarded as the finest in the world produced a negative chance-quality balance against elite opposition.

The Finishing Delta reinforces this picture. Against Tier 1 opponents Barcelona under-finished relative to their already-low xG, while Real Madrid remained close to or above their expected output. Read through the environment lens, Barcelona were not only creating lower-quality chances than their opponents against elite defences — the chances they did create were taken under sufficient pressure that they converted below expectation [HYPOTHESIS].

The opponent data completes the picture. Against Tier 1 opponents, Barcelona conceded a higher opponent xG than Real Madrid did. Combined with Barcelona's negative Quality Margin, this indicates that in Tier 1 matches Barcelona were being out-created in chance quality while simultaneously allowing elite opponents to generate dangerous opportunities of their own.

Against Tier 2 opponents the picture reverses. Barcelona's Field Tilt rose to 60.6% and their quality margin recovered to 0.99, demonstrating they could dominate weaker opposition comfortably. Real Madrid showed a similar but less pronounced pattern, with a Field Tilt of 55.2% and a quality margin of 2.14 against Tier 2 opponents — the highest figure across both teams and both tiers, reflecting Real Madrid's ability to completely overwhelm weaker opposition in shot quality terms.

The tier analysis establishes a clear early pattern: Real Madrid maintained meaningful performance levels against elite opposition while Barcelona's attacking and defensive metrics deteriorated sharply. This gulf between Tier 1 and Tier 2 performance is one of the central themes this study will track across all six seasons. This gulf — and in particular Barcelona's negative Quality Margin against elite opponents — is one of the central themes this study will track across all six seasons.

Shot Map Analysis

Figure 1 presents the shot maps for Real Madrid across all 2012-2013 matches, and Figure 2 presents the equivalent maps for Barcelona.

Real Madrid Spatial Profile

Real Madrid's shot distribution reveals a structured and balanced attacking pattern across both tiers. Against Tier 1 opponents, 46.7% of shots originated from the box and 48.5% from the edge zone, reflecting a team capable of generating attempts from multiple depth zones rather than being restricted to any single area. Laterally, 74.9% of shots came from the central corridor, with left and right channels contributing equally at 12.6% each — indicating a disciplined central attacking structure that did not rely on any single flank.

The goal rate data confirms that Box_Central was Real Madrid's primary source of goals — 60 shots in this zone against Tier 1 opponents produced 8 goals at a conversion rate of 13.3%, the highest of any zone. Edge_Central contributed a further 3 goals from 58 shots at 5.2%, serving as a meaningful secondary source. Against Tier 2 opponents, Box_Central's conversion rate rose sharply to 40.7% — 27 shots producing 11 goals — reflecting the expected efficiency advantage against weaker defences.

All 34 high xG shots identified in the dataset originated from Box_Central. As noted in the methodology section, this reflects the distance-based nature of the custom xG model rather than a tactical insight, since proximity to goal will always generate higher xG values regardless of the model used. The meaningful spatial insight therefore lies in the goal rate by zone rather than the xG distribution.

Shot Analysis: Real Madrid (Red) vs Opponents (Blue)
Champions League 2012-2013

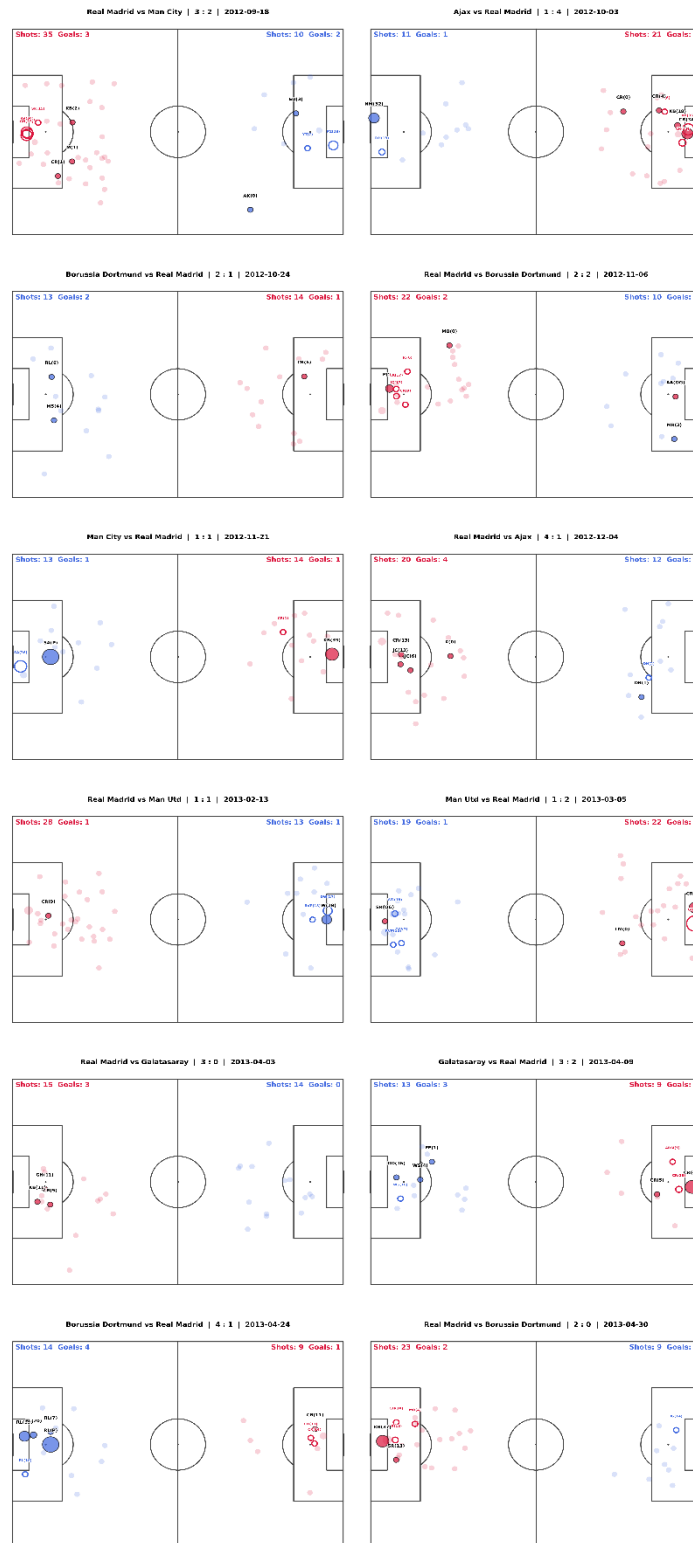


Figure 1 Real Madrid Shot Map — UCL 2012-2013

Barcelona Spatial Profile

Barcelona's spatial profile reveals a markedly different pattern, particularly against Tier 1 opponents. While the lateral distribution remained broadly central across both tiers — 77.2% central against Tier 1 and 78.5% against Tier 2 — the depth distribution shifted significantly. Against Tier 1 opponents, only 38.6% of Barcelona's shots came from inside the box, compared to 48.8% against Tier 2. The proportion from the edge zone rose to 57.9% against Tier 1 — meaning that against elite opposition, Barcelona were generating the majority of their attempts from outside the penalty area.

This shift in depth distribution has a direct impact on goal creation. Barcelona's Box_Central conversion rate against Tier 1 opponents was 11.1% — 18 shots producing 2 goals. However, the composition of those 2 goals is critically important. The highest xG shot in Barcelona's entire Tier 1 campaign was Xavi's penalty against PSG, with an xG of 0.760. A penalty is by definition a set piece situation rather than a product of open play spatial organization. Removing the penalty, Barcelona's remaining Tier 1 Box_Central goal came from open play, while the high xG shots list is otherwise dominated by Marc Bartra — a central defender — whose box appearances reflect defensive set pieces rather than structured attacking penetration. Barcelona's open play box penetration against elite opposition was therefore even more limited than the raw numbers suggest.

Against Tier 2 opponents Barcelona's spatial profile improved — 48.8% box shots, Box_Central conversion rate of 20.0%, and 10 goals from 50 attempts in that zone. This confirms that Barcelona could generate structured box penetration against weaker opposition, but that structure broke down when facing elite defences that could effectively organise against their attacking patterns.

**Shot Analysis: Barcelona (Blue) vs Opponents (Red)
Champions League 2012-2013**

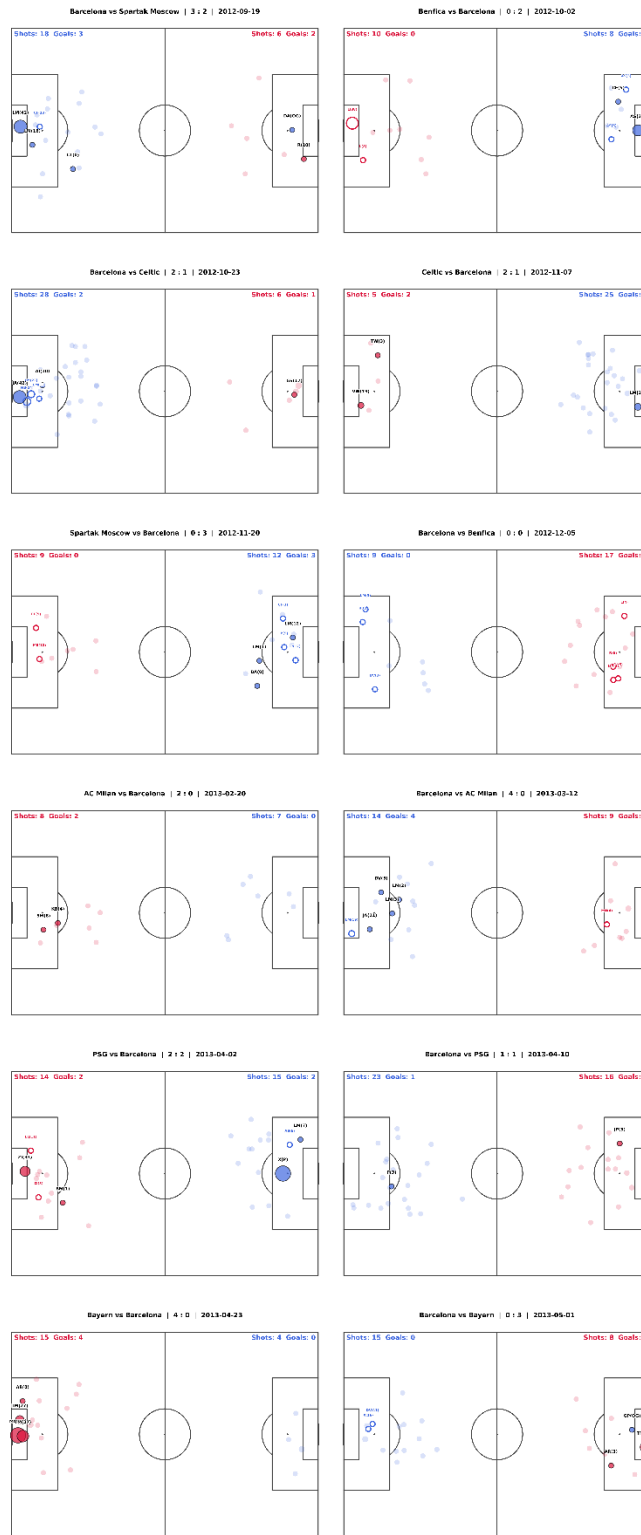


Figure 2 Barcelona Shot Map — UCL 2012-2013

Comparative Spatial Insight

The zone data establishes a clear spatial contrast between the two clubs. Real Madrid maintained consistent box penetration across both tiers — 46.7% against Tier 1 and 56.9% against Tier 2 — with Box_Central as a reliable and productive source of goals in both contexts. Barcelona's box penetration dropped meaningfully against Tier 1 opponents, and their limited Tier 1 box goals were heavily influenced by a penalty and defensive set piece contributions rather than sustained open play attacking structure.

The edge zone dominance in Barcelona's Tier 1 profile — 57.9% of shots — coincides with a sharp drop in central box entry against elite opposition. Edge_Central produced only 1 goal from 24 Tier 1 attempts at a 4.2% conversion rate. This edge-weighted Tier 1 distribution is one of the earliest and clearest signals in the dataset of Barcelona's difficulty generating high-value chances against the highest quality opposition [DATA]. Whether it reflects opponent compression, build-up structure, or final-third entry is a question for Modules 6 and 8 [HYPOTHESIS].

Table 10 Zone Shot Distribution — Real Madrid and Barcelona, UCL 2012-2013

Depth Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box	78	46.7%	37	56.9%	22	38.6%	59	48.8%
Edge	81	48.5%	25	38.5%	33	57.9%	58	47.9%
Long Range	8	4.8%	3	4.6%	2	3.5%	4	3.3%
TOTAL	167	100%	65	100%	57	100%	121	100%

Lateral Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Left	21	12.6%	10	15.4%	8	14.0%	18	14.9%
Central	125	74.9%	46	70.8%	44	77.2%	95	78.5%
Right	21	12.6%	9	13.8%	5	8.8%	8	6.6%
TOTAL	167	100%	65	100%	57	100%	121	100%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Left	10	6.0%	4	6.2%	2	3.5%	8	6.6%
Box_Central	60	35.9%	27	41.5%	18	31.6%	50	41.3%
Box_Right	8	4.8%	6	9.2%	2	3.5%	1	0.8%
Edge_Left	11	6.6%	5	7.7%	6	10.5%	10	8.3%
Edge_Central	58	34.7%	17	26.2%	24	42.1%	41	33.9%
Edge_Right	12	7.2%	3	4.6%	3	5.3%	7	5.8%
Long Range_Left	0	0.0%	1	1.5%	0	0.0%	0	0.0%
Long Range_Central	7	4.2%	2	3.1%	2	3.5%	4	3.3%
Long Range_Right	1	0.6%	0	0.0%	0	0.0%	0	0.0%
TOTAL	167	100%	65	100%	57	100%	121	100%

Table 11 Zone xG and Goals by Zone — Real Madrid and Barcelona, UCL 2012-2013

Depth Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box	78	0.125	9	11.5%	37	0.114	11	29.7%	22	0.118	2	9.1%	59	0.109	10	16.9%
Edge	81	0.010	4	4.9%	25	0.009	2	8.0%	33	0.013	1	3.0%	58	0.011	5	8.6%
Long Range	8	0.002	0	0.0%	3	0.002	0	0.0%	2	0.001	0	0.0%	4	0.003	0	0.0%

Lateral Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Left	21	0.020	1	4.8%	10	0.012	0	0.0%	8	0.019	0	0.0%	18	0.015	1	5.6%
Central	125	0.079	11	8.8%	46	0.089	13	28.3%	44	0.063	3	6.8%	95	0.071	14	14.7%
Right	21	0.015	1	4.8%	9	0.025	0	0.0%	5	0.021	0	0.0%	8	0.010	0	0.0%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box_Left	10	0.038	1	10.0%	4	0.022	0	0.0%	2	0.058	0	0.0%	8	0.025	0	0.0%
Box_Central	60	0.152	8	13.3%	27	0.146	11	40.7%	18	0.134	2	11.1%	50	0.124	10	20.0%
Box_Right	8	0.029	0	0.0%	6	0.033	0	0.0%	2	0.037	0	0.0%	1	0.039	0	0.0%
Edge_Left	11	0.003	0	0.0%	5	0.006	0	0.0%	6	0.006	0	0.0%	10	0.006	1	10.0%
Edge_Central	58	0.012	3	5.2%	17	0.010	2	11.8%	24	0.015	1	4.2%	41	0.012	4	9.8%
Edge_Right	12	0.007	1	8.3%	3	0.010	0	0.0%	3	0.011	0	0.0%	7	0.006	0	0.0%

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Long Range_Left	0	-	0	-	1	0.002	0	0.0%	0	-	0	-	0	-	0	-
Long Range_Central	7	0.002	0	0.0%	2	0.002	0	0.0%	2	0.001	0	0.0%	4	0.003	0	0.0%
Long Range_Right	1	0.001	0	0.0%	0	-	0	-	0	-	0	-	0	- 3333	0	-

Player Highlights

Table 16 presents the season average shot metrics for all players with a shot share above 5% for both Real Madrid and Barcelona across the 2012-2013 UEFA Champions League campaign.

Real Madrid's attacking load was heavily concentrated on Cristiano Ronaldo, who led the squad with a 36.9% shot share and 51.4% goal share — meaning he was responsible for more than half of all Real Madrid goals in the campaign. His xG per match of 0.413 and rationality of 0.063 reflect a player generating high volume from reasonable positions. Karim Benzema provided the most consistent support with 30.2% goal share and the highest rationality among the regular contributors at 0.087, reflecting disciplined shot selection from central positions at 54.2% box shot rate. Ángel Di María carried the second highest shot share at 16.8% but contributed zero goals and a low rationality of 0.025 — 43.3% of his attempts came from the right channel with only 31.5% from the box, indicating a high proportion of speculative wide attempts.

Marcelo appeared with a 9.5% shot share and 16.7% goal share entirely from open play left channel runs — all 3.0 shots per match were classified as open play with zero set pieces [DATA]. His attacking contribution reflects genuine tactical involvement as an overlapping full back rather than set piece positioning. Sergio Ramos contributed a 10.1% shot share with 0.8 set piece shots per match confirmed in the data [DATA], however his single goal in 2012-2013 was classified as open play [DATA] — suggesting his set piece involvement generated attempts but did not directly produce goals in this season."

Barcelona's player profile presents a more complex picture. Lionel Messi led with a 29.6% shot share and 33.3% goal share — a significant concentration but lower than Ronaldo's dominance at Real Madrid. Marc Bartra appeared second with a 23.5% shot share and 38.8% xG share despite zero goals. The data confirms that Bartra averaged 0.67 set piece

shots per match against Tier 1 opponents [DATA], meaning a meaningful portion of his box appearances originated from corners and set pieces. However, he also recorded open play shots — the full extent to which his shot contribution reflects set piece positioning versus tactical involvement cannot be determined from this data alone [CANNOT CONFIRM]. What the data does confirm is that his presence as the second highest shot contributor is analytically significant regardless of origin — it indicates that Barcelona's recognized attacking players were generating fewer attempts than expected, pushing a central defender up the shot share ranking.

Gerard Piqué recorded zero set piece shots across the season [DATA], meaning his box appearances were classified as open play. The tactical context of those attempts — whether positional overlaps, second ball situations, or genuine attacking runs — cannot be confirmed from the shot data [CANNOT CONFIRM].

Alexis Sánchez contributed 20.8% shot share with 12.5% goal share, while Andrés Iniesta, Xavi, and David Villa provided moderate contributions. The wider distribution across Barcelona's squad suggests a more collectivist attacking structure, though the presence of a central defender at second place in shot share raises questions about the depth of genuine attacking output that the data alone cannot fully resolve [CANNOT CONFIRM].

Xavi's single goal in 2012-2013 came from a penalty [DATA] — the only set piece derived goal Barcelona scored in this campaign — confirming Barcelona's near complete dependency on open play for their goals, in contrast to Real Madrid's broader set piece involvement."

Table 12 Player Season Average — Shot Analysis 2012-2013 (Shot Share > 5%)

Player	Team	Shots	Open Play	Set Piece	Goals	xG	Rationality	Box %	Left %	Right %	Shot Share %	xG Share %	Goal Share %
Cristiano Ronaldo	Real Madrid	6.75	6.58	0.08	1.00	0.413	0.063	51.0%	20.9%	6.3%	36.9%	36.7%	51.4%
Ángel Di María	Real Madrid	3.11	3.11	0.00	0.00	0.071	0.025	31.5%	4.1%	43.3%	16.8%	8.0%	0.0%
Karim Benzema	Real Madrid	2.75	2.75	0.00	0.63	0.282	0.087	54.2%	18.8%	24.0%	14.7%	21.2%	30.2%
José Callejón	Real Madrid	2.33	2.33	0.00	0.67	0.162	0.060	50.0%	0.0%	33.3%	11.0%	12.2%	16.7%
Sergio Ramos	Real Madrid	2.17	1.50	0.67	0.17	0.231	0.098	83.3%	8.3%	20.8%	10.1%	15.0%	8.3%
Marcelo	Real Madrid	3.00	3.00	0.00	0.50	0.083	0.028	60.0%	100.0%	0.0%	9.5%	4.2%	16.7%
Sami Khedira	Real Madrid	1.71	1.71	0.00	0.00	0.118	0.064	61.9%	0.0%	26.2%	9.2%	9.1%	0.0%
Kaká	Real Madrid	1.75	1.75	0.00	0.25	0.070	0.052	50.0%	37.5%	0.0%	8.2%	4.2%	6.3%
Gonzalo Higuaín	Real Madrid	1.67	1.50	0.17	0.17	0.262	0.136	83.3%	5.6%	0.0%	8.1%	16.3%	5.6%
Raphaël Varane	Real Madrid	1.50	1.50	0.00	0.00	0.214	0.134	83.3%	0.0%	0.0%	7.8%	16.7%	0.0%
Michael Essien	Real Madrid	1.50	1.50	0.00	0.00	0.020	0.010	12.5%	37.5%	0.0%	7.3%	6.1%	0.0%
Luka Modrić	Real Madrid	1.22	1.22	0.00	0.11	0.016	0.013	5.6%	11.1%	16.7%	6.9%	1.4%	5.6%
Mesut Özil	Real Madrid	1.50	1.33	0.17	0.17	0.055	0.047	38.9%	0.0%	22.2%	6.8%	4.0%	8.3%
Lionel Messi	Barcelona	4.10	4.10	0.00	0.80	0.237	0.052	48.2%	11.4%	27.9%	29.6%	25.0%	33.3%
Marc Bartra	Barcelona	2.00	1.33	0.67	0.00	0.311	0.156	100.0%	0.0%	0.0%	23.5%	38.8%	0.0%
Alexis Sánchez	Barcelona	3.00	3.00	0.00	0.25	0.306	0.112	81.3%	27.1%	8.3%	20.8%	28.9%	12.5%
David Villa	Barcelona	2.86	2.86	0.00	0.14	0.085	0.030	47.6%	33.3%	17.9%	18.2%	14.3%	3.6%

Player	Team	Shots	Open Play	Set Piece	Goals	xG	Rationality	Box %	Left %	Right %	Shot Share %	xG Share %	Goal Share %
Andrés Iniesta	Barcelona	2.75	2.75	0.00	0.13	0.073	0.021	19.3%	40.2%	0.0%	15.2%	8.6%	6.3%
Xavi	Barcelona	2.13	2.00	0.00	0.13	0.140	0.061	25.0%	39.6%	10.4%	13.1%	13.8%	6.3%
Pedro	Barcelona	2.00	2.00	0.00	0.13	0.114	0.051	62.5%	25.0%	25.0%	12.4%	13.0%	12.5%
Adriano	Barcelona	2.67	2.67	0.00	0.00	0.027	0.010	0.0%	33.3%	26.7%	12.3%	2.2%	0.0%
Gerard Piqué	Barcelona	1.00	1.00	0.00	0.00	0.094	0.094	100.0%	0.0%	0.0%	11.5%	17.8%	0.0%
Cesc Fàbregas	Barcelona	1.80	1.80	0.00	0.20	0.063	0.041	70.0%	40.0%	25.0%	11.2%	6.7%	10.0%
Dani Alves	Barcelona	1.40	1.40	0.00	0.20	0.014	0.008	10.0%	20.0%	50.0%	8.6%	1.9%	6.7%
Jordi Alba	Barcelona	1.25	1.25	0.00	0.50	0.143	0.139	62.5%	50.0%	0.0%	6.4%	11.1%	18.8%

Player Tier Comparison — Season 2012-2013

Table 13 presents the player-level tier breakdown for all players with a shot share above 5%.

The tier comparison reveals the most analytically significant player-level finding of the season. Lionel Messi's shot profile collapsed dramatically against Tier 1 opponents — from 4.75 shots per match against Tier 2 to just 1.50 against Tier 1, with xG dropping from 0.279 to 0.068. Critically, all of Messi's Tier 1 shots were classified as open play [DATA] — meaning elite opponents were not merely restricting his set piece involvement but suppressing his open play threat entirely. His 0.50 goals per match against Tier 1 came from a very small sample, making this figure unreliable as a performance indicator [CANNOT CONFIRM].

Marc Bartra's shot share against Tier 1 opponents rose to 31.7% — the highest of any Barcelona player against elite opposition. Against Bayern Munich specifically, 1 of his 2 shots in Leg 1 was a confirmed set piece [DATA] and 1 of his 2 shots in Leg 2 was also a confirmed set piece [DATA]. The remaining open play attempts reflect positional box presence that cannot be further classified [CANNOT CONFIRM]. The structural implication remains clear — with Messi suppressed to 1.50 shots per Tier 1 match, a central defender became Barcelona's most prominent shot contributor against the best teams in Europe.

Cristiano Ronaldo showed no meaningful decline against Tier 1 opponents — 6.88 shots per match against Tier 1 compared to 6.50 against Tier 2, with 6.63 open play shots per Tier 1 match [DATA] confirming his volume was driven by genuine attacking involvement rather than set piece situations. Karim Benzema similarly maintained consistent shot volume and rationality across both tiers. The contrast between the two clubs at the player level mirrors the team-level tier finding precisely — Real Madrid's key contributors maintained performance levels against elite opposition while Barcelona's primary attacking player was significantly suppressed, with shot burden partially absorbed by a central defender whose contributions were at least partially set piece derived.

Table 13 Player Tier Comparison — Shot Analysis 2012-2013 (Shot Share > 5%)

Player	Team	T1 Shots	T1 OP	T1 SP	T1 Goals	T1 xG	T1 Rat.	T1 Box%	T1 Shot%	T2 Shots	T2 OP	T2 SP	T2 Goals	T2 xG	T2 Rat.	T2 Box%	T2 Shot%
Cristiano Ronaldo	Real Madrid	6.88	6.63	0.13	0.63	0.388	0.058	51.6%	34.2%	6.50	6.50	0.00	1.75	0.464	0.073	49.6%	42.5%
Ángel Di María	Real Madrid	3.83	3.83	0.00	0.00	0.071	0.019	13.9%	18.4%	1.67	1.67	0.00	0.00	0.071	0.038	66.7%	13.4%
Karim Benzema	Real Madrid	2.75	2.75	0.00	0.75	0.273	0.087	45.8%	13.4%	2.75	2.75	0.00	0.50	0.292	0.088	62.5%	15.9%
José Callejón	Real Madrid	4.00	4.00	0.00	0.00	0.277	0.069	50.0%	18.2%	1.50	1.50	0.00	1.00	0.105	0.055	50.0%	7.4%
Sergio Ramos	Real Madrid	2.20	1.40	0.80	0.20	0.237	0.097	90.0%	10.2%	2.00	2.00	0.00	0.00	0.203	0.102	50.0%	9.5%
Marcelo	Real Madrid	5.00	5.00	0.00	1.00	0.137	0.027	20.0%	14.3%	1.00	1.00	0.00	0.00	0.028	0.028	100.0%	4.8%
Sami Khedira	Real Madrid	1.83	1.83	0.00	0.00	0.137	0.075	72.2%	9.9%	1.00	1.00	0.00	0.00	0.004	0.004	0.0%	5.0%
Kaká	Real Madrid	1.50	1.50	0.00	0.00	0.098	0.082	75.0%	6.7%	2.00	2.00	0.00	0.50	0.042	0.021	25.0%	9.8%
Gonzalo Higuaín	Real Madrid	2.00	2.00	0.00	0.00	0.351	0.163	75.0%	7.7%	1.00	0.50	0.50	0.50	0.083	0.083	100.0%	8.9%
Raphaël Varane	Real Madrid	1.50	1.50	0.00	0.00	0.214	0.132	75.0%	7.6%	1.50	1.50	0.00	0.00	0.216	0.137	100.0%	8.3%

Player	Team	T1 Shots	T1 OP	T1 SP	T1 Goals	T1 xG	T1 Rat.	T1 Box%	T1 Shot%	T2 Shots	T2 OP	T2 SP	T2 Goals	T2 xG	T2 Rat.	T2 Box%	T2 Shot%
Michael Essien	Real Madrid	1.67	1.67	0.00	0.00	0.025	0.013	16.7%	8.1%	1.00	1.00	0.00	0.00	0.002	0.002	0.0%	4.8%
Luka Modrić	Real Madrid	1.14	1.14	0.00	0.14	0.015	0.014	0.0%	6.5%	1.50	1.50	0.00	0.00	0.019	0.011	25.0%	8.4%
Mesut Özil	Real Madrid	1.60	1.40	0.20	0.20	0.063	0.054	46.7%	6.8%	1.00	1.00	0.00	0.00	0.013	0.013	0.0%	6.7%
Lionel Messi	Barcelona	1.50	1.50	0.00	0.50	0.068	0.045	75.0%	19.2%	4.75	4.75	0.00	0.88	0.279	0.054	41.5%	32.2%
Marc Bartra	Barcelona	2.00	1.00	1.00	0.00	0.280	0.140	100.0%	31.7%	2.00	2.00	0.00	0.00	0.374	0.187	100.0%	7.1%
Alexis Sánchez	Barcelona	4.00	4.00	0.00	0.00	0.110	0.028	25.0%	26.7%	2.67	2.67	0.00	0.33	0.371	0.140	100.0%	18.9%
David Villa	Barcelona	4.00	4.00	0.00	0.00	0.128	0.032	25.0%	22.1%	2.40	2.40	0.00	0.20	0.069	0.029	56.7%	16.7%
Andrés Iniesta	Barcelona	3.67	3.67	0.00	0.00	0.112	0.022	23.8%	19.0%	2.20	2.20	0.00	0.20	0.050	0.021	16.7%	13.0%
Xavi	Barcelona	2.00	1.67	0.00	0.33	0.317	0.143	44.4%	11.8%	2.20	2.20	0.00	0.00	0.034	0.013	13.3%	13.9%
Pedro	Barcelona	3.00	3.00	0.00	0.50	0.081	0.027	16.7%	16.5%	1.67	1.67	0.00	0.00	0.126	0.059	77.8%	11.0%
Adriano	Barcelona	2.00	2.00	0.00	0.00	0.030	0.015	0.0%	13.3%	3.00	3.00	0.00	0.00	0.026	0.008	0.0%	11.8%
Gerard Piqué	Barcelona	1.00	1.00	0.00	0.00	0.093	0.092	100.0%	14.7%	1.00	1.00	0.00	0.00	0.095	0.095	100.0%	9.4%
Cesc Fàbregas	Barcelona	2.50	2.50	0.00	0.00	0.043	0.022	50.0%	12.1%	1.33	1.33	0.00	0.33	0.077	0.053	83.3%	10.6%
Dani Alves	Barcelona	1.50	1.50	0.00	0.00	0.024	0.013	25.0%	7.7%	1.33	1.33	0.00	0.33	0.007	0.004	0.0%	9.3%
Jordi Alba	Barcelona	1.00	1.00	0.00	0.00	0.004	0.004	0.0%	6.7%	1.33	1.33	0.00	0.67	0.189	0.184	83.3%	6.2%

Critical Match Analysis — Semi-Finals

The Semi-Final stage proved decisive for both clubs. Table 14 and Table 15 present the player-level shot data for Real Madrid's two legs against Borussia Dortmund, and Table 16 and Table 17 present the equivalent data for Barcelona against Bayern Munich.

Table 14 Player Shot Data — Borussia Dortmund vs Real Madrid, Semi-Final Leg 1, Lost 4:1, 2013-04-24

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Robert Lewandowski	Borussia Dortmund	7	4	1.598	0.228	85.7%	50.0%
Ilkay Gündogan	Borussia Dortmund	2	0	0.019	0.010	0.0%	14.3%
Jakub Blaszczykowski	Borussia Dortmund	2	0	0.101	0.050	100.0%	14.3%
Marco Reus	Borussia Dortmund	2	0	0.064	0.032	50.0%	14.3%
Marcel Schmelzer	Borussia Dortmund	1	0	0.006	0.006	0.0%	7.1%
TEAM TOTAL	Borussia Dortmund	14	4	1.788	0.128	-	100%
Cristiano Ronaldo	Real Madrid	5	1	0.598	0.120	80.0%	55.6%
Luka Modrić	Real Madrid	1	0	0.005	0.005	0.0%	11.1%
Raphaël Varane	Real Madrid	1	0	0.171	0.171	100.0%	11.1%
Sami Khedira	Real Madrid	1	0	0.022	0.022	100.0%	11.1%
Xabi Alonso	Real Madrid	1	0	0.004	0.004	0.0%	11.1%
TEAM TOTAL	Real Madrid	9	1	0.800	0.089	-	100%

Table 15 Player Shot Data — Real Madrid vs Borussia Dortmund, Semi-Final Leg 2, Won 2:0, 2013-04-30

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	6	0	0.375	0.062	50.0%	26.1%
Sergio Ramos	Real Madrid	4	1	0.606	0.152	100.0%	17.4%
Ángel Di María	Real Madrid	3	0	0.036	0.012	0.0%	13.0%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Karim Benzema	Real Madrid	3	1	0.428	0.143	66.7%	13.0%
Sami Khedira	Real Madrid	2	0	0.018	0.009	0.0%	8.7%
Gonzalo Higuaín	Real Madrid	1	0	0.103	0.103	100.0%	4.3%
Michael Essien	Real Madrid	1	0	0.002	0.002	0.0%	4.3%
Mesut Özil	Real Madrid	1	0	0.044	0.044	100.0%	4.3%
Kaká	Real Madrid	1	0	0.132	0.132	100.0%	4.3%
Raphaël Varane	Real Madrid	1	0	0.035	0.035	0.0%	4.3%
TEAM TOTAL	Real Madrid	23	2	1.780	0.077	-	100%
Robert Lewandowski	Borussia Dortmund	5	0	0.364	0.073	100.0%	55.6%
Ilkay Gündogan	Borussia Dortmund	1	0	0.141	0.141	100.0%	11.1%
Jakub Blaszczykowski	Borussia Dortmund	1	0	0.066	0.066	100.0%	11.1%
Lukasz Piszczek	Borussia Dortmund	1	0	0.019	0.019	0.0%	11.1%
Marco Reus	Borussia Dortmund	1	0	0.004	0.004	0.0%	11.1%
TEAM TOTAL	Borussia Dortmund	9	0	0.593	0.066	-	100%

Real Madrid vs Borussia Dortmund

Real Madrid's Semi-Final elimination by Borussia Dortmund was defined almost entirely by a single player in a single match. In Leg 1 away at Dortmund, Robert Lewandowski produced one of the most dominant individual attacking performances in UEFA Champions League Semi-Final history — 7 shots, 4 goals, xG 1.598, with 85.7% of his attempts from inside the box. He alone accounted for 50% of Dortmund's total shots and generated more xG in one match than Barcelona's entire team would generate against Bayern across two legs. Real Madrid's response was limited to 9 shots in total, with

Cristiano Ronaldo contributing 5 of them for 1 goal at an xG of 0.598. The remaining four Real Madrid contributors combined for just 4 shots with a total xG of 0.202 — offering no meaningful attacking support. Real Madrid were tactically contained to a degree that concentrated all attacking responsibility on a single player, and the deficit created in this match proved insurmountable.

Leg 2 at the Bernabéu told a completely different story. Real Madrid responded with a significantly more distributed and purposeful attacking performance — 10 players contributed shots, generating 23 attempts with a combined xG of 1.780. Sergio Ramos contributed 4 shots and 1 goal with an xG of 0.606, Karim Benzema added 3 shots and 1 goal with an xG of 0.428, and Cristiano Ronaldo led with 6 shots and an xG of 0.375 despite not scoring. The breadth of contribution — with forwards, midfielders, and defenders all generating meaningful attempts — reflected a far more cohesive and structured attacking display. Borussia Dortmund were effectively contained to 9 shots with a total xG of 0.593, Lewandowski generating 5 of them but converting none. Real Madrid won the match comfortably but could not overcome the four-goal deficit from the first leg. The two-match contrast raises an important analytical question that this study cannot fully answer from shot data alone — whether Real Madrid's Leg 1 performance represented a tactical failure, a preparation failure, or simply the effect of an extraordinary individual performance that no defensive system could have reliably prevented.

Table 16 Player Shot Data — Bayern Munich vs Barcelona, Semi-Final Leg 1, Lost 4:0, 2013-04-23

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Arjen Robben	Bayern Munich	5	1	0.295	0.059	60.0%	33.3%
Thomas Müller	Bayern Munich	5	2	0.849	0.170	80.0%	33.3%
Dante	Bayern Munich	1	0	0.119	0.119	100.0%	6.7%
Franck Ribéry	Bayern Munich	1	0	0.034	0.034	100.0%	6.7%
Javi Martínez	Bayern Munich	1	0	0.210	0.210	100.0%	6.7%
Mario Gomez	Bayern Munich	1	1	0.519	0.519	100.0%	6.7%
Philipp Lahm	Bayern Munich	1	0	0.004	0.004	0.0%	6.7%
TEAM TOTAL	Bayern Munich	15	4	2.030	0.135	-	100%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Marc Bartra	Barcelona	2	0	0.308	0.154	100.0%	50.0%
Gerard Piqué	Barcelona	1	0	0.118	0.118	100.0%	25.0%
Lionel Messi	Barcelona	1	0	0.045	0.045	100.0%	25.0%
TEAM TOTAL	Barcelona	4	0	0.470	0.118	-	100%

Table 17 Player Shot Data — Barcelona vs Bayern Munich, Semi-Final Leg 2, Lost 0:3, 2013-05-01

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
David Villa	Barcelona	4	0	0.147	0.037	25.0%	26.7%
Pedro	Barcelona	3	0	0.035	0.012	0.0%	20.0%
Adriano	Barcelona	2	0	0.030	0.015	0.0%	13.3%
Andrés Iniesta	Barcelona	2	0	0.029	0.014	0.0%	13.3%
Marc Bartra	Barcelona	2	0	0.251	0.126	100.0%	13.3%
Thiago Alcântara	Barcelona	1	0	0.168	0.168	100.0%	6.7%
Xavi	Barcelona	1	0	0.162	0.162	100.0%	6.7%
TEAM TOTAL	Barcelona	15	0	0.822	0.055	-	100%
Arjen Robben	Bayern Munich	3	1	0.103	0.034	66.7%	37.5%
Thomas Müller	Bayern Munich	2	1	0.362	0.181	100.0%	25.0%
David Alaba	Bayern Munich	1	0	0.005	0.005	0.0%	12.5%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Philipp Lahm	Bayern Munich	1	0	0.015	0.015	0.0%	12.5%
Rafinha	Bayern Munich	1	0	0.052	0.052	100.0%	12.5%
TEAM TOTAL	Bayern Munich	8	2	0.537	0.067	-	100%

Note: Bayern's team total shows two goals against a final score of 0:3. Their third goal was an own goal, which under the player-attribution convention set out in 4.1 is credited to no shooter and appears in neither team's shot data.

Barcelona vs Bayern Munich

Barcelona's elimination by Bayern Munich is among the most analytically striking results in this dataset, and the player-level data explains why. In Leg 1 at the Allianz Arena, Bayern generated 15 shots from 7 different players with a total xG of 2.030 — a figure reflecting sustained, high-quality attacking pressure across the entire match. Thomas Müller alone contributed 5 shots and 2 goals with an xG of 0.849, Arjen Robben added 5 shots and 1 goal, and Mario Gomez converted a single high-quality opportunity with an xG of 0.519. Bayern's attack was distributed, purposeful, and lethal from multiple directions simultaneously.

Barcelona's response was reduced to 4 shots from 3 players — the lowest single-match total in their entire 2012-2013 campaign. Marc Bartra, a central defender, accounted for 50% of Barcelona's shots. Lionel Messi, the focal point of Barcelona's entire attacking system and widely regarded as the best player in the world at this time, was limited to a single shot with an xG of only 0.045. This was not a statistical coincidence — it was a reflection of Bayern's systematic tactical suppression of Barcelona's creative structure. With Messi neutralised and no alternative attacking mechanism capable of generating meaningful threat, Barcelona's xG of 0.470 across the entire match represents a complete attacking collapse against elite opposition.

Leg 2 at the Camp Nou produced a superficially more competitive shot count — Barcelona generated 15 shots from 7 contributors — but the underlying quality data tells a different story. Barcelona's total xG across 15 shots was only 0.822, a rationality of 0.055 per shot reflecting an extremely low average shot quality. Critically, Lionel Messi does not appear in the shot data at all in Leg 2. Barcelona's leading contributors were David Villa with 4 shots and an xG of 0.147, Pedro with 3 shots and an xG of only 0.035, and Adriano and Iniesta combining for 4 shots with a total xG of 0.059. These are peripheral contributors generating

peripheral attempts — Barcelona were shooting in volume from positions that offered no genuine goal threat. Bayern, by contrast, needed only 8 shots to score twice, with Robben and Müller again the primary contributors at an efficient combined xG of 0.465 and a rationality of 0.067.

The two-match aggregate tells a clear story. Bayern generated 23 shots with a combined xG of 2.567 and scored 6 goals. Barcelona generated 19 shots with a combined xG of 1.292 and scored 0. Bayern's average rationality across both matches was 0.112 — more than double Barcelona's 0.068. The structural suppression of Messi across 180 minutes was complete, and Barcelona could not construct an alternative attacking pathway. This result was not an upset — the shot data suggests it was a systematic tactical dismantling that Barcelona had no answer for at any point across the tie.

Key Observations — Season 2012-2013

The 2012-2013 season establishes several early themes that form the foundation of this comparative study. The evidence across season overview, tier comparison, spatial distribution, and critical match analysis consistently points in the same direction — Real Madrid demonstrated greater structural resilience against elite opposition while Barcelona showed a pattern of performance deterioration when facing the highest quality opponents.

At the season level, Real Madrid outperformed Barcelona across every primary attacking metric — shot volume, goals, xG, rationality, and Field Tilt. The gap was not marginal. Real Madrid averaged 19.3 shots per match compared to Barcelona's 14.8, and their Quality Margin of 0.38 indicated a larger surplus of chance quality over opponents than Barcelona's 0.12. Both teams showed similar shot accuracy at approximately 37%, confirming that the differentiator was attacking volume and chance creation rather than finishing precision.

The tier breakdown reveals the most significant divergence. Real Madrid maintained a positive Quality Margin against Tier 1 opponents and a Field Tilt above 60%, demonstrating that their attacking structure remained functional even against elite defences. Barcelona's Quality Margin against Tier 1 opponents was negative — elite opponents generated more chance quality than Barcelona did — and their Field Tilt dropped below 50%, meaning those opponents also achieved greater territorial dominance against Barcelona than Barcelona achieved against them. That Barcelona were out-created in chance quality by elite opposition, despite their reputation as the finest team in the world, is a finding that warrants careful attention as the study progresses across subsequent seasons.

The spatial analysis adds important context to the tier findings. Real Madrid maintained consistent box penetration across both tiers — approximately 47% of shots from inside the

box against Tier 1 opponents — with Box_Central as a reliable and productive source of goals at a 13.3% conversion rate. Barcelona's box penetration dropped to 38.6% against Tier 1 opponents, with the majority of shots shifting to the edge zone at 57.9%. Barcelona's two Tier 1 Box_Central goals were heavily influenced by a penalty and a defender's set piece attempt rather than sustained open play penetration, suggesting their genuine open play box access against elite opposition was even more limited than the raw figures indicate.

The critical match analysis provides the sharpest illustration of these patterns. Real Madrid's elimination by Borussia Dortmund was driven primarily by a single extraordinary individual performance — Lewandowski's 7-shot, 4-goal display in Leg 1 — rather than a sustained structural failure. Real Madrid recovered in Leg 2 with a distributed, purposeful attacking performance from 10 contributors. Barcelona's elimination by Bayern Munich, by contrast, reflected a systematic structural collapse across 180 minutes. Bayern's tactical suppression of Messi reduced Barcelona to 4 shots in Leg 1 — the lowest single-match total in their campaign — with a defender accounting for 50% of those attempts. In Leg 2, Barcelona generated 15 shots but with a combined xG of only 0.822 and Messi absent from the shot data entirely. Barcelona had no alternative attacking mechanism capable of generating genuine goal threat when their primary system was neutralized.

It is important to acknowledge what the data cannot confirm at this stage. Whether Barcelona's Tier 1 vulnerability reflected tactical choices, opponent-specific game plans, squad depth limitations, or deeper structural issues in their playing system cannot be determined from a single season of shot data alone. These questions will be revisited across the subsequent five seasons and across all five analytical pillars before any conclusions are drawn.

4.3 Season 2013-2014

Season Overview

The 2013-2014 season reveals a significant tactical evolution for Real Madrid compared to the previous campaign. Despite generating fewer shots per match — down from 19.3 to 16.2 — Real Madrid scored substantially more goals, rising from 2.2 to 3.2 per match. This improvement was not captured by the distance-based xG model, which remained broadly unchanged at 1.28 compared to 1.25 the previous season. The more meaningful quality indicators tell a clearer story: accuracy jumped from 37.4% to 48.0%, rationality improved from 0.067 to 0.079, and box shot penetration rose from 51.6% to 55.8%. Since the custom xG model in this study is based solely on shot distance, metrics such as accuracy, rationality, and box shot percentage serve as important complementary indicators of shot quality that the xG model alone cannot fully capture. Real Madrid were shooting less but from significantly better positions, with greater precision, and against less defensive resistance — the hallmarks of a counter-attack oriented system.

The Finishing Delta rose sharply from 0.91 to 1.87 — meaning Real Madrid scored, on average, 1.87 goals per match above their distance-based xG. Because the xG model cannot see defensive pressure, this large and growing Finishing Delta is the statistical fingerprint of Real Madrid's counter-attacking evolution: shots taken against stretched, retreating defences convert far above their distance-based expectation. The Quality Margin, by contrast, held steady at 0.38 — Real Madrid's surplus of chance quality over opponents was essentially unchanged from the previous season. The story of 2013-2014 is therefore not that Real Madrid created better positions than their opponents than before, but that they converted their chances in what the shot data suggests were less congested conditions [HYPOTHESIS]. Field Tilt dropping from 58.7% to 53.7% is consistent with this: Real Madrid conceded territory deliberately and struck in the space it created. The shot pillar records the Finishing Delta and the territorial concession; whether the two are causally linked in the way the counter-attack reading proposes is examined in Modules 6 and 8.

Barcelona also improved across most metrics in 2013-2014. xG rose from 0.84 to 1.11, rationality matched Real Madrid at 0.079, box shot penetration improved from 50.0% to 56.7%. Field Tilt rose sharply from 56.8% to 64.5%, and opponent shots and xG both fell, indicating tighter defensive control. Barcelona's Quality Margin improved from 0.12 to 0.58 — a genuine increase in their chance-quality advantage over opponents, driven partly by conceding fewer and lower-quality shots (opponent xG fell from 0.72 to 0.54). Their Finishing Delta also rose, from 0.66 to 0.99. On the quality metrics Barcelona had a materially better season than 2012-2013. However Barcelona's accuracy was virtually

unchanged at 36.5% compared to 37.0% the previous season — a notable contrast to Real Madrid's sharp improvement. Barcelona generated more possession and more territorial dominance, but their finishing execution did not improve meaningfully.

Comparing the two clubs, the divergence in 2013-2014 lay in the nature of their improvement. Real Madrid's gain was concentrated in Finishing Delta — an environment-driven surge tied to counter-attacking — while their Quality Margin held flat. Barcelona's gain was spread across Quality Margin and opponent suppression, without a comparable environment effect. A full cross-season comparison across all six seasons is presented in Module 9.

Table 18 Season Overview — Shot Analysis 2013-2014

Metric	Real Madrid	Barcelona
Avg Shots per Match	16.2	14.8
Avg Goals per Match	3.2	2.1
xG per Match	1.28	1.11
Rationality (xG/Shot)	0.079	0.079
Box Shot %	55.8%	56.7%
Accuracy %	48.0%	36.5%
Big Misses per Match	0.38	0.10
Opp Shots per Match	13.7	8.0
Opp xG per Match	0.90	0.54
Finishing Delta (Goals – xG)	1.87	0.99
Quality Margin (xG – Opp xG)	0.38	0.58
Field Tilt %	53.7%	64.5%

Tier Comparison

Table 18 presents the tier breakdown separating performance against Tier 1 (Elite, Elo > 1850) and Tier 2 (Standard, Elo ≤ 1850) opponents.

The tier breakdown for 2013-2014 reinforces and deepens the tactical picture established in the season overview. Against Tier 1 opponents, Real Madrid's Field Tilt dropped to 48.3% — below the 50% threshold, meaning elite opponents generated more shots than Real Madrid in those matches. Yet Real Madrid's Finishing Delta against Tier 1 opponents was strongly positive at 1.20 — they scored well above their distance-based xG against elite sides. Interpreted through the environment lens, this is the counter-attack signature at its clearest: absorbing pressure, conceding territory, and converting the resulting low-congestion chances at a rate the distance model cannot anticipate [HYPOTHESIS]. The pillar locates the anomaly; the mechanism behind it is handed to Modules 6 and 8.

Against Tier 2 opponents, Real Madrid adopted a different approach entirely. Field Tilt rose to 60.0%, indicating territorial dominance against weaker sides. Their Finishing Delta of 2.65 against Tier 2 opponents reflects ruthless conversion — Real Madrid scored well over two goals per match above their distance-based xG, the fingerprint of a team striking against opponents who committed forward and left space behind. Their Quality Margin against Tier 2 sides was a healthy 0.79.

Barcelona's Tier 1 profile presents a striking contrast in style, though not in the direction the raw goal figures first suggest. Field Tilt against elite opponents was 58.1% — actually higher than Real Madrid's 48.3% — meaning Barcelona outshot their Tier 1 opponents territorially. On chance quality, the two clubs were closer than their results imply: Barcelona's Quality Margin against Tier 1 opponents was +0.24, while Real Madrid's was only +0.04. Against elite opposition Real Madrid barely out-created their opponents in xG terms — they were, in effect, level. What separated the two clubs was not chance-quality creation but conversion environment. Real Madrid's Tier 1 Finishing Delta was 1.20 against Barcelona's 0.39 — Real Madrid scored far above their distance-based xG while Barcelona scored only modestly above theirs. The accuracy figures point to the same gap — Real Madrid achieved 43.5% accuracy against Tier 1 opponents compared to Barcelona's 35.8%, despite Barcelona recording a higher box shot percentage of 54.8% against 49.3% for Real Madrid. Barcelona were penetrating the box more frequently against elite opponents but converting those opportunities into on-target attempts at a significantly lower rate. The distance-based xG model does not capture this difference, because it cannot see the defensive pressure around each shot; accuracy, rationality, and Finishing Delta together suggest Real Madrid were taking their Tier 1 chances in materially less congested conditions than Barcelona were [HYPOTHESIS]. This is a reading the shot data is consistent with rather than one it can establish; construction and defensive structure are examined in Modules 6 and 8.

Against Tier 2 opponents Barcelona's Field Tilt rose to 68.8% and their Finishing Delta to 1.38, reflecting comfortable dominance against weaker opposition. Their Tier 2 Quality

Margin of 0.80 was virtually identical to Real Madrid's 0.79 — against standard opposition the two clubs created near-identical surpluses of chance quality. The divergence between them was concentrated entirely at Tier 1, and within Tier 1 it was concentrated in conversion environment rather than chance creation.

The tier comparison establishes a pattern that mirrors 2012-2013 with an important evolution. In 2012-2013 Barcelona's Quality Margin against Tier 1 opponents was negative — they were out-created by elite sides. In 2013-2014 Barcelona actually edged their Tier 1 Quality Margin positive, and above Real Madrid's — a real improvement in chance-quality creation against elite opposition. What did not improve was the outcome: Barcelona were eliminated a round earlier. What changed in 2013-2014 was the identity of the elite opponent — from Bayern Munich's aggressive pressing to Atletico Madrid's deep defensive block — yet the outcome for Barcelona was the same. Real Madrid, by contrast, converted their chances against elite opponents at a rate Barcelona could not match, despite creating no greater quality of chance. This gap — conversion environment rather than chance creation — is flagged as a key theme for the cross-season synthesis in Module 9, where the counter-attack interpretation is examined against the possession-vs-block trap that Barcelona fell into and that Real Madrid would themselves encounter in 2014-2015.

Table 19 Tier Comparison — Shot Analysis 2013-2014

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Avg Shots per Match	14.6	18.2	14.0	15.3
Avg Goals per Match	2.3	4.2	1.25	2.7
xG per Match	1.09	1.51	0.86	1.28
Rationality (xG/Shot)	0.077	0.082	0.068	0.086
Box Shot %	49.3%	63.4%	54.8%	57.9%
Accuracy %	43.5%	53.2%	35.8%	36.9%
Big Misses per Match	0.29	0.50	0.25	0.00
Opp Shots per Match	15.6	11.5	10.0	6.7
Opp xG per Match	1.05	0.72	0.61	0.49

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Finishing Delta (Goals – xG)	1.20	2.65	0.39	1.38
Quality Margin (xG – Opp xG)	0.04	0.79	0.24	0.80
Field Tilt %	48.3%	60.0%	58.1%	68.8%

Shot Map Analysis — Spatial Distribution 2013-2014

Figure 3 and Figure 4 present the shot maps for Real Madrid and Barcelona respectively across all 2013-2014 UEFA Champions League matches. The spatial analysis draws on the zone distribution and zone xG data presented in Tables 20 and 21.

Real Madrid Spatial Profile

Real Madrid's spatial distribution in 2013-2014 maintained the disciplined central structure observed in the previous season, with 69.6% of Tier 1 shots coming from the central corridor — a slight widening from 74.9% in 2012-2013 but still reflecting a predominantly central attacking approach. Box penetration against Tier 1 opponents improved from 46.7% to 50.0%, a meaningful increase given Real Madrid's counter-attack orientation this season. The hypothesis that counter-attacking against elite opponents who committed forward created more box access is consistent with the data — opponents leaving space behind their defensive line may have facilitated easier central penetration — though this cannot be confirmed from shot data alone [HYPOTHESIS]. Box_Central remained the primary goal zone with a 31.7% conversion rate against Tier 1, consistent with the previous season.

Against Tier 2 opponents Real Madrid's box penetration rose further to 65.1%, reflecting their territorial dominance against weaker sides. The lateral distribution remained consistently central at 76.1%, with no significant channel bias in either tier. Real Madrid's spatial profile confirms a structured, disciplined attacking system that adapted depth of penetration based on opponent quality while maintaining central lateral discipline throughout.

Barcelona Spatial Profile

Barcelona's spatial profile shows the most significant change between seasons. Against Tier 1 opponents, box penetration rose sharply from 38.6% in 2012-2013 to 53.6% in 2013-

2014 — the most notable spatial improvement in Barcelona's data across both seasons. Edge zone shooting dropped correspondingly from 57.9% to 46.4%, suggesting Barcelona found more routes into the box against elite opposition than in the previous season. However despite this improved penetration, Box_Central conversion rate against Tier 1 was only 16.7% — lower than Real Madrid's 31.7% — confirming that spatial access did not translate into finishing quality.

The most analytically interesting spatial finding for Barcelona in 2013-2014 is the right channel shift against Tier 1 opponents. Right channel shots rose from 8.8% in 2012-2013 to 21.4%, while the central corridor share dropped from 77.2% to 66.1%. The player tier comparison data confirms that Lionel Messi's lateral profile shifted against Tier 1 opponents — recording 0% left channel and 10% right channel against elite sides compared to 11.1% left and 5.6% right against Tier 2 [DATA]. This suggests Messi adjusted his positioning against elite opposition, moving toward the right channel rather than his more natural central and left-leaning positions against weaker sides. Whether this shift was tactically deliberate or a response to elite opponents' defensive organisation cannot be confirmed from the shot data alone [CANNOT CONFIRM].

**Shot Analysis: Real Madrid (Red) vs Opponents (Blue)
Champions League 2013-2014**

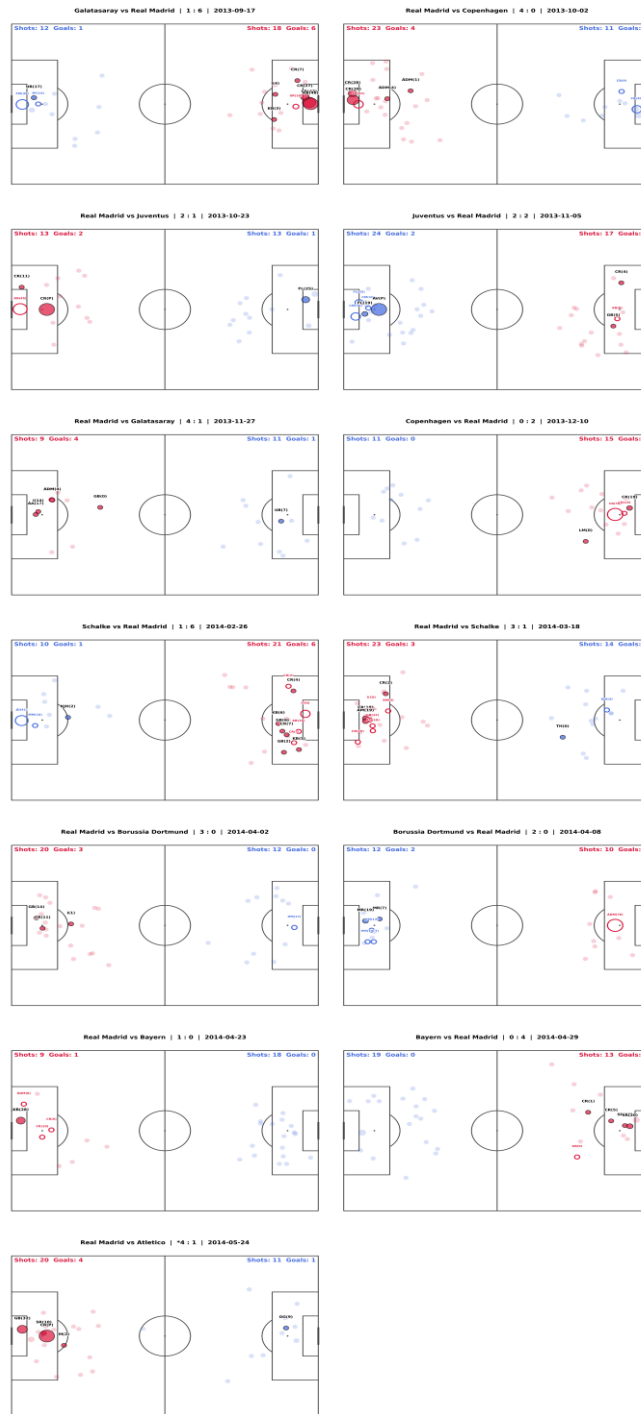


Figure 3 Real Madrid Shot Map — UCL 2013-2014

Barcelona Spatial Profile

Barcelona's spatial profile shows the most significant change between seasons. Against Tier 1 opponents, box penetration rose sharply from 38.6% in 2012-2013 to 53.6% in 2013-2014 — the most notable spatial improvement in Barcelona's data across both seasons. Edge zone shooting dropped correspondingly from 57.9% to 46.4%, suggesting Barcelona found more routes into the box against elite opposition than in the previous season. However despite this improved penetration, Box_Central conversion rate against Tier 1 was only 16.7% — lower than Real Madrid's 31.7% — confirming that spatial access did not translate into finishing quality.

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Shot Analysis: Barcelona (Blue) vs Opponents (Red)
Champions League 2013-2014

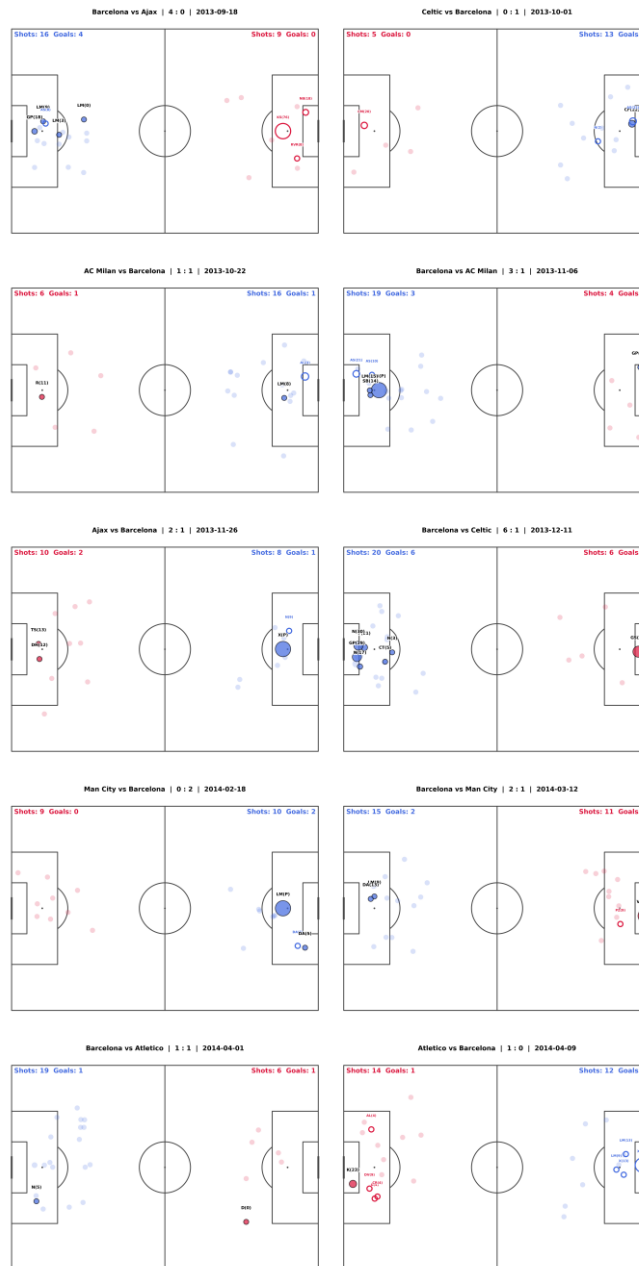


Figure 4 Barcelona Shot Map — UCL 2013-2014

Comparative Spatial Insight

The 2013-2014 spatial data reveals an interesting convergence and divergence between the two clubs. Both teams achieved similar box penetration rates against Tier 1 opponents — 50.0% for Real Madrid and 53.6% for Barcelona — a narrowing of the gap from the previous season where Real Madrid led 46.7% to 38.6%. However Real Madrid's Box_Central conversion rate of 31.7% against Tier 1 was nearly double Barcelona's 16.7%, confirming that spatial access alone does not explain the quality gap. Real Madrid converted box opportunities at a significantly higher rate despite similar penetration levels — a finding that the distance-based xG model cannot fully explain and that points to finishing quality differences beyond shot location [CANNOT CONFIRM].

Table 20 Zone Shot Distribution — Real Madrid and Barcelona, UCL 2013-2014

Depth Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box	51	50.0%	71	65.1%	30	53.6%	54	58.7%
Edge	47	46.1%	35	32.1%	26	46.4%	34	37.0%
Long Range	4	3.9%	3	2.8%	0	0.0%	4	4.3%
TOTAL	102	100%	109	100%	56	100%	92	100%

Lateral Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Left	17	16.7%	15	13.8%	7	12.5%	10	10.9%
Central	71	69.6%	83	76.1%	37	66.1%	75	81.5%
Right	14	13.7%	11	10.1%	12	21.4%	7	7.6%
TOTAL	102	100%	109	100%	56	100%	92	100%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Left	4	3.9%	4	3.7%	3	5.4%	6	6.5%
Box_Central	41	40.2%	60	55.0%	24	42.9%	45	48.9%

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Right	6	5.9%	7	6.4%	3	5.4%	3	3.3%
Edge_Left	11	10.8%	9	8.3%	4	7.1%	4	4.3%
Edge_Central	28	27.5%	22	20.2%	13	23.2%	26	28.3%
Edge_Right	8	7.8%	4	3.7%	9	16.1%	4	4.3%
Long Range_Left	2	2.0%	2	1.8%	0	0.0%	0	0.0%
Long Range_Central	2	2.0%	1	0.9%	0	0.0%	4	4.3%
Long Range_Right	0	0.0%	0	0.0%	0	0.0%	0	0.0%
TOTAL	102	100%	109	100%	56	100%	92	100%

Table 21 Zone xG and Goals by Zone — Real Madrid and Barcelona, UCL 2013-2014

Depth Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box	51	0.1404	13	25.5%	71	0.1224	22	31.0%	30	0.1072	5	16.7%	54	0.1345	13	24.1%
Edge	47	0.0096	3	6.4%	35	0.0111	3	8.6%	26	0.0079	0	0.0%	34	0.0127	3	8.8%
Long Range	4	0.0012	0	0.0%	3	0.0011	0	0.0%	0	0	0	0.0%	4	0.0022	0	0.0%

Lateral Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Left	17	0.0090	0	0.0%	15	0.0115	1	6.7%	7	0.0150	0	0.0%	10	0.0218	0	0.0%
Central	71	0.1015	16	22.5%	83	0.1049	22	26.5%	37	0.0854	4	10.8%	75	0.0989	16	21.3%
Right	14	0.0183	0	0.0%	11	0.0185	2	18.2%	12	0.0130	1	8.3%	7	0.0096	0	0.0%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box_Left	4	0.0198	0	0.0%	4	0.0304	1	25.0%	3	0.0266	0	0.0%	6	0.0321	0	0.0%
Box_Central	41	0.1678	13	31.7%	60	0.1398	19	31.7%	24	0.1255	4	16.7%	45	0.1559	13	28.9%
Box_Right	6	0.0339	0	0.0%	7	0.0259	2	28.6%	3	0.0408	1	33.3%	3	0.0190	0	0.0%
Edge_Left	11	0.0067	0	0.0%	9	0.0054	0	0.0%	4	0.0064	0	0.0%	4	0.0064	0	0.0%
Edge_Central	28	0.0117	3	10.7%	22	0.0144	3	13.6%	13	0.0112	0	0.0%	26	0.0153	3	11.5%
Edge_Right	8	0.0065	0	0.0%	4	0.0056	0	0.0%	9	0.0037	0	0.0%	4	0.0025	0	0.0%
Long Range_Left	2	0.0005	0	0.0%	2	0.0009	0	0.0%	0	0	0	0.0%	0	0	0	0.0%
Long Range_Central	2	0.0020	0	0.0%	1	0.0014	0	0.0%	0	0	0	0.0%	4	0.0022	0	0.0%
Long Range_Right	0	0	0	0.0%	0	0	0	0.0%	0	0	0	0.0%	0	0	0	0.0%

Player Highlights — Season 2013-2014

Table 22 presents the season average shot metrics for all players with a shot share above 5% for both Real Madrid and Barcelona across the 2013-2014 UEFA Champions League campaign.

Real Madrid's attacking structure evolved significantly from the previous season. Cristiano Ronaldo remained the primary threat at 41.3% shot share and 46.2% goal share, improving his goals per match from 1.00 to 1.55. Critically, Ronaldo's goal contribution became more diverse — 14 open play goals, 1 set piece, and 2 penalties [DATA] — reflecting a broader range of scoring mechanisms compared to his exclusively open play contribution in 2012-2013. Gareth Bale emerged as a genuine second contributor at 20.6% shot share and 16.7% goal share, all 6 of his goals coming from open play [DATA] with a strong right channel bias at 45.2% — a direct wide attacking role that complemented Ronaldo's more central positioning. Álvaro Morata appeared as a new third contributor at 13.8% shot share with 93.3% box shots and zero set piece attempts [DATA] — a pure penalty box presence. Karim Benzema maintained the highest rationality among regular contributors at 0.172 despite a slight drop in shot share to 13.3%. Sergio Ramos contributed 2 open play goals and 1 set piece goal [DATA], confirming his dual role as both an open play and set piece

threat from central defensive positions. Ángel Di María contributed 3 goals entirely from open play [DATA], all coming against Tier 2 opponents.

Real Madrid's goal type profile in 2013-2014 represents a meaningful evolution from 2012-2013 — broader contributor base, more diverse goal types, and penalties added as an additional weapon. This diversity of scoring mechanisms across multiple players and goal types is a structural advantage that Barcelona did not replicate.

Barcelona's attacking structure also evolved with Neymar's arrival as second contributor at 27.5% shot share alongside Lionel Messi's 32.9%. However the goal type data reveals an important nuance — of Messi's 8 goals, only 6 came from open play with 2 from penalties [DATA]. While Messi remained Barcelona's primary threat, his open play goal contribution was more modest than the raw total suggests. Neymar contributed 4 goals entirely from open play [DATA] but at a significantly lower efficiency than his shot share would imply. Xavi's single goal came from a penalty [DATA] — consistent with his 2012-2013 pattern — confirming zero open play goals despite a 15.3% shot share. Barcelona recorded zero set piece goals across the entire campaign [DATA], maintaining their near complete dependency on open play for goal creation. This stands in direct contrast to Real Madrid, who diversified their scoring mechanisms across open play, set pieces, and penalties.

The structural difference in goal type diversity between the two clubs is flagged as a key analytical finding for the cross-season synthesis in Module 9. Real Madrid's ability to generate goals from multiple sources — including set pieces and penalties — adds a dimension of attacking flexibility that Barcelona's open play dependency does not provide.

Player Tier Comparison — Season 2013-2014

Table 23 presents the player-level tier breakdown using the same player list as Table 22.

Cristiano Ronaldo's tier split reveals an important dimension of Real Madrid's elite match performance. Against Tier 1 opponents Ronaldo scored 7 goals — 5 from open play and 2 from penalties [DATA]. The penalty contribution against elite opponents is analytically significant — penalties represent high-value set piece opportunities that Real Madrid were able to generate and convert in critical matches. Against Tier 2 opponents Ronaldo scored 10 goals, all from open play with 1 set piece [DATA], reflecting his natural open play dominance against weaker sides. Gareth Bale maintained consistent output across both tiers — 3 goals against Tier 1 and 3 against Tier 2, all from open play [DATA] — confirming his counter-attack role was equally effective regardless of opponent quality. Sergio Ramos contributed all 3 of his season goals against Tier 1 opponents — 2 open play and 1 set piece [DATA] — a striking finding that positions him as a genuine elite match contributor rather than a Tier 2 accumulator. Ángel Di María and Álvaro Morata both scored zero goals

against Tier 1 opponents, with all their contributions coming against Tier 2 sides [DATA], confirming their roles as depth contributors rather than elite-level performers.

The goal type tier breakdown for Real Madrid reveals a team with multiple scoring mechanisms against elite opposition — open play through Ronaldo and Bale, penalties through Ronaldo, and set pieces through Ramos. This structural diversity of goal sources against Tier 1 opponents is one of the most significant findings in the 2013-2014 player data.

Barcelona's tier breakdown tells a contrasting story. Lionel Messi scored 2 goals against Tier 1 opponents — 1 from open play and 1 from a penalty [DATA]. His single open play goal against elite opposition is a more modest contribution than his overall season figures suggest, particularly given his 4.5 shots per match against Tier 1. Neymar contributed 1 open play goal against Tier 1 opponents [DATA], confirming his limited impact against elite sides despite his strong Tier 2 record of 3 goals. Dani Alves was Barcelona's most productive non-Messi contributor against Tier 1 opponents with 2 open play goals [DATA] — an unexpected finding from a right back that reflects Barcelona's difficulty generating goals from their recognized attacking players against elite defences. Barcelona recorded zero set piece goals against Tier 1 opponents [DATA], and their total Tier 1 goal contribution came from just three players — Messi, Neymar, and Alves — with a combined open play total of only 4 goals across all Tier 1 matches.

The tier comparison establishes a clear structural contrast. Real Madrid generated goals against elite opponents from multiple sources and multiple players, with penalties and set pieces supplementing open play contribution. Barcelona's Tier 1 goal creation was narrow — concentrated on three players with minimal open play depth and zero set piece contribution. When Messi's open play output was limited to a single goal against elite opposition, Barcelona had insufficient alternative mechanisms to compensate. This structural vulnerability is consistent with the pattern observed in 2012-2013 and is flagged for the cross-season synthesis in Module 9.

Table 22 Player Season Average — Shot Analysis 2013-2014 (Shot Share > 5%)

Player	Team	Shots	OP	SP	Goals	OP G	SP G	FK G	Pen G	xG	Rat.	Box %	Left %	Right %	Shot %	xG %	Goal %
Cristiano Ronaldo	Real Madrid	6.91	6.00	0.36	1.55	14	1	0	2	0.625	0.093	54.3%	29.6%	17.2%	41.3%	43.3%	46.2%
Gareth Bale	Real Madrid	3.20	3.20	0.00	0.60	6	0	0	0	0.173	0.052	61.3%	5.8%	45.2%	20.6%	14.3%	16.7%
Álvaro Morata	Real Madrid	2.67	2.67	0.00	0.33	1	0	0	0	0.201	0.051	93.3%	16.7%	33.3%	13.8%	10.2%	11.1%
Karim Benzema	Real Madrid	2.11	2.11	0.00	0.56	5	0	0	0	0.257	0.172	68.5%	13.0%	11.1%	13.3%	21.5%	18.5%
Ángel Di María	Real Madrid	2.00	1.90	0.00	0.30	3	0	0	0	0.150	0.065	45.7%	35.0%	16.4%	13.1%	13.5%	7.5%
Isco	Real Madrid	1.75	1.75	0.00	0.38	3	0	0	0	0.105	0.074	58.3%	25.0%	0.0%	11.2%	9.8%	9.4%
Sergio Ramos	Real Madrid	1.86	1.43	0.43	0.43	2	1	0	0	0.216	0.111	85.7%	7.1%	7.1%	10.2%	16.0%	10.7%
Luka Modrić	Real Madrid	1.25	1.25	0.00	0.25	1	0	0	0	0.010	0.007	0.0%	0.0%	37.5%	7.9%	0.6%	12.5%
Lionel Messi	Barcelona	4.71	4.43	0.00	1.14	6	0	0	2	0.456	0.107	58.6%	4.8%	8.1%	32.9%	42.0%	48.8%
Neymar	Barcelona	4.11	4.11	0.00	0.44	4	0	0	0	0.253	0.057	70.8%	26.5%	12.8%	27.5%	20.5%	16.7%
Xavi	Barcelona	2.13	1.88	0.00	0.13	0	0	0	1	0.202	0.143	33.8%	5.6%	6.3%	15.3%	19.4%	12.5%
Andrés Iniesta	Barcelona	1.67	1.67	0.00	0.00	0	0	0	0	0.045	0.023	22.2%	47.2%	13.9%	10.7%	5.9%	0.0%
Dani Alves	Barcelona	1.50	1.50	0.00	0.33	2	0	0	0	0.045	0.033	38.9%	0.0%	83.3%	10.3%	6.0%	16.7%
Cesc Fàbregas	Barcelona	1.29	1.14	0.14	0.14	1	0	0	0	0.081	0.060	64.3%	35.7%	7.1%	9.9%	9.7%	14.3%
Marc Bartra	Barcelona	1.25	1.00	0.25	0.00	0	0	0	0	0.082	0.066	62.5%	0.0%	37.5%	9.0%	9.0%	0.0%
Adriano	Barcelona	1.20	1.20	0.00	0.00	0	0	0	0	0.058	0.057	40.0%	70.0%	0.0%	7.1%	6.0%	0.0%

Table 23 Player Tier Comparison — Shot Analysis 2013-2014 (Shot Share > 5%)

Player	Team	T1 Sh	T1 OP	T1 SP	T1 G	T1 OPG	T1 SPG	T1 FKG	T1 PG	T1 xG	T1 Rat	T1 Box%	T1 Sh%	T2 Sh	T2 OP	T2 SP	T2 G	T2 OPG	T2 SPG	T2 FKG	T2 PG	T2 xG	T2 Rat	T2 Box%
Cristiano Ronaldo	Real Madrid	6.00	4.83	0.33	1.17	5	0	0	2	0.509	0.088	50.2%	41.1%	8.00	7.40	0.40	2.00	9	1	0	0	0.765	0.099	59.2%
Gareth Bale	Real Madrid	3.60	3.60	0.00	0.60	3	0	0	0	0.204	0.049	51.0%	23.2%	2.80	2.80	0.00	0.60	3	0	0	0	0.143	0.055	71.7%
Álvaro Morata	Real Madrid	2.00	2.00	0.00	0.00	0	0	0	0	0.064	0.032	100.0%	15.4%	3.00	3.00	0.00	0.50	1	0	0	0	0.269	0.061	90.0%
Karim Benzema	Real Madrid	1.80	1.80	0.00	0.20	1	0	0	0	0.201	0.170	56.7%	13.1%	2.50	2.50	0.00	1.00	4	0	0	0	0.328	0.175	83.3%
Ángel Di María	Real Madrid	1.50	1.33	0.00	0.00	0	0	0	0	0.158	0.081	33.3%	12.3%	2.75	2.75	0.00	0.75	3	0	0	0	0.138	0.042	64.3%
Isco	Real Madrid	2.00	2.00	0.00	0.33	1	0	0	0	0.026	0.013	16.7%	13.3%	1.60	1.60	0.00	0.40	2	0	0	0	0.152	0.110	83.3%
Sergio Ramos	Real Madrid	1.75	1.50	0.25	0.75	2	1	0	0	0.162	0.093	75.0%	10.6%	2.00	1.33	0.67	0.00	0	0	0	0	0.288	0.134	100.0%
Luka Modrić	Real Madrid	1.50	1.50	0.00	0.00	0	0	0	0	0.014	0.008	0.0%	10.2%	1.00	1.00	0.00	0.50	1	0	0	0	0.006	0.006	0.0%
Lionel Messi	Barcelona	4.50	4.25	0.00	0.50	1	0	0	1	0.396	0.080	56.7%	34.9%	5.00	4.67	0.00	2.00	5	0	0	1	0.537	0.143	61.1%
Neymar	Barcelona	2.67	2.67	0.00	0.33	1	0	0	0	0.122	0.060	86.7%	16.0%	4.83	4.83	0.00	0.50	3	0	0	0	0.319	0.056	62.9%
Xavi	Barcelona	3.00	3.00	0.00	0.00	0	0	0	0	0.261	0.117	56.7%	23.3%	1.60	1.20	0.00	0.20	0	0	0	1	0.166	0.159	20.0%
Andrés Iniesta	Barcelona	2.00	2.00	0.00	0.00	0	0	0	0	0.055	0.024	27.8%	12.9%	1.33	1.33	0.00	0.00	0	0	0	0	0.035	0.022	16.7%
Dani Alves	Barcelona	2.00	2.00	0.00	0.67	2	0	0	0	0.087	0.064	77.8%	14.2%	1.00	1.00	0.00	0.00	0	0	0	0	0.002	0.002	0.0%
Cesc Fàbregas	Barcelona	1.33	1.33	0.00	0.00	0	0	0	0	0.017	0.015	50.0%	9.6%	1.25	1.00	0.25	0.25	1	0	0	0	0.130	0.094	75.0%
Marc Bartra	Barcelona	1.50	1.00	0.50	0.00	0	0	0	0	0.097	0.066	75.0%	11.0%	1.00	1.00	0.00	0.00	0	0	0	0	0.067	0.067	50.0%
Adriano	Barcelona	0.00	0.00	0.00	0.00	0	0	0	0	0.000	0.000	0.0%	0.0%	1.20	1.20	0.00	0.00	0	0	0	0	0.058	0.057	40.0%

Critical Match Analysis — 2013-2014

The 2013-2014 critical match analysis covers five Real Madrid matches across the Quarter-Final, Semi-Final, and Final, and two Barcelona matches in the Quarter-Final against

Atletico Madrid. Tables 24 through 30 present the player-level shot data for both teams across all seven matches.

Real Madrid vs Borussia Dortmund — Quarter-Final

Real Madrid's Quarter-Final against Borussia Dortmund carried particular significance — it was a rematch against the side that had eliminated them in the previous season's Semi-Final. Leg 1 at the Bernabéu delivered a commanding performance. Real Madrid generated 20 shots from 7 different contributors, with Bale, Ronaldo, and Isco each scoring. The attacking load was broadly distributed — no single player dominated shot share — reflecting a structured, multi-threat offensive display. Borussia Dortmund generated 12 shots with an xG of 0.412 but converted none, with Aubameyang carrying 33.3% of their shot burden without return.

Leg 2 in Dortmund produced a reversal in territorial terms. Real Madrid generated only 10 shots against Dortmund's 12, and the score ended 2:0 to the hosts. The critical data point is Ángel Di María's xG of 0.766 from 2 shots — the highest single-player xG in any Real Madrid critical match this season — yet he converted neither. This represents a significant missed opportunity that could have altered the tie. Marco Reus scored both Dortmund goals from only 3 shots at a combined xG of 0.272 — efficient but fortunate finishing. Real Madrid progressed on aggregate 3:2, but the Leg 2 performance exposed a vulnerability when their attacking structure was disrupted away from home.

Real Madrid vs Bayern Munich — Semi-Final

The Semi-Final against Bayern Munich produced two of the most analytically contrasting performances in Real Madrid's entire campaign. Leg 1 at the Bernabéu was a study in defensive discipline. Bayern dominated shots 18 to 9 with a total xG of 1.049 — nearly double Real Madrid's 0.599 — yet failed to score. Robben alone generated 6 shots with zero goals. Real Madrid converted their most clinical opportunity of the match: Benzema's single shot carrying an xG of 0.290 and producing the only goal. Real Madrid won 1:0 having been outshot and out-chanced — a result that reflected exceptional defensive organisation and ruthless efficiency rather than attacking dominance [DATA].

Leg 2 in Munich produced one of the most extraordinary away performances in Real Madrid's modern European history. Real Madrid won 4:0 in the Allianz Arena — a result that reversed the pattern of Leg 1 entirely. Bayern generated 19 shots with a total xG of only 0.640, reflecting a team creating volume but from poor positions — their rationality of 0.034 was among the lowest recorded across all critical matches in this study. Real Madrid generated only 13 shots but converted 4, led by Ronaldo's 2 goals from 7 shots and Ramos' 2 goals from 2 shots at an extraordinary combined conversion rate. Ramos' two goals —

both from 100% box positions — confirmed his elite match contribution identified in the player tier analysis, and are particularly significant given his 0 goals against Tier 2 opponents this season [DATA].

Real Madrid vs Atletico Madrid — Final

The Final against Atletico Madrid at the Estadio da Luz in Lisbon is the defining match of the 2013-2014 season. Real Madrid generated 20 shots with a total xG of 2.059 — the highest single-match xG figure in any critical match across both teams this season — against Atletico's 11 shots and 0.652 xG. The quality gap was substantial and reflected in the final score of 4:1 after extra time.

Ronaldo led with 9 shots and 1 goal — his xG of 1.226 from the penalty spot included — while Bale contributed 4 shots and 1 goal and Ramos converted his famous header from 1 shot. Marcelo added a fourth goal from a low-xG position. The breadth of Real Madrid's goal contribution across four different players — a forward, a wide player, a central defender, and a full back — reflects the structural diversity identified throughout the season analysis. Atletico's only goal came from Diego Godín, a central defender, from a corner — a set piece that Real Madrid conceded in the 36th minute that allowed Atletico to hold on until the 93rd minute. Real Madrid's equaliser from Ramos in the 93rd minute, also from a set piece, is one of the most dramatic moments in the competition's history and underscores the analytical importance of set piece contribution identified earlier in this module [DATA].

Barcelona vs Atletico Madrid — Quarter-Final

Barcelona's Quarter-Final elimination by Atletico Madrid across two legs is one of the most analytically significant results in this study. The contrast with Real Madrid's Final victory against the same opponent provides a direct comparative lens.

In Leg 1 at the Camp Nou, Barcelona generated 19 shots against Atletico's 6 — a Field Tilt of 76% that would typically suggest overwhelming dominance. Yet the xG figures tell a different story: Barcelona's 0.674 against Atletico's 0.160. Barcelona were generating volume but from poor positions — a rationality of only 0.035 across 19 shots, the lowest figure in any critical match across both teams this season. Messi generated 5 shots and 0 goals with an xG of 0.290, while Neymar scored from 5 shots at an xG of 0.193. Atletico scored from a single low-xG opportunity — Diego converting from 0.003 xG — a goal that reflects Atletico's ability to maximise minimal opportunities while Barcelona squandered their territorial dominance.

Leg 2 in Madrid produced Barcelona's most analytically stark performance of the season. Barcelona generated 12 shots with a total xG of 1.091 — a reasonable quality figure — yet converted none. The critical data point is Xavi's 2 shots carrying a combined xG of 0.594 —

the highest single-player xG figure in any Barcelona critical match this season — yet both were unconverted. Messi generated 4 shots and 0 goals from an xG of 0.244. Koke scored Atletico's winning goal from a single shot with an xG of 0.238 — a high-value opportunity created and converted with clinical efficiency. Barcelona's inability to convert despite generating genuine quality opportunities in Leg 2 is a finding that the shot data cannot fully explain — whether it reflects finishing failure, goalkeeper performance, or defensive organisation by Atletico [CANNOT CONFIRM]. What the data confirms is that Atletico restricted Barcelona to a low shot quality profile in Leg 1 while converting their own minimal opportunities, and held firm in Leg 2 despite Barcelona generating their highest-quality chances of the tie [DATA].

Table 24 Player Shot Data — Real Madrid vs Borussia Dortmund, Quarter-Final Leg 1, Won 3:0, 2014-04-02

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Gareth Bale	Real Madrid	5	1	0.264	0.053	80.0%	25.0%
Cristiano Ronaldo	Real Madrid	4	1	0.208	0.052	50.0%	20.0%
Isco	Real Madrid	3	1	0.031	0.010	0.0%	15.0%
Karim Benzema	Real Madrid	3	0	0.140	0.047	33.3%	15.0%
Pepe	Real Madrid	2	0	0.175	0.088	100.0%	10.0%
Sergio Ramos	Real Madrid	2	0	0.131	0.066	50.0%	10.0%
Fábio Coentrão	Real Madrid	1	0	0.003	0.003	0.0%	5.0%
TEAM TOTAL	Real Madrid	20	3	0.952	0.048	-	100%
Pierre-Emerick Aubameyang	Borussia Dortmund	4	0	0.104	0.026	75.0%	33.3%
Henrikh Mkhitaryan	Borussia Dortmund	2	0	0.212	0.106	100.0%	16.7%
Marco Reus	Borussia Dortmund	2	0	0.013	0.006	0.0%	16.7%
Kevin Großkreutz	Borussia Dortmund	1	0	0.006	0.006	0.0%	8.3%
Julian Schieber	Borussia Dortmund	1	0	0.015	0.015	100.0%	8.3%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Mats Hummels	Borussia Dortmund	1	0	0.061	0.061	100.0%	8.3%
Nuri Sahin	Borussia Dortmund	1	0	0.001	0.001	0.0%	8.3%
TEAM TOTAL	Borussia Dortmund	12	0	0.412	0.034	-	100%

Table 25 Player Shot Data — Borussia Dortmund vs Real Madrid, Quarter-Final Leg 2, Lost 2:0, 2014-04-08

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Marco Reus	Borussia Dortmund	3	2	0.272	0.091	66.7%	25.0%
Henrikh Mkhitaryan	Borussia Dortmund	3	0	0.289	0.096	100.0%	25.0%
Kevin Großkreutz	Borussia Dortmund	2	0	0.210	0.105	100.0%	16.7%
Lukasz Piszczek	Borussia Dortmund	1	0	0.032	0.032	100.0%	8.3%
Manuel Friedrich	Borussia Dortmund	1	0	0.156	0.156	100.0%	8.3%
Mats Hummels	Borussia Dortmund	1	0	0.116	0.116	100.0%	8.3%
Robert Lewandowski	Borussia Dortmund	1	0	0.079	0.079	100.0%	8.3%
TEAM TOTAL	Borussia Dortmund	12	2	1.153	0.096	-	100%
Gareth Bale	Real Madrid	4	0	0.175	0.044	50.0%	40.0%
Isco	Real Madrid	2	0	0.036	0.018	50.0%	20.0%
Karim Benzema	Real Madrid	2	0	0.028	0.014	0.0%	20.0%
Ángel Di María	Real Madrid	2	0	0.766	0.383	50.0%	20.0%
TEAM TOTAL	Real Madrid	10	0	1.005	0.100	-	100%

Table 26 Player Shot Data — Real Madrid vs Bayern Munich, Semi-Final Leg 1, Won 1:0, 2014-04-23

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	5	0	0.219	0.044	60.0%	55.6%
Ángel Di María	Real Madrid	2	0	0.088	0.044	50.0%	22.2%
Karim Benzema	Real Madrid	1	1	0.290	0.290	100.0%	11.1%
Xabi Alonso	Real Madrid	1	0	0.002	0.002	0.0%	11.1%
TEAM TOTAL	Real Madrid	9	1	0.599	0.067	-	100%
Arjen Robben	Bayern Munich	6	0	0.131	0.022	33.3%	33.3%
Toni Kroos	Bayern Munich	3	0	0.081	0.027	33.3%	16.7%
Philipp Lahm	Bayern Munich	2	0	0.125	0.062	100.0%	11.1%
Thomas Müller	Bayern Munich	2	0	0.107	0.054	50.0%	11.1%
Bastian Schweinsteiger	Bayern Munich	1	0	0.138	0.138	100.0%	5.6%
Mario Götze	Bayern Munich	1	0	0.057	0.057	100.0%	5.6%
Javi Martínez	Bayern Munich	1	0	0.173	0.173	100.0%	5.6%
Dante	Bayern Munich	1	0	0.057	0.057	100.0%	5.6%
Mario Mandzukic	Bayern Munich	1	0	0.181	0.181	100.0%	5.6%
TEAM TOTAL	Bayern Munich	18	0	1.049	0.058	-	100%

Table 27 Player Shot Data — Bayern Munich vs Real Madrid, Semi-Final Leg 2, Won 4:0, 2014-04-29

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	7	2	0.247	0.035	28.6%	53.8%
Sergio Ramos	Real Madrid	2	2	0.356	0.178	100.0%	15.4%
Fábio Coentrão	Real Madrid	1	0	0.228	0.228	100.0%	7.7%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Gareth Bale	Real Madrid	1	0	0.005	0.005	0.0%	7.7%
Raphaël Varane	Real Madrid	1	0	0.138	0.138	100.0%	7.7%
Ángel Di María	Real Madrid	1	0	0.011	0.011	0.0%	7.7%
TEAM TOTAL	Real Madrid	13	4	0.984	0.076	-	100%
Arjen Robben	Bayern Munich	5	0	0.033	0.007	0.0%	26.3%
David Alaba	Bayern Munich	5	0	0.055	0.011	20.0%	26.3%
Franck Ribéry	Bayern Munich	2	0	0.058	0.029	100.0%	10.5%
Toni Kroos	Bayern Munich	2	0	0.008	0.004	0.0%	10.5%
Thomas Müller	Bayern Munich	2	0	0.078	0.039	50.0%	10.5%
Javi Martínez	Bayern Munich	1	0	0.117	0.117	100.0%	5.3%
Mario Mandzukic	Bayern Munich	1	0	0.244	0.244	100.0%	5.3%
Mario Götze	Bayern Munich	1	0	0.047	0.047	100.0%	5.3%
TEAM TOTAL	Bayern Munich	19	0	0.640	0.034	-	100%

Table 28 Player Shot Data — Real Madrid vs Atletico Madrid, Final, Won 4:1 AET, 2014-05-24

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	9	1	1.226	0.136	55.6%	45.0%
Gareth Bale	Real Madrid	4	1	0.476	0.119	75.0%	20.0%
Ángel Di María	Real Madrid	2	0	0.071	0.036	100.0%	10.0%
Sergio Ramos	Real Madrid	1	1	0.103	0.103	100.0%	5.0%
Marcelo	Real Madrid	1	1	0.027	0.027	0.0%	5.0%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Isco	Real Madrid	1	0	0.011	0.011	0.0%	5.0%
Luka Modrić	Real Madrid	1	0	0.003	0.003	0.0%	5.0%
Raphaël Varane	Real Madrid	1	0	0.142	0.142	100.0%	5.0%
TEAM TOTAL	Real Madrid	20	4	2.059	0.103	-	100%
Diego Godín	Atletico Madrid	3	1	0.246	0.082	66.7%	27.3%
Raúl García	Atletico Madrid	3	0	0.028	0.009	0.0%	27.3%
Adrián López	Atletico Madrid	2	0	0.253	0.126	100.0%	18.2%
José Sosa	Atletico Madrid	1	0	0.003	0.003	0.0%	9.1%
Koke	Atletico Madrid	1	0	0.080	0.080	100.0%	9.1%
Tiago	Atletico Madrid	1	0	0.043	0.043	100.0%	9.1%
TEAM TOTAL	Atletico Madrid	11	1	0.652	0.059	-	100%

Table 29 Player Shot Data — Barcelona vs Atletico Madrid, Quarter-Final Leg 1, Drew 1:1, 2014-04-01

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Neymar	Barcelona	5	1	0.193	0.039	60.0%	26.3%
Lionel Messi	Barcelona	5	0	0.290	0.058	60.0%	26.3%
Dani Alves	Barcelona	3	0	0.019	0.006	33.3%	15.8%
Andrés Iniesta	Barcelona	2	0	0.082	0.041	50.0%	10.5%
Marc Bartra	Barcelona	1	0	0.069	0.069	100.0%	5.3%
Cesc Fàbregas	Barcelona	1	0	0.005	0.005	0.0%	5.3%
Jordi Alba	Barcelona	1	0	0.010	0.010	0.0%	5.3%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Sergio Busquets	Barcelona	1	0	0.006	0.006	0.0%	5.3%
TEAM TOTAL	Barcelona	19	1	0.674	0.035	-	100%
David Villa	Atletico Madrid	4	0	0.149	0.037	50.0%	66.7%
Diego	Atletico Madrid	1	1	0.003	0.003	0.0%	16.7%
Tiago	Atletico Madrid	1	0	0.008	0.008	0.0%	16.7%
TEAM TOTAL	Atletico Madrid	6	1	0.160	0.027	-	100%

Table 30 Player Shot Data — Atletico Madrid vs Barcelona, Quarter-Final Leg 2, Lost 1:0, 2014-04-09

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Raúl García	Atletico Madrid	4	0	0.084	0.021	25.0%	28.6%
David Villa	Atletico Madrid	3	0	0.224	0.075	100.0%	21.4%
Gabi	Atletico Madrid	2	0	0.062	0.031	50.0%	14.3%
Koke	Atletico Madrid	1	1	0.238	0.238	100.0%	7.1%
Adrián López	Atletico Madrid	1	0	0.041	0.041	100.0%	7.1%
Diego	Atletico Madrid	1	0	0.035	0.035	100.0%	7.1%
Cristian Rodríguez	Atletico Madrid	1	0	0.048	0.048	100.0%	7.1%
Diego Godín	Atletico Madrid	1	0	0.061	0.061	100.0%	7.1%
TEAM TOTAL	Atletico Madrid	14	1	0.793	0.057	-	100%
Lionel Messi	Barcelona	4	0	0.244	0.061	50.0%	33.3%
Xavi	Barcelona	2	0	0.594	0.297	100.0%	16.7%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Marc Bartra	Barcelona	2	0	0.125	0.062	50.0%	16.7%
Cesc Fàbregas	Barcelona	2	0	0.016	0.008	50.0%	16.7%
Andrés Iniesta	Barcelona	1	0	0.006	0.006	0.0%	8.3%
Neymar	Barcelona	1	0	0.107	0.107	100.0%	8.3%
TEAM TOTAL	Barcelona	12	0	1.091	0.091	-	100%

Key Observations — Season 2013-2014

The 2013-2014 season provides the richest and most analytically significant dataset in the study period. Several key themes emerge from the combined evidence across season overview, tier analysis, spatial distribution, player data, and critical match analysis.

At the season level, Real Madrid's tactical evolution toward counter-attack against elite opponents — trading Field Tilt for a large positive Finishing Delta while holding their Quality Margin steady — proved decisive. The team demonstrated genuine flexibility, dominating territorially against Tier 2 opponents while absorbing pressure and exploiting space against elite sides. Barcelona improved across most season-level metrics but their improvements were broad rather than transformative, and their accuracy remained stagnant at 36.5%.

The critical match analysis provides the sharpest evidence. Real Madrid defeated three tactically contrasting opponents — BVB and Bayern (aggressive pressing) and Atletico (deep defensive block) — demonstrating adaptability that no single tactical system could neutralise. Barcelona, by contrast, were eliminated by the same Atletico side that Real Madrid defeated in the Final, unable to break down a deep defensive structure despite generating 19 shots in Leg 1. The contrast is not merely about quality — it is about tactical adaptability. Real Madrid could win ugly (Leg 1 vs Bayern, 1:0 with 9 shots), win spectacularly (Leg 2 vs Bayern, 4:0 away), and win under pressure (Final vs Atletico, 4:1 AET). Barcelona could not find a way past one disciplined defensive block across 180 minutes.

The player data adds structural depth to this contrast. Real Madrid's goal diversity — across multiple players and multiple goal types including open play, set pieces, and penalties — provided attacking solutions that could not be fully suppressed by any single defensive approach. Barcelona's near complete open play dependency, combined with

Messi's modest open play output against Tier 1 opponents (1 open play goal), left them structurally exposed when their primary system was neutralised.

The spatial analysis adds a further dimension to the cross-season comparison. Both clubs improved box penetration against Tier 1 opponents from 2012-2013 to 2013-2014 — Real Madrid from 46.7% to 50.0% and Barcelona from 38.6% to 53.6%. Barcelona's improvement was actually larger in absolute terms, suggesting they found more routes into the box against elite opposition than in the previous season. However the Box_Central conversion rate tells a completely different story. Real Madrid's conversion rate in this zone against Tier 1 opponents nearly tripled from 13.3% to 31.7%, while Barcelona's improved only modestly from 11.1% to 16.7%. The gap between the two clubs in 2013-2014 was therefore not about spatial access — Barcelona actually penetrated the box more frequently than Real Madrid against elite opponents — but about the ability to convert those box opportunities into goals. This conversion gap, rather than territorial or spatial dominance, is identified as a key structural differentiator between the two clubs and is flagged for the cross-season synthesis in Module 9.

4.4 Season 2014-2015

The 2014-2015 UEFA Champions League season inverted the outcome of the study period for the first time. Barcelona won the competition, defeating Juventus 3:1 in the Final, while Real Madrid were eliminated at the Semi-Final stage by the same Juventus side. On the surface this appears a straightforward reversal of fortune. The shot data tells a more layered story — one in which Real Madrid, by several measures, performed *better* than Barcelona across the season yet won less, and in which the tactical archetypes each club happened to draw shaped their paths as much as their own quality did.

Season Overview

The headline figures resist the simple reading that the winning team was the better-performing one. Barcelona scored more goals per match than Real Madrid — 2.38 against 1.92 — yet generated meaningfully *lower* chance quality, with an xG of 1.00 per match against Real Madrid's 1.42. Real Madrid's Quality Margin, the surplus of chance quality created over opponents, was 0.89 against Barcelona's 0.35: by this measure Real Madrid dominated their opponents in chance creation more decisively than Barcelona did. Yet Barcelona's Finishing Delta — goals scored above distance-based xG — was 1.39, nearly triple Real Madrid's 0.50.

Read through the environment principle established in this module, the two clubs present opposite profiles. Real Madrid created the better positions and finished them modestly. Barcelona created lesser positions by the distance model and finished them ruthlessly. Since the xG model is blind to defensive pressure, Barcelona's very high Finishing Delta is consistent with a front line converting chances in conditions of low defensive congestion — the transition and open-space finishing associated with their attacking trident — while Real Madrid's modest Finishing Delta, despite superior chance quality and a Field Tilt of 62.3%, is consistent with shots taken against organised, retreating defences [HYPOTHESIS]. Barcelona's higher accuracy of 48.2% against Real Madrid's 42.3% reinforces this: Barcelona were hitting the target with a substantially larger share of their attempts despite generating lower distance-based xG, exactly the signature of shooting in less congested conditions.

Real Madrid's Field Tilt of 62.3% is itself notable. In 2013-2014 Real Madrid had deliberately ceded territory against elite opponents, dropping their Tier 1 Field Tilt below 50% in the counter-attacking approach that delivered the title. In 2014-2015 they were the more territorially dominant side overall — a shift away from the counter-attacking identity of the previous season. Whether this reflected a deliberate evolution toward possession or

was imposed by the opponents they faced is examined in the tier analysis, where the distinction proves central.

Table 31 Season Overview — Shot Analysis 2014-2015

Metric	Real Madrid	Barcelona
Avg Shots per Match	18.7	14.8
Avg Goals per Match	1.9	2.4
xG per Match	1.42	1.00
Rationality (xG/Shot)	0.075	0.069
Box Shot %	52.7%	60.2%
Accuracy %	42.3%	48.2%
Big Misses per Match	0.33	0.15
Opp Shots per Match	10.8	9.9
Opp xG per Match	0.52	0.65
Finishing Delta (Goals – xG)	0.50	1.39
Quality Margin (xG – Opp xG)	0.89	0.35
Field Tilt %	62.3%	60.0%

Note: Goals are player-attributed throughout; own goals are excluded for both clubs and their opponents alike, per the convention set out in 4.1. In 2014-2015 this affects two goals: one scored for Real Madrid against FC Basel (group stage, Tier 2), reducing their recorded total from five to four in that match, and one own goal by PSG in the quarter-final first leg (see Table 41). Real Madrid's Tier 1 figures are unaffected.

Tier Comparison

The tier breakdown is where the 2014-2015 season delivers its most important evidence, and where the opponent archetypes each club drew become decisive.

Against Tier 1 opponents, Real Madrid presented a profile that inverts the conventional reading of their elimination. They generated the higher chance quality — a Quality Margin of 0.54 against their elite opponents, meaning they created substantially more xG than those opponents created against them. They dominated territory, with a Field Tilt of 69.0%. Yet they scored only 0.75 goals per Tier 1 match, because their Finishing Delta was negative at –0.54 — Real Madrid scored well *below* their distance-based xG against elite sides. Their

Tier 1 accuracy collapsed to 31.7%, the lowest figure in any tier for either club this season. Real Madrid were creating good positions, dominating the ball, and failing to convert.

Read through the environment principle, this is the signature of a team shooting into congestion. The distance model rated Real Madrid's Tier 1 chances highly, but the negative Finishing Delta and collapsed accuracy are consistent with those chances being taken against organised, retreating defences the model cannot see [HYPOTHESIS]. This is the possession-versus-block trap, and it is the same profile that eliminated Barcelona against Atletico Madrid in 2013-2014 — high territorial dominance, good distance-based xG, and an inability to convert against a set defence [HYPOTHESIS].

A substantial part of the explanation may lie in the opponents Real Madrid drew. Both of their Tier 1 opponents — Atletico Madrid and Juventus — are widely regarded as among the most disciplined defensive sides in Europe of that period [RESEARCH], a characterisation the shot data cannot itself establish. Across all four Tier 1 legs, Real Madrid's opponents ceded territory: Field Tilt reached 79.3% against Atletico at home and 76.7% against Juventus at home. A team that had won the previous season by counter-attacking against opponents who committed forward now faced two opponents who refused to commit forward at all. The counter-attacking approach that delivered the 2013-2014 title requires an opponent willing to be counter-attacked; the shot data records the ceded territory and the conversion drop, but whether the two are linked through the opponents' defensive shape is a question for Module 8 [HYPOTHESIS].

Barcelona's Tier 1 profile is almost the mirror image. Their Quality Margin against elite opponents was only 0.13 — they barely out-created their Tier 1 opponents in xG terms, far below Real Madrid's 0.54. Their Field Tilt against elite sides was 53.3%, modest by comparison with Real Madrid's 69.0%. Yet Barcelona scored 2.33 goals per Tier 1 match against Real Madrid's 0.75, because their Finishing Delta was +1.37 — they finished dramatically above their distance-based xG — and their Tier 1 accuracy was 50.4%, the highest figure for either club in any tier. Barcelona created less against elite opponents but converted far more.

The opponent draw explains much of this contrast. Barcelona's Tier 1 opponents in this campaign were Paris Saint-Germain, Manchester City and Bayern Munich — sides more commonly associated with attacking, forward-committing play than with deep defending [RESEARCH], though the shot pillar cannot verify how each set up. These are precisely the conditions under which a fast, direct attacking line converts in low-congestion space. The clearest single illustration is the Bayern away leg, where Barcelona registered a Field Tilt of just 20.8% — they were overwhelmingly out-shot — yet scored twice from five shots. That is not territorial dominance; it is ruthless transition finishing against an opponent who came

forward. The pattern is consistent with Real Madrid facing opponents who ceded less space and Barcelona facing opponents who ceded more; establishing that lies with Modules 6 and 8 [HYPOTHESIS].

Against Tier 2 opponents the two clubs converge, as in previous seasons. Both recorded strong Finishing Deltas — 1.02 for Real Madrid, 1.43 for Barcelona — and comfortable territorial control. The divergence that defined the season was concentrated entirely at Tier 1, and within Tier 1 it was concentrated not in who created the better chances, but in the defensive environment each club's chances were taken in.

The tier comparison therefore poses the central question of the season, which the critical match analysis and the Juventus Test that follow are designed to examine: was Real Madrid's elimination a decline in quality, or the consequence of drawing precisely the opponent archetype — the deep defensive block — against which their game was least suited, in a format where a single tie allows no room to recover?

Table 32 Tier Comparison — Shot Analysis 2014-2015

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Avg Shots per Match	19.0	18.5	13.7	17.3
Avg Goals per Match	0.75	2.5	2.33	2.5
xG per Match	1.29	1.48	0.97	1.07
Rationality (xG/Shot)	0.069	0.078	0.072	0.063
Box Shot %	49.8%	54.2%	60.9%	58.7%
Accuracy %	31.7%	47.6%	50.4%	43.3%
Big Misses per Match	0.50	0.25	0.11	0.25
Opp Shots per Match	8.3	12.1	11.8	5.8
Opp xG per Match	0.75	0.41	0.83	0.23
Finishing Delta (Goals – xG)	-0.54	1.02	1.37	1.43
Quality Margin (xG – Opp xG)	0.54	1.07	0.13	0.84
Field Tilt %	69.0%	59.0%	53.3%	75.0%

Shot Map Analysis — Spatial Distribution 2014-2015**Real Madrid Spatial Profile**

Real Madrid's spatial distribution in 2014-2015 shows a team penetrating centrally and in volume, yet failing to convert in the zone that matters most. Against Tier 1 opponents, 48.7% of shots came from inside the box and 70.1% from the central corridor — a concentrated, central attacking profile consistent with a side dominating the ball against deep defences. The distribution is not the problem. The conversion is. In Box_Central against Tier 1 opponents, Real Madrid took 43 shots and scored 7 — a goal rate of 16.3%. From the edge of the box centrally, they took 34 shots and scored none. Real Madrid were reaching the most valuable zones repeatedly and converting them at roughly half the rate they would manage against weaker sides, where the same Box_Central zone yielded a 28.0% goal rate.

This is the spatial expression of the possession-versus-block trap identified in the tier analysis. High central penetration paired with low central conversion is the signature of shooting into a congested, well-organised box — the distance model records the shots as high value, but the defensive density the model cannot see may suppress their conversion [HYPOTHESIS]. The shot pillar records the penetration-conversion mismatch; the defensive mechanism behind it is examined in Module 8. A notable secondary feature is Real Madrid's elevated right-channel share against Tier 1 opponents at 17.1%, up from 7.5% against Tier 2 — a lateral shift consistent with a team probing for openings against defences that denied them the centre.

Shot Analysis: Real Madrid (Red) vs Opponents (Blue)
Champions League 2014-2015

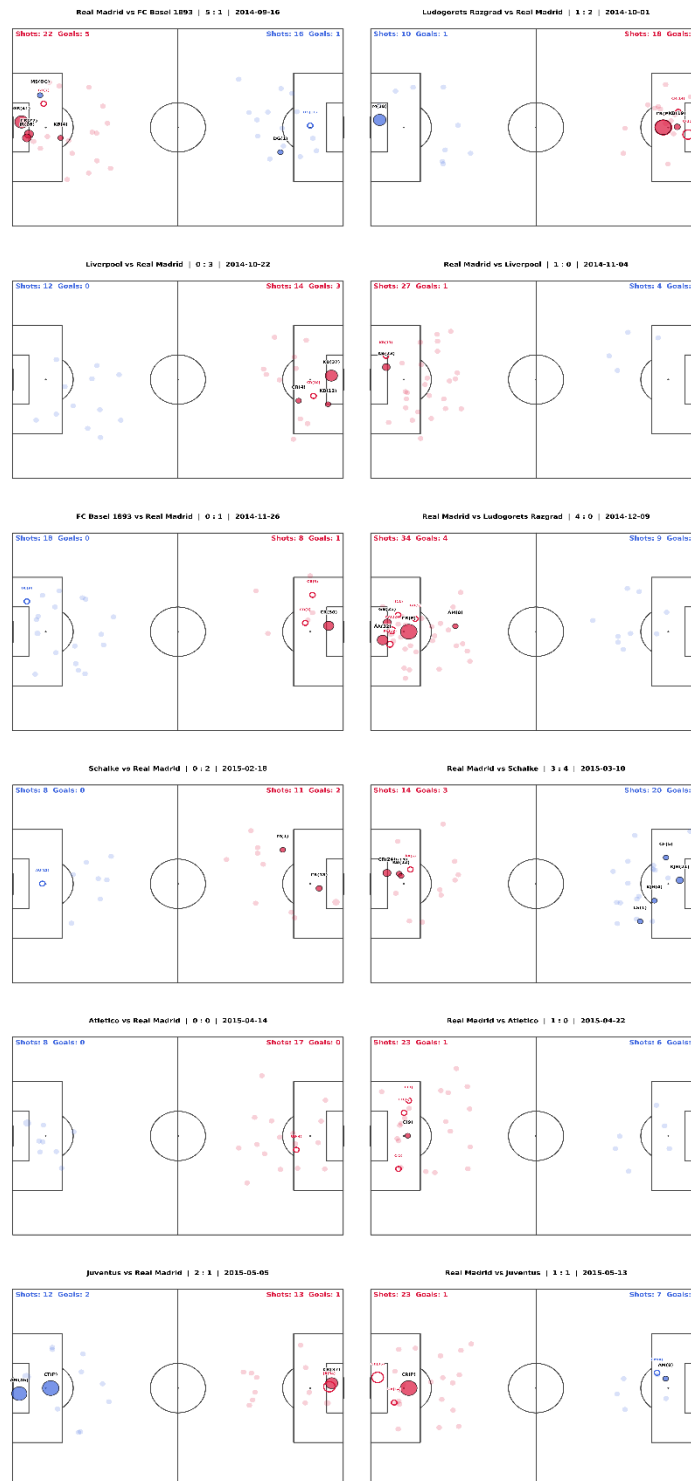


Figure 5 Real Madrid Shot Map — UCL 2014-2015]

Barcelona Spatial Profile

Barcelona's spatial profile against elite opposition is the more striking of the two, because it pairs extreme central concentration with high conversion. Against Tier 1 opponents, 59.3% of Barcelona's shots came from inside the box and 82.1% from the central corridor — the most centrally concentrated attacking profile of either club in any tier this season. More than half of all Barcelona's Tier 1 shots, 52.8%, came from Box_Central alone. And unlike Real Madrid, Barcelona converted them: 65 Box_Central shots against Tier 1 opponents yielded 18 goals, a 27.7% goal rate — nearly identical to their Tier 2 rate of 25.0%, and well above Real Madrid's 16.3% in the same zone against the same tier of opposition.

This is the environment effect made spatial. Barcelona's Box_Central conversion held up against elite opponents where Real Madrid's collapsed, despite Barcelona generating slightly lower average xG per shot in that zone (0.119 against Real Madrid's 0.140). The distance model rated Real Madrid's central chances as marginally *better* than Barcelona's, yet Barcelona scored them at nearly double the rate. The most plausible reading, consistent with the opponent-draw argument, is that Barcelona were taking their central chances against opponents who came forward and left the box less congested, while Real Madrid took theirs against deep blocks that packed it [HYPOTHESIS].

Shot Analysis: Barcelona (Blue) vs Opponents (Red)
Champions League 2014-2015

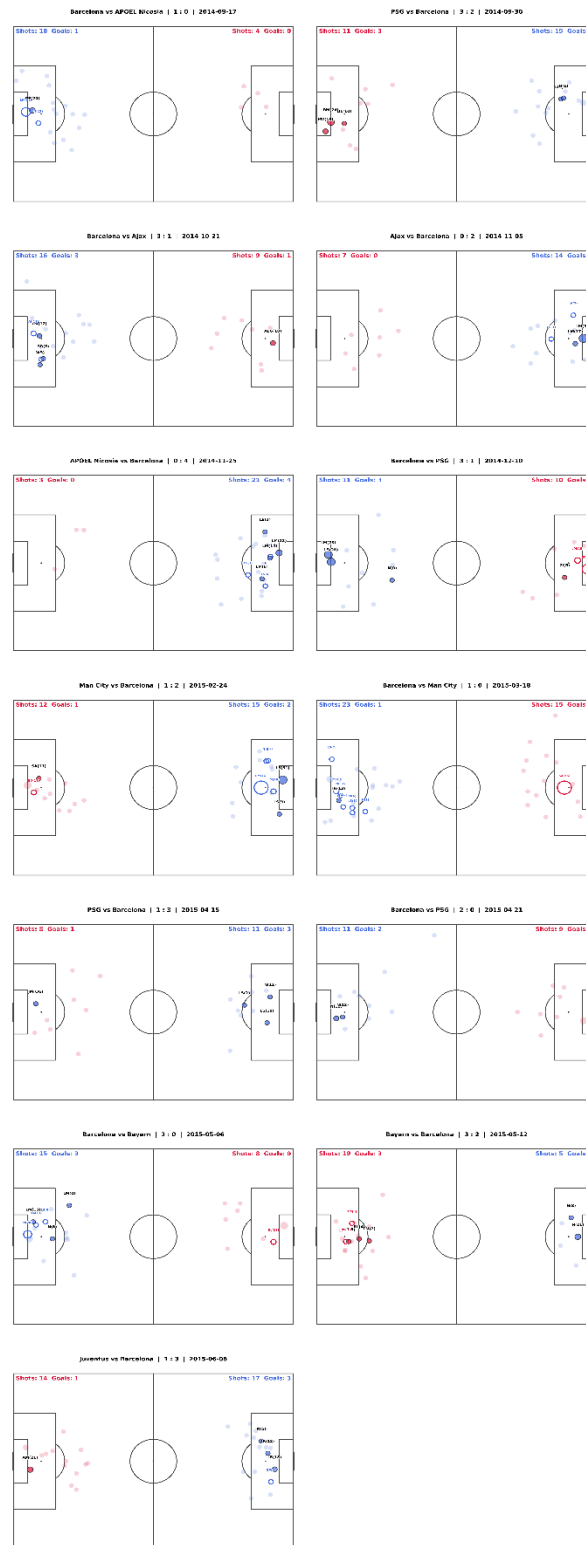


Figure 6 Barcelona Shot Map — UCL 2014-2015

Comparative Spatial Insight

The 2014-2015 zone data isolates the season's central finding with unusual clarity. Both clubs concentrated their attacks centrally against elite opposition. Both reached Box_Central in volume — indeed Barcelona did so more, taking 65 shots there against Tier 1 opposition to Real Madrid's 43. The divergence was entirely in conversion: 27.7% for Barcelona against 16.3% for Real Madrid, in the same zone, against the same tier, with Real Madrid holding the marginally higher average xG per shot.

This is the clearest spatial demonstration in the study so far that the distance-based xG model cannot, on its own, explain the gap between the two clubs. Spatial access to the box was not Real Madrid's problem in 2014-2015 — they penetrated centrally as effectively as they ever had. What separated the eventual champions from the eliminated semi-finalists was the defensive environment in which each club's central chances were taken, and the conversion that environment permitted. The Juventus Test that follows the critical match analysis holds the opponent constant and examines this same effect within a single fixture.

Table 33 Zone Shot Distribution — Real Madrid and Barcelona, UCL 2014-2015

Depth Zone

Zone	RM T1 Sh	RM T1 %	RM T2 Sh	RM T2 %	BA T1 Sh	BA T1 %	BA T2 Sh	BA T2 %
Box	57	48.7%	62	57.9%	73	59.3%	42	60.9%
Edge	53	45.3%	41	38.3%	48	39.0%	26	37.7%
Long Range	7	6.0%	4	3.7%	2	1.6%	1	1.4%
TOTAL	117	100%	107	100%	123	100%	69	100%

Lateral Zone

Zone	RM T1 Sh	RM T1 %	RM T2 Sh	RM T2 %	BA T1 Sh	BA T1 %	BA T2 Sh	BA T2 %
Left	15	12.8%	17	15.9%	14	11.4%	2	2.9%
Central	82	70.1%	82	76.6%	101	82.1%	60	87.0%
Right	20	17.1%	8	7.5%	8	6.5%	7	10.1%
TOTAL	117	100%	107	100%	123	100%	69	100%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Sh	RM T1 %	RM T2 Sh	RM T2 %	BA T1 Sh	BA T1 %	BA T2 Sh	BA T2 %
Box_Left	7	6.0%	7	6.5%	7	5.7%	1	1.4%
Box_Central	43	36.8%	50	46.7%	65	52.8%	36	52.2%
Box_Right	7	6.0%	5	4.7%	1	0.8%	5	7.2%
Edge_Left	8	6.8%	9	8.4%	7	5.7%	1	1.4%
Edge_Central	34	29.1%	29	27.1%	35	28.5%	23	33.3%
Edge_Right	11	9.4%	3	2.8%	6	4.9%	2	2.9%
LR_Left	0	0.0%	1	0.9%	0	0.0%	0	0.0%
LR_Central	5	4.3%	3	2.8%	1	0.8%	1	1.4%
LR_Right	2	1.7%	0	0.0%	1	0.8%	0	0.0%
TOTAL	117	100%	107	100%	123	100%	69	100%

Table 34 Zone xG and Goals by Zone — Real Madrid and Barcelona, UCL 2014-2015

Depth Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box	57	0.1143	7	12.3%	62	0.1544	14	22.6%	73	0.1105	18	24.7%	42	0.0938	10	23.8%
Edge	53	0.0099	0	0.0%	41	0.0097	2	4.9%	48	0.0130	3	6.2%	26	0.0124	0	0.0%
Long Range	7	0.0018	0	0.0%	4	0.0016	0	0.0%	2	0.0013	0	0.0%	1	0.0024	0	0.0%

Lateral Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Left	15	0.0212	0	0.0%	17	0.0147	1	5.9%	14	0.0231	0	0.0%	2	0.0228	1	50.0%
Central	82	0.0782	7	8.5%	82	0.1169	15	18.3%	101	0.0820	20	19.8%	60	0.0674	9	15.0%
Right	20	0.0159	0	0.0%	8	0.0173	0	0.0%	8	0.0107	1	12.5%	7	0.0248	0	0.0%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box_Left	7	0.0370	0	0.0%	7	0.0273	0	0.0%	7	0.0377	0	0.0%	1	0.0391	1	100.0%
Box_Central	43	0.1397	7	16.3%	50	0.1850	14	28.0%	65	0.1194	18	27.7%	36	0.1039	9	25.0%
Box_Right	7	0.0356	0	0.0%	5	0.0259	0	0.0%	1	0.0426	0	0.0%	5	0.0324	0	0.0%
Edge_Left	8	0.0073	0	0.0%	9	0.0064	1	11.1%	7	0.0084	0	0.0%	1	0.0065	0	0.0%
Edge_Central	34	0.0117	0	0.0%	29	0.0114	1	3.4%	35	0.0149	2	5.7%	23	0.0132	0	0.0%
Edge_Right	11	0.0060	0	0.0%	3	0.0032	0	0.0%	6	0.0072	1	16.7%	2	0.0059	0	0.0%
Long Range_Left	0	0	0	0.0%	1	0.0015	0	0.0%	0	0	0	0.0%	0	0	0	0.0%
Long Range_Central	5	0.0020	0	0.0%	3	0.0017	0	0.0%	1	0.0026	0	0.0%	1	0.0024	0	0.0%
Long Range_Right	2	0.0012	0	0.0%	0	0	0	0.0%	1	0	0	0.0%	0	0	0	0.0%

Player Highlights — Season 2014-2015

Real Madrid's shooting load remained concentrated on Cristiano Ronaldo at 33.6% shot share and 45.8% goal share, though both figures represent a decline from his 41.3% shot share the previous season. Gareth Bale was the second contributor at 19.2% shot share, but converted only 2 goals for a 5.6% goal share — a marked drop from his 16.7% goal share in 2013-2014 despite a comparable shooting volume. Chicharito ranked third at 15.3% shot share with the highest box concentration in either squad at 90.0%, yet returned a single goal. Karim Benzema was the season's most efficient Real Madrid contributor, converting 6 goals from only 13.1% of the team's shots — a 34.4% goal share from roughly an eighth of the attempts, at the squad's highest rationality of 0.115.

Barcelona's profile is the most distributed of any club-season in the study to this point. Three players carried near-identical shooting loads: Messi at 30.0%, Suárez at 29.8%, and Neymar at 23.1% — together 82.9% of all Barcelona shots. Critically, their goal shares were similarly balanced at 27.6%, 32.3%, and 37.9% respectively. Neymar's figures are the most striking: the lowest shot share of the three at 23.1% but the highest goal share at 37.9%, converting 10 goals from 3.00 shots per match.

The goal-type data records a complete reversal of the pattern identified in 2013-2014. In that season Real Madrid drew goals from open play, set pieces, and penalties while Barcelona were near-exclusively open play, and this study flagged Real Madrid's mechanism diversity as a structural advantage. In 2014-2015 Barcelona scored all 30 of their goals from open play [DATA] — no set pieces, no free kicks, no penalties — and scored more than Real Madrid, who took 3 of Ronaldo's 10 goals from the penalty spot. Mechanism diversity did not determine the outcome in this season. The observation from 2013-2014 therefore holds as a description of that season rather than as a general principle, and its status will be reassessed across the full six-season sample in Module 9.

Player Tier Comparison — Season 2014-2015

The tier split produces the most lopsided player-level distribution in the study so far.

Neymar scored 9 of his 10 season goals against Tier 1 opponents [DATA], from an identical 3.00 shots per match in both tiers. His Tier 1 goal share was 42.6% — Neymar alone accounted for over two-fifths of Barcelona's goals against elite opposition. Luis Suárez shows the same pattern: 6 of his 7 goals came against Tier 1 opponents, from 5.00 shots per Tier 1 match at an 81.0% box concentration. Messi's split was more even at 4 Tier 1 goals to 6 Tier 2. Together the three forwards accounted for 19 of Barcelona's 21 Tier 1 goals, all from open play.

Two contextual factors qualify this distribution before it is interpreted. The first is fixture composition. Barcelona's group contained Paris Saint-Germain, an opponent classified Tier 1 under the Elo threshold, meaning Barcelona met the same elite side four times across the group stage and the Quarter-Finals. Across those four matches Barcelona's forward line scored all ten of the club's goals, with Neymar contributing five, Suárez three and Messi two.¹ A substantial share of Neymar's Tier 1 return therefore came against a single opponent, and the split reflects the shape of Barcelona's draw as much as a general elevation against elite opposition. The second is sample balance: Barcelona played nine Tier 1 matches to Real Madrid's four, so Barcelona's forwards simply had more elite-opposition fixtures in which to score.

With those qualifications, the underlying observation remains. Neymar's shooting volume was identical across tiers at 3.00 shots per match, yet his conversion differed by an order of magnitude. Whatever changed between his Tier 1 and Tier 2 appearances, it was not how often he shot. The difference lies in the quality or construction of the chances he received, which the shot data records but cannot explain. Chance construction is examined in Module 6.

Real Madrid's tier split runs the opposite way. Gareth Bale scored zero goals against Tier 1 opponents from 3.50 shots per match, with both his season goals coming against Tier 2 sides. Chicharito managed 1 Tier 1 goal from 3.00 shots per match at 100% box concentration — every one of his elite-opposition attempts came from inside the box. Ronaldo scored 3 against Tier 1, of which one was a penalty, leaving 2 open-play goals against elite opposition from 5.50 shots per match. Benzema matched Ronaldo's Tier 1 open-play return of 3 goals from fewer than half the attempts, at the squad's highest Tier 1 rationality of 0.135.

Real Madrid's total Tier 1 output was 7 goals across 4 matches, 6 of them from open play. Barcelona's was 21 goals across 9 matches, all from open play. Normalised per match, that is 1.75 for Real Madrid against 2.33 for Barcelona — a gap that widens when chance quality is taken into account, since Real Madrid generated a Quality Margin of +0.54 against elite opponents to Barcelona's +0.13.

The player data therefore reinforces the team-level anomaly rather than resolving it. Real Madrid's forwards took more shots from better-rated positions against elite opposition and scored fewer of them. Barcelona's forwards took fewer shots from lower-rated positions and scored more. The shot pillar establishes that this gap exists and locates it in the forward line of both squads; it cannot determine whether the cause lies in chance construction, defensive pressure at the moment of the shot, or finishing execution. These questions are addressed in Modules 6 and 8.

Table 35 Player Season Average — Shot Analysis 2014-2015 (Shot Share > 5%)

Player	Team	Shots	OP	SP	FK	Pen	Goals	OP G	SP G	Pen G	xG	Rat.	Box %	Left %	Right %	Shot %	xG %	Goal %
Cristiano Ronaldo	Real Madrid	5.92	4.25	0.08	1.25	0.33	0.83	7	0	3	0.562	0.105	46.3%	34.1%	9.1%	33.6%	39.9%	45.8%
Gareth Bale	Real Madrid	3.78	3.44	0.11	0.22	0.00	0.22	2	0	0	0.317	0.096	68.4%	10.5%	19.9%	19.2%	19.1%	5.6%
Chicharito	Real Madrid	3.40	3.20	0.20	0.00	0.00	0.20	1	0	0	0.282	0.096	90.0%	15.7%	13.6%	15.3%	16.4%	20.0%
Karim Benzema	Real Madrid	2.38	2.38	0.00	0.00	0.00	0.75	6	0	0	0.240	0.115	77.1%	9.4%	16.7%	13.1%	21.9%	34.4%
Luka Modrić	Real Madrid	1.75	1.75	0.00	0.00	0.00	0.00	0	0	0	0.020	0.010	0.0%	0.0%	8.3%	9.4%	2.2%	0.0%
Toni Kroos	Real Madrid	1.75	1.75	0.00	0.00	0.00	0.00	0	0	0	0.018	0.011	18.8%	6.3%	6.3%	9.2%	1.2%	0.0%
Marcelo	Real Madrid	1.60	1.60	0.00	0.00	0.00	0.20	1	0	0	0.025	0.013	16.7%	53.3%	0.0%	8.9%	2.4%	10.0%
James Rodríguez	Real Madrid	1.57	1.43	0.00	0.14	0.00	0.14	1	0	0	0.117	0.103	50.0%	7.1%	23.8%	8.9%	10.1%	3.6%
Sergio Ramos	Real Madrid	1.40	1.20	0.00	0.20	0.00	0.00	0	0	0	0.137	0.090	80.0%	20.0%	0.0%	8.2%	8.7%	0.0%
Isco	Real Madrid	1.56	1.56	0.00	0.00	0.00	0.00	0	0	0	0.040	0.022	31.5%	11.1%	25.9%	8.2%	3.0%	0.0%
Raphaël Varane	Real Madrid	1.50	1.25	0.25	0.00	0.00	0.00	0	0	0	0.211	0.135	100.0%	0.0%	0.0%	7.1%	16.6%	0.0%
Álvaro Arbeloa	Real Madrid	1.67	1.67	0.00	0.00	0.00	0.33	1	0	0	0.245	0.100	66.7%	0.0%	33.3%	6.5%	9.6%	8.3%
Lionel Messi	Barcelona	4.31	3.15	0.08	1.00	0.08	0.77	10	0	0	0.345	0.076	53.8%	9.6%	11.1%	30.0%	30.7%	27.6%
Luis Suárez	Barcelona	4.75	4.63	0.13	0.00	0.00	0.88	7	0	0	0.380	0.077	71.1%	27.7%	14.4%	29.8%	34.1%	32.3%
Neymar	Barcelona	3.00	3.00	0.00	0.00	0.00	0.91	10	0	0	0.226	0.077	78.8%	29.5%	2.3%	23.1%	30.1%	37.9%
Dani Alves	Barcelona	1.80	1.20	0.00	0.60	0.00	0.00	0	0	0	0.041	0.038	40.0%	0.0%	33.3%	11.5%	3.6%	0.0%
Ivan Rakitić	Barcelona	1.45	1.36	0.00	0.09	0.00	0.18	2	0	0	0.064	0.037	33.3%	9.1%	25.8%	10.3%	5.7%	12.1%
Sandro Ramírez	Barcelona	1.67	1.67	0.00	0.00	0.00	0.33	1	0	0	0.074	0.050	88.9%	33.3%	22.2%	9.9%	8.2%	11.1%
Andrés Iniesta	Barcelona	1.50	1.25	0.25	0.00	0.00	0.00	0	0	0	0.074	0.060	33.3%	16.7%	0.0%	9.3%	8.8%	0.0%
Pedro	Barcelona	1.50	1.50	0.00	0.00	0.00	0.00	0	0	0	0.076	0.058	50.0%	12.5%	25.0%	9.0%	7.4%	0.0%
Jordi Alba	Barcelona	1.00	1.00	0.00	0.00	0.00	0.00	0	0	0	0.112	0.112	100.0%	20.0%	0.0%	5.8%	11.1%	0.0%

Table 36 Player Tier Comparison — Shot Analysis 2014-2015 (Shot Share > 5%)

Player	Team	T1 Sh	T1 OP	T1 G	T1 OPG	T1 PG	T1 xG	T1 Rat	T1 Box%	T1 Sh%	T2 Sh	T2 OP	T2 G	T2 OPG	T2 PG	T2 xG	T2 Rat	T2 Box%	T2 Sh%
Cristiano Ronaldo	Real Madrid	5.50	3.83	3	2	1	0.319	0.082	37.8%	28.6%	6.33	4.67	7	5	2	0.805	0.128	54.9%	38.7%
Gareth Bale	Real Madrid	3.50	3.25	0	0	0	0.271	0.092	64.3%	16.7%	4.00	3.60	2	2	0	0.353	0.100	71.7%	21.2%
Chicharito	Real Madrid	3.00	2.67	1	1	0	0.240	0.102	100.0%	14.1%	4.00	4.00	0	0	0	0.346	0.086	75.0%	17.0%
Karim Benzema	Real Madrid	2.75	2.75	3	3	0	0.347	0.135	85.4%	14.0%	2.00	2.00	3	3	0	0.134	0.094	68.8%	12.3%
Luka Modrić	Real Madrid	1.67	1.67	0	0	0	0.016	0.008	0.0%	9.5%	2.00	2.00	0	0	0	0.030	0.015	0.0%	9.1%
Toni Kroos	Real Madrid	2.20	2.20	0	0	0	0.018	0.008	10.0%	11.6%	1.00	1.00	0	0	0	0.017	0.017	33.3%	5.2%
Marcelo	Real Madrid	2.00	2.00	0	0	0	0.036	0.017	27.8%	10.3%	1.00	1.00	1	1	0	0.008	0.008	0.0%	6.8%
James Rodríguez	Real Madrid	1.67	1.50	0	0	0	0.092	0.076	41.7%	9.6%	1.00	1.00	1	1	0	0.268	0.268	100.0%	4.5%
Sergio Ramos	Real Madrid	1.50	1.50	0	0	0	0.170	0.113	100.0%	6.5%	1.33	1.00	0	0	0	0.116	0.075	66.7%	9.4%
Isco	Real Madrid	1.60	1.60	0	0	0	0.034	0.017	26.7%	7.8%	1.50	1.50	0	0	0	0.048	0.028	37.5%	8.7%
Raphaël Varane	Real Madrid	1.33	1.00	0	0	0	0.172	0.126	100.0%	7.4%	2.00	2.00	0	0	0	0.327	0.163	100.0%	5.9%
Álvaro Arbeloa	Real Madrid	1.00	1.00	0	0	0	0.018	0.018	0.0%	3.7%	2.00	2.00	1	1	0	0.359	0.141	100.0%	8.0%
Lionel Messi	Barcelona	3.67	2.56	4	4	0	0.262	0.070	44.9%	27.8%	5.75	4.50	6	6	0	0.529	0.090	73.8%	35.0%
Luis Suárez	Barcelona	5.00	4.83	6	6	0	0.451	0.091	81.0%	32.5%	4.00	4.00	1	1	0	0.167	0.035	41.7%	21.5%
Neymar	Barcelona	3.00	3.00	9	9	0	0.237	0.081	79.6%	24.4%	3.00	3.00	1	1	0	0.173	0.060	75.0%	17.4%
Dani Alves	Barcelona	1.33	1.33	0	0	0	0.061	0.061	66.7%	10.3%	2.50	1.00	0	0	0	0.011	0.004	0.0%	13.4%
Ivan Rakitić	Barcelona	1.50	1.38	2	2	0	0.070	0.041	33.3%	11.3%	1.33	1.33	0	0	0	0.049	0.026	33.3%	7.6%
Sandro Ramírez	Barcelona	1.00	1.00	0	0	0	0.052	0.052	100.0%	6.7%	2.00	2.00	1	1	0	0.085	0.049	83.4%	11.5%
Andrés Iniesta	Barcelona	1.67	1.33	0	0	0	0.036	0.018	11.1%	10.3%	1.00	1.00	0	0	0	0.186	0.186	100.0%	6.2%
Pedro	Barcelona	1.00	1.00	0	0	0	0.009	0.009	0.0%	6.7%	1.67	1.67	0	0	0	0.098	0.074	66.7%	9.7%
Jordi Alba	Barcelona	1.00	1.00	0	0	0	0.110	0.110	100.0%	5.6%	1.00	1.00	0	0	0	0.114	0.113	100.0%	6.0%

Critical Match Analysis — 2014-2015

Nine matches are examined for this season: four for Real Madrid and five for Barcelona. They are presented in order of decisiveness rather than chronology, with the Juventus Semi-Final first, since it determined Real Madrid's elimination and provides the direct comparison with Barcelona's Final against the same opponent.

Before the individual matches, one aggregate figure frames the section. Across their four critical matches Real Madrid generated 5.15 xG and scored 3 goals. Across their five critical matches Barcelona generated 3.82 xG and scored 13. Real Madrid created a third more chance quality in the matches that decided their season and scored less than a quarter of the goals [DATA].

Real Madrid vs Juventus — Semi-Final, Leg 1 (Away, Lost 2:1)

This is the decisive match of Real Madrid's season. Juventus generated 12 shots for 1.615 xG at a rationality of 0.135 — the highest opponent chance quality Real Madrid conceded in any Tier 1 match during the campaign. Carlos Tevez alone accounted for 0.843 xG from 5 shots, and Álvaro Morata converted from 0.469 xG across 2 attempts. Real Madrid registered a comparable shot count of 13 for 1.198 xG, with Ronaldo scoring from 0.393 xG.

The distribution of Real Madrid's remaining attempts is notable. James Rodríguez recorded a single shot worth 0.347 xG and Gareth Bale a single shot worth 0.204 — two of the highest-value individual chances in the match, both unconverted. Toni Kroos took 3 shots for a combined 0.011 xG. Real Madrid were not starved of quality opportunities; they created several and converted one.

The two-goal deficit established here shaped everything that followed. Real Madrid entered the home leg needing to score, against an opponent with every incentive to defend a lead.

Real Madrid vs Juventus — Semi-Final, Leg 2 (Home, Drew 1:1)

The home leg is the most complete expression of the profile identified throughout this season. Real Madrid took 23 shots for 2.200 xG — the highest single-match xG total either club recorded in any critical match this season — and scored once. Their accuracy was 21.7%. Juventus took 7 shots for 0.303 xG and scored once.

Gareth Bale led with 7 shots for 0.690 xG without scoring; Karim Benzema added 4 shots for 0.322 xG without scoring. Ronaldo converted from 0.770 xG across 3 attempts, the highest individual xG figure in the match. Juventus were held to their lowest shot count of the tie, conceding territory almost entirely — and still drew.

Real Madrid dominated the match by every volume and territorial measure and were eliminated. The shot data establishes that the chances existed by distance-based measures and were not converted. It cannot establish why. Chance construction is examined in Module 6; the density of the Juventus defensive structure in Module 8.

Real Madrid vs Atletico Madrid — Quarter-Final (Drew 0:0 away, Won 1:0 home)

The Atletico tie shows the same shape at lower intensity. In the away leg Real Madrid took 17 shots to Atletico's 8, yet Atletico generated the higher xG total — 0.858 against 0.681 — at more than double the rationality, 0.107 against 0.040. Real Madrid's leading shooter, Ronaldo, took 5 shots for a combined 0.121 xG with only 20% from inside the box. Neither side scored.

The home leg was resolved by Chicharito, whose 7 shots for 0.482 xG at 100% box concentration produced the single goal of the tie. Ronaldo took 6 shots for 0.091 xG — a volume of low-value attempts that mirrors the away leg. Real Madrid took 23 shots for 1.075 xG and won 1:0; Atletico managed 6 shots for 0.233 xG.

Across both legs Real Madrid generated 40 shots for 1.756 xG and scored once. Against two deep-defending opponents in succession, the pattern was consistent.

Barcelona vs Paris Saint-Germain — Quarter-Final (Won 3:1 away, Won 2:0 home)

Barcelona's Quarter-Final inverts every ratio. In the away leg Barcelona took 11 shots for 0.463 xG and scored 3 — a return of more than six times their distance-based expectation. Suárez scored twice from 3 shots worth 0.151 xG combined; Neymar converted from 0.260 xG. Messi took 3 shots, none from inside the box, for 0.043 xG. PSG managed 8 shots for 0.241 xG and scored none.

The home leg followed the same pattern more modestly: 11 shots for 0.623 xG and 2 goals, both from Neymar, whose 3 attempts carried 0.392 xG at 100% box concentration. Only four Barcelona players registered shots in the away leg and five in the home leg — an extremely narrow shooting base producing five goals across the tie.

Barcelona vs Bayern Munich — Semi-Final (Won 3:0 home, Lost 3:2 away)

The home leg produced Barcelona's highest critical-match xG of the season at 1.252 from 15 shots, converted into 3 goals. Suárez took the most shots at 6 for 0.552 xG without scoring; Messi scored twice from 4 shots worth 0.169 xG combined, at 25% box concentration.

The away leg is the more analytically striking. Bayern took 19 shots for 1.122 xG and scored 3. Barcelona took 5 shots for 0.482 xG and scored 2 — both by Neymar, from 2 attempts at

100% box concentration worth 0.300 xG. Barcelona registered a Field Tilt of 20.8% in this match, the lowest recorded by either club in any critical match across the study to date, and still scored twice. Only three Barcelona players attempted a shot.

Barcelona lost the match and progressed 5:3 on aggregate. The two away goals, scored from five total attempts, were decisive.

Barcelona vs Juventus — Final (Won 3:1)

Barcelona took 17 shots for 1.001 xG and scored 3. Juventus took 14 shots for 0.754 xG and scored 1. Ivan Rakitić converted from a single shot worth 0.124 xG; Suárez scored from 7 attempts worth 0.497 xG at 85.7% box concentration; Neymar converted from 3 shots worth 0.093 xG combined. Messi took 3 shots and did not score.

The figure that connects this match to the rest of the section is Juventus's shot count. Barcelona permitted Juventus 14 shots — double the 7 that Real Madrid permitted them in the Bernabéu semi-final leg. Barcelona won a more open match against an opponent widely regarded as the most disciplined defensive side in the competition.

Table 37 Player Shot Data — Juventus vs Real Madrid, Semi-Final Leg 1, Lost 2:1, 2015-05-05

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Carlos Tevez	Juventus	5	1	0.843	0.169	60.0%	41.7%
Stefano Sturaro	Juventus	2	0	0.013	0.006	0.0%	16.7%
Álvaro Morata	Juventus	2	1	0.469	0.234	50.0%	16.7%
Claudio Marchisio	Juventus	1	0	0.004	0.004	0.0%	8.3%
Fernando Llorente	Juventus	1	0	0.263	0.263	100.0%	8.3%
Stephan Lichtsteiner	Juventus	1	0	0.024	0.024	100.0%	8.3%
TEAM TOTAL	Juventus	12	2	1.615	0.135	-	100%
Cristiano Ronaldo	Real Madrid	3	1	0.393	0.131	66.7%	23.1%
Toni Kroos	Real Madrid	3	0	0.011	0.004	0.0%	23.1%
Marcelo	Real Madrid	2	0	0.065	0.032	50.0%	15.4%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Chicharito	Real Madrid	1	0	0.106	0.106	100.0%	7.7%
Isco	Real Madrid	1	0	0.002	0.002	0.0%	7.7%
Gareth Bale	Real Madrid	1	0	0.204	0.204	100.0%	7.7%
James Rodríguez	Real Madrid	1	0	0.347	0.347	100.0%	7.7%
Raphaël Varane	Real Madrid	1	0	0.070	0.070	100.0%	7.7%
TEAM TOTAL	Real Madrid	13	1	1.198	0.092	-	100%

Table 38 Player Shot Data — Real Madrid vs Juventus, Semi-Final Leg 2, Drew 1:1, 2015-05-13

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Gareth Bale	Real Madrid	7	0	0.690	0.099	57.1%	30.4%
Karim Benzema	Real Madrid	4	0	0.322	0.080	75.0%	17.4%
Toni Kroos	Real Madrid	3	0	0.030	0.010	0.0%	13.0%
Cristiano Ronaldo	Real Madrid	3	1	0.770	0.257	33.3%	13.0%
Sergio Ramos	Real Madrid	2	0	0.227	0.114	100.0%	8.7%
Chicharito	Real Madrid	1	0	0.132	0.132	100.0%	4.3%
Isco	Real Madrid	1	0	0.020	0.020	100.0%	4.3%
James Rodríguez	Real Madrid	1	0	0.004	0.004	0.0%	4.3%
Marcelo	Real Madrid	1	0	0.005	0.005	0.0%	4.3%
TEAM TOTAL	Real Madrid	23	1	2.200	0.096	-	100%
Arturo Vidal	Juventus	2	0	0.097	0.048	50.0%	28.6%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Claudio Marchisio	Juventus	2	0	0.055	0.028	50.0%	28.6%
Álvaro Morata	Juventus	2	1	0.098	0.049	50.0%	28.6%
Paul Pogba	Juventus	1	0	0.053	0.053	100.0%	14.3%
TEAM TOTAL	Juventus	7	1	0.303	0.043	-	100%

Table 39 Player Shot Data — Atletico Madrid vs Real Madrid, Quarter-Final Leg 1, Drew 0:0, 2015-04-14

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Antoine Griezmann	Atletico Madrid	2	0	0.184	0.092	50.0%	25.0%
Diego Godín	Atletico Madrid	2	0	0.247	0.124	100.0%	25.0%
Arda Turan	Atletico Madrid	1	0	0.114	0.114	100.0%	12.5%
Guilherme Siqueira	Atletico Madrid	1	0	0.023	0.023	100.0%	12.5%
Mario Mandzukic	Atletico Madrid	1	0	0.051	0.051	100.0%	12.5%
Mario Suárez	Atletico Madrid	1	0	0.238	0.238	100.0%	12.5%
TEAM TOTAL	Atletico Madrid	8	0	0.858	0.107	-	100%
Cristiano Ronaldo	Real Madrid	5	0	0.121	0.024	20.0%	29.4%
Gareth Bale	Real Madrid	3	0	0.092	0.031	66.7%	17.6%
Luka Modrić	Real Madrid	3	0	0.036	0.012	0.0%	17.6%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
James Rodríguez	Real Madrid	2	0	0.072	0.036	50.0%	11.8%
Dani Carvajal	Real Madrid	1	0	0.027	0.027	100.0%	5.9%
Karim Benzema	Real Madrid	1	0	0.157	0.157	100.0%	5.9%
Raphaël Varane	Real Madrid	1	0	0.170	0.170	100.0%	5.9%
Toni Kroos	Real Madrid	1	0	0.007	0.007	0.0%	5.9%
TEAM TOTAL	Real Madrid	17	0	0.681	0.040	-	100%

Table 40 Player Shot Data — Real Madrid vs Atletico Madrid, Quarter-Final Leg 2, Won 1:0, 2015-04-22

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Chicharito	Real Madrid	7	1	0.482	0.069	100.0%	30.4%
Cristiano Ronaldo	Real Madrid	6	0	0.091	0.015	16.7%	26.1%
Isco	Real Madrid	2	0	0.017	0.008	0.0%	8.7%
James Rodríguez	Real Madrid	2	0	0.006	0.003	0.0%	8.7%
Raphaël Varane	Real Madrid	2	0	0.276	0.138	100.0%	8.7%
Toni Kroos	Real Madrid	2	0	0.011	0.006	0.0%	8.7%
Pepe	Real Madrid	1	0	0.079	0.079	100.0%	4.3%
Sergio Ramos	Real Madrid	1	0	0.112	0.112	100.0%	4.3%
TEAM TOTAL	Real Madrid	23	1	1.075	0.047	-	100%
Koke	Atletico Madrid	2	0	0.131	0.066	100.0%	33.3%
Gabi	Atletico Madrid	1	0	0.005	0.005	0.0%	16.7%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Jesús Gámez	Atletico Madrid	1	0	0.006	0.006	0.0%	16.7%
Raúl García	Atletico Madrid	1	0	0.007	0.007	0.0%	16.7%
Tiago	Atletico Madrid	1	0	0.084	0.084	100.0%	16.7%
TEAM TOTAL	Atletico Madrid	6	0	0.233	0.039	-	100%

Table 41 Player Shot Data — Paris Saint-Germain vs Barcelona, Quarter-Final Leg 1, Won 3:1, 2015-04-15

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Edinson Cavani	PSG	3	0	0.215	0.072	100.0%	37.5%
Adrien Rabiot	PSG	2	0	0.012	0.006	0.0%	25.0%
Gregory van der Wiel	PSG	1	0	0.005	0.005	0.0%	12.5%
Javier Pastore	PSG	1	0	0.004	0.004	0.0%	12.5%
Yohan Cabaye	PSG	1	0	0.005	0.005	0.0%	12.5%
TEAM TOTAL	PSG	8	0	0.241	0.030	-	100%
Neymar	Barcelona	4	1	0.260	0.065	100.0%	36.4%
Lionel Messi	Barcelona	3	0	0.043	0.014	0.0%	27.3%
Luis Suárez	Barcelona	3	2	0.151	0.050	33.3%	27.3%
Ivan Rakitić	Barcelona	1	0	0.009	0.009	0.0%	9.1%
TEAM TOTAL	Barcelona	11	3	0.463	0.042	-	100%

Note: PSG's team total shows zero goals against a final score of 3:1. Their goal was an own goal, which under the player-attribution convention set out in 4.1 is credited to no shooter and appears in neither team's shot data.

Table 42 Player Shot Data — Barcelona vs Paris Saint-Germain, Quarter-Final Leg 2, Won 2:0, 2015-04-21

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Lionel Messi	Barcelona	4	0	0.115	0.029	25.0%	36.4%
Neymar	Barcelona	3	2	0.392	0.131	100.0%	27.3%
Dani Alves	Barcelona	2	0	0.003	0.002	0.0%	18.2%
Ivan Rakitić	Barcelona	1	0	0.004	0.004	0.0%	9.1%
Gerard Piqué	Barcelona	1	0	0.109	0.109	100.0%	9.1%
TEAM TOTAL	Barcelona	11	2	0.623	0.057	-	100%
Yohan Cabaye	PSG	2	0	0.020	0.010	0.0%	22.2%
Blaise Matuidi	PSG	1	0	0.258	0.258	100.0%	11.1%
Ezequiel Lavezzi	PSG	1	0	0.044	0.044	100.0%	11.1%
Edinson Cavani	PSG	1	0	0.123	0.123	100.0%	11.1%
Javier Pastore	PSG	1	0	0.050	0.050	100.0%	11.1%
Lucas Moura	PSG	1	0	0.022	0.022	0.0%	11.1%
Marco Verratti	PSG	1	0	0.050	0.050	100.0%	11.1%
Zlatan Ibrahimović	PSG	1	0	0.012	0.012	0.0%	11.1%
TEAM TOTAL	PSG	9	0	0.578	0.064	-	100%

Table 43 Player Shot Data — Barcelona vs Bayern Munich, Semi-Final Leg 1, Won 3:0, 2015-05-06

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Luis Suárez	Barcelona	6	0	0.552	0.092	66.7%	40.0%
Lionel Messi	Barcelona	4	2	0.169	0.042	25.0%	26.7%
Neymar	Barcelona	3	1	0.393	0.131	100.0%	20.0%
Dani Alves	Barcelona	1	0	0.132	0.132	100.0%	6.7%
Ivan Rakitić	Barcelona	1	0	0.005	0.005	0.0%	6.7%
TEAM TOTAL	Barcelona	15	3	1.252	0.083	-	100%
Robert Lewandowski	Bayern Munich	3	0	0.293	0.098	66.7%	37.5%
Bastian Schweinsteiger	Bayern Munich	2	0	0.284	0.142	50.0%	25.0%
Thiago Alcântara	Bayern Munich	2	0	0.019	0.010	0.0%	25.0%
Xabi Alonso	Bayern Munich	1	0	0.010	0.010	0.0%	12.5%
TEAM TOTAL	Bayern Munich	8	0	0.605	0.076	-	100%

Table 44 Player Shot Data — Bayern Munich vs Barcelona, Semi-Final Leg 2, Lost 3:2, 2015-05-12

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Thomas Müller	Bayern Munich	5	1	0.262	0.052	60.0%	26.3%
Robert Lewandowski	Bayern Munich	5	1	0.251	0.050	40.0%	26.3%
Medhi Benatia	Bayern Munich	4	1	0.406	0.102	100.0%	21.1%
Bastian Schweinsteiger	Bayern Munich	2	0	0.093	0.046	50.0%	10.5%
Thiago Alcântara	Bayern Munich	2	0	0.100	0.050	50.0%	10.5%
Juan Bernat	Bayern Munich	1	0	0.009	0.009	0.0%	5.3%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
TEAM TOTAL	Bayern Munich	19	3	1.122	0.059	-	100%
Lionel Messi	Barcelona	2	0	0.120	0.060	100.0%	40.0%
Neymar	Barcelona	2	2	0.300	0.150	100.0%	40.0%
Ivan Rakitić	Barcelona	1	0	0.063	0.063	100.0%	20.0%
TEAM TOTAL	Barcelona	5	2	0.482	0.096	-	100%

Table 45 Player Shot Data — Juventus vs Barcelona, Final, Won 3:1, 2015-06-06

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Carlos Tevez	Juventus	4	0	0.133	0.033	25.0%	28.6%
Álvaro Morata	Juventus	3	1	0.366	0.122	100.0%	21.4%
Claudio Marchisio	Juventus	3	0	0.017	0.006	0.0%	21.4%
Arturo Vidal	Juventus	2	0	0.020	0.010	0.0%	14.3%
Paul Pogba	Juventus	2	0	0.218	0.109	50.0%	14.3%
TEAM TOTAL	Juventus	14	1	0.754	0.054	-	100%
Luis Suárez	Barcelona	7	1	0.497	0.071	85.7%	41.2%
Lionel Messi	Barcelona	3	0	0.055	0.018	33.3%	17.6%
Neymar	Barcelona	3	1	0.093	0.031	33.3%	17.6%
Ivan Rakitić	Barcelona	1	1	0.124	0.124	100.0%	5.9%
Gerard Piqué	Barcelona	1	0	0.090	0.090	100.0%	5.9%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Dani Alves	Barcelona	1	0	0.049	0.049	100.0%	5.9%
Jordi Alba	Barcelona	1	0	0.092	0.092	100.0%	5.9%
TEAM TOTAL	Barcelona	17	3	1.001	0.059	-	100%

The Juventus Test

Both clubs faced Juventus in the same season, three weeks apart. Holding the opponent constant removes the opponent-archetype variable that complicates every other comparison in this module.

Table 46 The results of Juventus matches for both Real Madrid and Barcelona

	Real Madrid — SF Leg 2 (H)	Barcelona — Final
Shots	23	17
xG	2.200	1.001
Goals	1	3
Accuracy	21.7%	47.1%
Rationality	0.096	0.059
Box Shot %	52.2%	70.6%
Field Tilt	76.7%	54.8%
Juventus shots permitted	7	14
Juventus xG permitted	0.303	0.754

By every distance-based measure Real Madrid's performance was superior. They took more shots, generated more than double the xG, recorded a higher rationality, and restricted Juventus to half the shots Barcelona permitted. That performance produced one goal and elimination. Barcelona's produced three goals and the trophy.

The two profiles differ most in what each club permitted. Real Madrid held Juventus to 7 shots and 0.303 xG — a near-total territorial suppression. Barcelona permitted 14 shots and 0.754 xG, more than double on both counts, and scored three times while doing so. Barcelona's Box Shot percentage of 70.6% against Real Madrid's 52.2% indicates that a substantially larger share of Barcelona's attempts came from inside the penalty area despite their lower total xG.

This is the clearest demonstration available in the dataset that the distance-only xG model does not capture what separated these two performances. Two teams, one opponent, one season: the side generating less than half the expected goals scored three times as many. The shot pillar can establish that this gap exists and quantify it precisely. It cannot determine its cause. Whether the difference lies in how each club's chances were constructed, or in the defensive configuration Juventus adopted in each match, is examined in Modules 6 and 8 respectively.

Key Observations — Season 2014-2015

The 2014-2015 season is the first in the study period where the two clubs' outcomes inverted, and it is the season where the shot pillar's limits become clearest. Real Madrid outperformed Barcelona on most measures the pillar can capture, and were eliminated a round earlier.

Season level. Real Madrid generated more shots per match (18.7 to 14.8), higher chance quality (xG 1.42 to 1.00), and a larger Quality Margin over their opponents (0.89 to 0.35). Barcelona scored more (2.38 goals per match to 1.92) on a Finishing Delta of 1.39 against Real Madrid's 0.50, and recorded higher accuracy at 48.2% against 42.3%. On chance creation Real Madrid satisfied the first condition comfortably. On conversion they did not.

Tier level. The pattern sharpened against elite opposition. Real Madrid's Tier 1 Quality Margin of +0.54 was four times Barcelona's +0.13, and their Field Tilt of 69.0% substantially exceeded Barcelona's 53.3%. Yet Real Madrid's Tier 1 Finishing Delta was -0.54 against Barcelona's +1.37, and their Tier 1 accuracy fell to 31.7% against Barcelona's 50.4%. Real Madrid out-created and out-territored elite opponents while scoring 1.75 goals per Tier 1 match to Barcelona's 2.33. The second condition — robustness against quality — was satisfied in creation and failed in conversion.

Two contextual factors qualify the tier comparison and should be carried into Module 9 rather than treated as settled here. Real Madrid played four Tier 1 matches to Barcelona's nine, so Real Madrid's tier averages rest on a small sample. And all four of Real Madrid's Tier 1 legs came against deep-defending opponents — Atletico Madrid and Juventus — while Barcelona's Tier 1 fixtures were predominantly against open, attacking sides. The

archetypes each club drew differed materially, and the tactical consequence of that difference is a question for Modules 6 and 8.

Critical match level. This is where the season was decided, and where the aggregates proved not to be protective. Across four critical matches Real Madrid generated 5.15 xG and scored 3. Across five, Barcelona generated 3.82 xG and scored 13. The Juventus away leg was the single decisive result: Juventus produced 1.615 xG from 12 shots at a rationality of 0.135, the highest opponent chance quality Real Madrid conceded all season, and won 2:1. The deficit established there forced the home leg in which Real Madrid took 23 shots for 2.200 xG, held Juventus to 7 shots, and drew 1:1.

Judged against the three necessary conditions set out in Module 3, Real Madrid satisfied the season condition, satisfied the tier condition in creation but not conversion, and failed at the critical match level. The failure at the third level was sufficient on its own.

On diversity. This season requires a qualification to an observation made in 2013-2014. That season identified Real Madrid's ability to score from open play, set pieces and penalties as a structural advantage over Barcelona's near-exclusive reliance on open play. The 2014-2015 data does not support treating that as a general principle. Barcelona scored all 30 of their goals from open play — the most concentrated mechanism profile recorded in the study — and won the competition.

The distinction that survives is between two different forms of redundancy. Mechanism diversity concerns how goals are scored: open play, set piece, free kick, penalty. Personnel diversity concerns how many players carry the scoring load. Real Madrid in 2013-2014 had high mechanism diversity and low personnel diversity, with Ronaldo taking 41.3% of shots and 46.2% of goals. Barcelona in 2014-2015 had the lowest mechanism diversity in the study and the highest personnel diversity, with Messi, Suárez and Neymar holding goal shares of 27.6%, 32.3% and 37.9% respectively.

Both clubs, in their respective winning seasons, possessed a form of redundancy. Neither possessed both. This suggests the relevant property may not be diversity of mechanism specifically, but the presence of *some* alternative route to goal when a primary threat is neutralised — whether that alternative is another method or another player. Whether either form is genuinely protective, or whether the association is coincidental across two seasons, is tested against the full six-season sample in Module 9.

On format tolerance. One structural observation closes this season, and it reframes the apparent paradox that Real Madrid performed better by several measures yet went out. Unlike a domestic league, where performance is averaged across thirty-eight matches and superior chance creation reliably converts into points over time, the Champions League

knockout stage is a low-sample, low-tolerance format. A two-legged tie offers little room to absorb a single poor result, and the variance of one match can override a season's worth of stronger underlying performance. Real Madrid's elimination is precisely this case: across the Tier 1 sample they created higher chance quality than the eventual champions, yet a single leg decided the outcome against them. The distinction between performing well on average and performing reliably at the decisive moment is one this study returns to repeatedly, and it is the empirical heart of the quality-control framing developed in Module 9. In a competition this unforgiving, the team that wins is not always the team that plays better on average, but the team that fails least at the moment failure is most expensive.

Open questions carried forward. The shot pillar has established, with unusual precision this season, that a substantial conversion gap existed between two clubs whose chance creation ran in the opposite direction. It has located that gap in the forward line of both squads, quantified it at team, tier and match level, and isolated it under controlled conditions in the Juventus Test. It cannot explain it. Three questions pass to the pillars that follow: how each club was organised to build (Module 5), how their chances were constructed — through sustained circulation or through transition (Module 6), and how compressed the defensive structures they faced actually were (Module 8).

4.5 Season 2015-2016

The 2015-2016 UEFA Champions League produced Real Madrid's eleventh European title and Barcelona's second consecutive quarter-final elimination at the hands of Atletico Madrid. Real Madrid defeated Atletico in the final at the San Siro, drawing 1:1 after extra time before winning 5:3 on penalties. Barcelona, the reigning champions, were eliminated 3:2 on aggregate after losing the second leg in Madrid.

The structure of this season reproduces 2013-2014 precisely. In both campaigns Barcelona were eliminated by Atletico Madrid at the quarter-final stage and Real Madrid subsequently defeated Atletico in the final. The same opponent, at the same two stages, produced the same two outcomes two seasons apart. This provides the third same-opponent comparison in the study, after Atletico Madrid in 2013-2014 and Juventus in 2014-2015, and the second instance of this particular structure.

It is also the season in which the shot pillar produces its most counterintuitive result. Real Madrid won the competition while scoring two goals in five matches against elite opposition.

Season Overview

Table 47 presents the season average shot metrics for both teams across all matches in the 2015-2016 UEFA Champions League campaign.

Table 47 Season Overview — Shot Analysis 2015-2016

Metric	Real Madrid	Barcelona
Avg Shots per Match	21.2	16.8
Avg Goals per Match	2.08	2.20
xG per Match	1.39	1.50
Rationality (xG/Shot)	0.066	0.087
Box Shot %	58.9%	59.5%
Accuracy %	36.2%	37.0%
Big Misses per Match	0.23	0.30
Opp Shots per Match	9.2	10.4
Opp xG per Match	0.67	0.72
Finishing Delta (Goals – xG)	0.69	0.70

Metric	Real Madrid	Barcelona
Quality Margin (xG – Opp xG)	0.71	0.78
Field Tilt %	69.8%	63.3%

At season level the two clubs are almost indistinguishable. Barcelona scored marginally more per match, 2.20 against 2.08, and generated marginally higher chance quality at 1.50 xG against 1.39. Their Finishing Delta figures are effectively identical at 0.70 and 0.69, as are their Quality Margins at 0.78 and 0.71. Accuracy differed by less than a percentage point. Barcelona's rationality of 0.087 exceeded Real Madrid's 0.066, indicating a higher average xG per attempt, which is consistent with Real Madrid's substantially greater shot volume of 21.2 per match against Barcelona's 16.8 — a larger denominator distributed across a similar total.

Real Madrid held the clearer territorial advantage, with a Field Tilt of 69.8% against Barcelona's 63.3%, and conceded fewer shots per match, 9.2 against 10.4. Both figures are among the strongest recorded by either club in the study period to this point.

On these aggregates alone there is no basis for predicting that one club would win the competition and the other exit in the quarter-finals. The season averages satisfy the first of the three conditions set out in Module 3 for both clubs, and separate them by nothing of consequence. The divergence appears only when the sample is partitioned by opponent quality.

Tier Comparison

Table 48 presents the tier breakdown separating performance against Tier 1 (Elite, Elo > 1850) and Tier 2 (Standard, Elo ≤ 1850) opponents. Real Madrid played five Tier 1 matches and eight Tier 2; Barcelona played four Tier 1 and six Tier 2.

Table 48 Tier Comparison — Shot Analysis 2015-2016

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Avg Shots per Match	16.8	23.9	16.5	17.0
Avg Goals per Match	0.40	3.13	1.75	2.50
xG per Match	1.14	1.54	1.22	1.70
Rationality (xG/Shot)	0.067	0.066	0.073	0.096
Box Shot %	55.7%	60.9%	58.0%	60.5%
Accuracy %	36.5%	36.1%	34.9%	38.3%

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Big Misses per Match	0.20	0.25	0.00	0.50
Opp Shots per Match	10.4	8.4	11.0	10.0
Opp xG per Match	0.62	0.70	0.82	0.65
Finishing Delta (Goals – xG)	–0.74	1.59	0.53	0.80
Quality Margin (xG – Opp xG)	0.52	0.84	0.39	1.05
Field Tilt %	63.9%	73.5%	61.3%	64.6%

The tier split separates what the season averages concealed. Real Madrid's Tier 1 goal output was 0.40 per match — two goals across five matches against elite opposition, the lowest figure recorded by either club in any tier across the study period. Their Tier 1 Finishing Delta was –0.74, meaning they scored roughly three quarters of a goal per match below their distance-based expectation. Their Quality Margin against the same opponents was +0.52, indicating they generated more chance quality than those opponents did.

That combination — a positive Quality Margin paired with a sharply negative Finishing Delta — is the diagnostic pattern identified in Module 4.4. In 2014-2015 Real Madrid recorded a Tier 1 Quality Margin of +0.54 and a Finishing Delta of –0.54, and were eliminated in the semi-finals. In 2015-2016 the same profile appears in a slightly more extreme form, with a Quality Margin of +0.52 and a Finishing Delta of –0.74, and the club won the competition. Two seasons, near-identical Tier 1 shooting profiles, opposite outcomes.

The difference is on the defensive side of the same table. Real Madrid conceded 0.62 opponent xG per Tier 1 match in 2015-2016, against 0.75 the previous season, and conceded one goal in total across the five matches. The Tier 1 results were 0:0 and 1:0 against Paris Saint-Germain, 0:0 and 1:0 against Manchester City, and 1:1 against Atletico Madrid in the final. A team scoring twice in five elite matches won the competition because it conceded once.

This is the shot pillar observing a defensive outcome through the only lens available to it — opponent shot volume and opponent chance quality. It can establish that Real Madrid's elite opponents were restricted to 10.4 shots and 0.62 xG per match. It cannot establish how. The defensive mechanism is examined in Module 8.

Barcelona's Tier 1 profile is the more conventional of the two and, on the attacking metrics, the stronger. They scored 1.75 goals per Tier 1 match against Real Madrid's 0.40, recorded a positive Finishing Delta of 0.53 where Real Madrid's was negative, and generated marginally higher Tier 1 xG at 1.22 against 1.14. Their Quality Margin of 0.39 was lower than Real Madrid's 0.52, and

their opponent xG conceded was higher at 0.82 against 0.62. Barcelona out-scored Real Madrid against elite opposition by a factor of more than four and were eliminated two rounds earlier.

Against Tier 2 opponents both clubs performed comfortably. Real Madrid's Tier 2 Finishing Delta of 1.59 and Field Tilt of 73.5% reflect substantial dominance, and their Tier 2 goal output of 3.13 per match stands against 0.40 at Tier 1 — the largest tier gap recorded by either club in the study. Barcelona's Tier 2 figures show a smaller but consistent advantage, with a Quality Margin of 1.05 and 2.50 goals per match.

Real Madrid's tier gap warrants emphasis. A goal output of 3.13 per match against standard opposition and 0.40 against elite opposition is a ratio of nearly eight to one. Whatever the club's attacking structure was doing against weaker opponents, it did not survive contact with elite defences.

Shot Map Analysis

Figure 7 presents the shot maps for Real Madrid across all 2015-2016 matches, and Figure 8 presents the equivalent maps for Barcelona. The spatial analysis draws on the zone distribution and zone xG data presented in Tables 49 and 50.

Real Madrid Spatial Profile

Real Madrid's spatial distribution against Tier 1 opponents shows a team reaching the box without difficulty. Box shots accounted for 53.6% of Tier 1 attempts and the central corridor for 71.4%, both broadly consistent with their Tier 2 figures of 58.6% and 74.3%. Access to the penalty area was not materially restricted by opponent quality.

Conversion was. In Box_Central, the most productive zone in every season of this study, Real Madrid took 34 shots against Tier 1 opponents and scored one — a goal rate of 2.9%. Against Tier 2 opponents the identical zone produced 21 goals from 84 shots, a rate of 25.0%. The average xG per shot in that zone was higher against elite opposition, at 0.1482 against 0.1272, so the model rated the Tier 1 chances as marginally better than the Tier 2 chances that were converted at nearly nine times the rate.

Across all zones Real Madrid scored two Tier 1 goals: one from Box_Left, from four attempts, and one from Box_Central, from thirty-four. Every attempt from the edge of the box and beyond — 39 shots in total — produced nothing. This is the sharpest zone-level conversion collapse recorded in the study.

Shot Analysis: Real Madrid (Red) vs Opponents (Blue)
Champions League 2015-2016

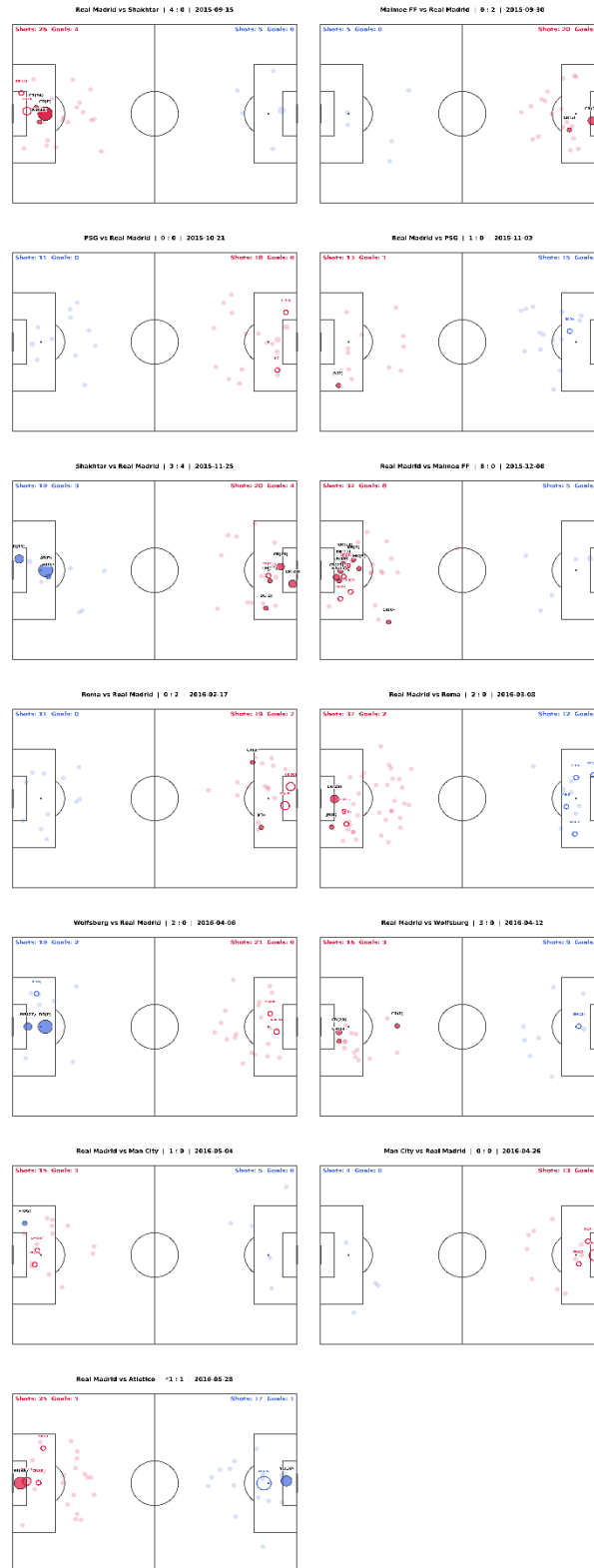


Figure 7 Real Madrid Shot Map — UCL 2015-2016

Barcelona Spatial Profile

Barcelona's Tier 1 spatial profile shows the opposite problem. Box shots accounted for only 47.1% of their Tier 1 attempts against 66.4% at Tier 2, and edge shots rose to 52.9% from 31.3%. Barcelona were being pushed out of the penalty area by elite opposition — the majority of their shots against Tier 1 opponents came from outside the box, a distribution not seen from either club in any other tier or season in the study to this point.

In absolute terms the numbers are small: 34 Tier 1 shots in total, of which 16 from the box and 13 from Box_Central. That sample supported two goals, both from Box_Central, at a rate of 15.4% — five times Real Madrid's rate in the same zone against the same tier. Barcelona's problem was not conversion. It was access.

Their lateral distribution reinforces this. Against Tier 1 opponents 79.4% of Barcelona's shots came from the central corridor and only 5.9% from the right channel, against 85.8% and 6.7% at Tier 2. The attack was funnelled centrally at both tiers, but at Tier 1 it was funnelled centrally and backwards, into the edge of the box.

Shot Analysis: Barcelona (Blue) vs Opponents (Red)
Champions League 2015-2016

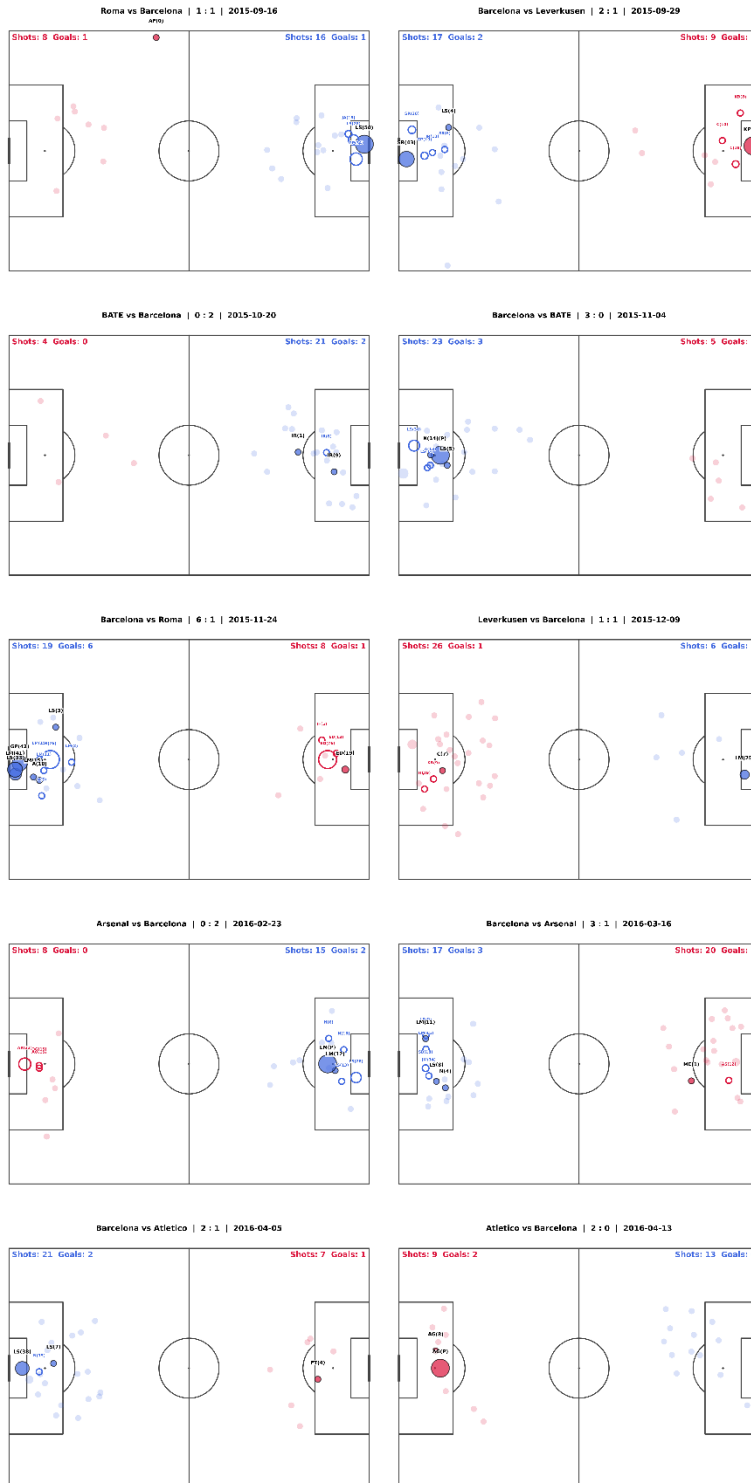


Figure 8 Barcelona Shot Map — UCL 2015-2016

Comparative Spatial Insight

The two clubs failed against elite opposition in opposite ways. Real Madrid reached Box_Central 34 times against Tier 1 opponents and scored once. Barcelona reached it 13 times and scored twice. Real Madrid had the access and not the conversion; Barcelona had the conversion and not the access.

Neither pattern is explicable from shot data. A distance-only model cannot distinguish a shot taken in space from one taken in congestion, and both clubs' Tier 1 zone profiles carry average xG values comparable to or higher than their Tier 2 equivalents. The spatial record establishes where the shots came from and how many were converted. Whether Real Madrid's central attempts were taken under greater defensive pressure, and whether Barcelona's displacement to the edge of the box reflects an opponent structure rather than a choice, are questions for Modules 6 and 8.

Table 49 Zone Shot Distribution — Real Madrid and Barcelona, UCL 2015-2016

Depth Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box	45	53.6%	112	58.6%	16	47.1%	89	66.4%
Edge	36	42.9%	72	37.7%	18	52.9%	42	31.3%
Long Range	3	3.6%	7	3.7%	0	0.0%	3	2.2%
TOTAL	84	100%	191	100%	34	100%	134	100%

Lateral Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Left	11	13.1%	32	16.8%	5	14.7%	10	7.5%
Central	60	71.4%	142	74.3%	27	79.4%	115	85.8%
Right	13	15.5%	17	8.9%	2	5.9%	9	6.7%
TOTAL	84	100%	191	100%	34	100%	134	100%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Left	4	4.8%	20	10.5%	2	5.9%	4	3.0%
Box_Central	34	40.5%	84	44.0%	13	38.2%	78	58.2%

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Right	7	8.3%	8	4.2%	1	2.9%	7	5.2%
Edge_Left	7	8.3%	11	5.8%	3	8.8%	6	4.5%
Edge_Central	23	27.4%	52	27.2%	14	41.2%	34	25.4%
Edge_Right	6	7.1%	9	4.7%	1	2.9%	2	1.5%
Long Range_Left	0	0.0%	1	0.5%	0	0.0%	0	0.0%
Long Range_Central	3	3.6%	6	3.1%	0	0.0%	3	2.2%
Long Range_Right	0	0.0%	0	0.0%	0	0.0%	0	0.0%
TOTAL	84	100%	191	100%	34	100%	134	100%

Table 50 Zone xG and Goals by Zone — Real Madrid and Barcelona, UCL 2015-2016

Depth Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box	45	0.1202	2	4.4%	112	0.1034	22	19.6%	16	0.1039	2	12.5%	89	0.1422	19	21.3%
Edge	36	0.0077	0	0.0%	72	0.0101	3	4.2%	18	0.0106	0	0.0%	42	0.0128	1	2.4%
Long Range	3	0.0027	0	0.0%	7	0.0017	0	0.0%	0	0	0	0.0%	3	0.0012	0	0.0%

Lateral Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Left	11	0.0181	1	9.1%	32	0.0215	2	6.2%	5	0.0126	0	0.0%	10	0.0152	0	0.0%
Central	60	0.0874	1	1.7%	142	0.0794	22	15.5%	27	0.0647	2	7.4%	115	0.1106	20	17.4%
Right	13	0.0193	0	0.0%	17	0.0212	1	5.9%	2	0.0210	0	0.0%	9	0.0360	0	0.0%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box_Left	4	0.0387	1	25.0%	20	0.0301	0	0.0%	2	0.0227	0	0.0%	4	0.0281	0	0.0%
Box_Central	34	0.1482	1	2.9%	84	0.1272	21	25.0%	13	0.1215	2	15.4%	78	0.1568	19	24.4%
Box_Right	7	0.0305	0	0.0%	8	0.0361	1	12.5%	1	0.0378	0	0.0%	7	0.0446	0	0.0%
Edge_Left	7	0.0063	0	0.0%	11	0.0076	2	18.2%	3	0.0058	0	0.0%	6	0.0066	0	0.0%

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Edge_Central	23	0.0085	0	0.0%	52	0.0110	1	1.9%	14	0.0121	0	0.0%	34	0.0143	1	2.9%
Edge_Right	6	0.0063	0	0.0%	9	0.0079	0	0.0%	1	0.0042	0	0.0%	2	0.0060	0	0.0%
Long Range_Left	0	0	0	0.0%	1	0.0011	0	0.0%	0	0	0	0.0%	0	0	0	0.0%
Long Range_Central	3	0.0027	0	0.0%	6	0.0018	0	0.0%	0	0	0	0.0%	3	0.0012	0	0.0%
Long Range_Right	0	0	0	0.0%	0	0	0	0.0%	0	0	0	0.0%	0	0	0	0.0%

Player Highlights

Table 51 presents the season average shot metrics for all players with a shot share above 5% for both clubs.

Table 51 Player Season Average — Shot Analysis 2015-2016 (Shot Share > 5%)

Player	Team	Shots	OP	SP	Goals	xG	Rat.	Box %	Left %	Right %	Shot %	xG %	Goal %
Cristiano Ronaldo	Real Madrid	7.92	6.33	0.08	1.33	0.662	0.078	59.1%	25.6%	15.3%	36.3%	43.8%	39.6%
Gareth Bale	Real Madrid	3.75	3.25	0.13	0.00	0.219	0.054	52.6%	28.5%	10.6%	20.0%	16.9%	0.0%
Karim Benzema	Real Madrid	3.38	3.13	0.25	0.50	0.329	0.091	84.0%	13.3%	18.8%	15.7%	20.5%	7.8%
Isco	Real Madrid	2.71	2.57	0.14	0.00	0.127	0.039	47.1%	18.6%	26.2%	13.8%	12.5%	0.0%
James Rodríguez	Real Madrid	2.67	2.67	0.00	0.33	0.110	0.040	50.0%	8.3%	16.7%	9.2%	8.2%	16.7%
Sergio Ramos	Real Madrid	1.40	1.00	0.40	0.20	0.216	0.175	100.0%	0.0%	0.0%	8.7%	18.6%	20.0%
Casemiro	Real Madrid	1.88	1.88	0.00	0.00	0.093	0.052	33.3%	4.2%	20.8%	8.3%	7.6%	0.0%
Toni Kroos	Real Madrid	1.56	1.56	0.00	0.00	0.018	0.009	9.3%	24.1%	20.4%	7.8%	2.1%	0.0%
Marcelo	Real Madrid	1.88	1.75	0.13	0.00	0.046	0.023	50.0%	72.9%	0.0%	7.4%	3.3%	0.0%
Lucas Vázquez	Real Madrid	1.25	1.25	0.00	0.00	0.046	0.037	62.5%	0.0%	37.5%	6.1%	3.4%	0.0%
Jesé	Real Madrid	1.13	1.13	0.00	0.13	0.055	0.049	68.8%	12.5%	62.5%	6.0%	4.0%	6.3%
Luka Modrić	Real Madrid	1.33	1.33	0.00	0.17	0.052	0.047	38.9%	0.0%	22.2%	5.9%	4.0%	4.2%
Danilo	Real Madrid	1.67	1.67	0.00	0.00	0.042	0.021	33.3%	0.0%	83.3%	5.6%	2.0%	0.0%
Mateo Kovačić	Real Madrid	1.25	1.25	0.00	0.25	0.027	0.022	25.0%	0.0%	50.0%	5.5%	1.9%	3.1%
Lionel Messi	Barcelona	5.57	4.86	0.00	0.86	0.491	0.090	60.7%	20.8%	7.7%	36.1%	38.3%	38.1%
Neymar	Barcelona	3.89	3.44	0.00	0.33	0.392	0.084	67.0%	26.3%	0.0%	20.6%	20.0%	11.1%

Player	Team	Shots	OP	SP	Goals	xG	Rat.	Box %	Left %	Right %	Shot %	xG %	Goal %
Luis Suárez	Barcelona	3.33	3.11	0.11	0.89	0.418	0.121	85.2%	6.5%	17.6%	18.5%	26.2%	38.9%
Sandro Ramírez	Barcelona	1.67	1.67	0.00	0.00	0.109	0.040	33.3%	0.0%	11.1%	13.0%	8.5%	0.0%
Ivan Rakitić	Barcelona	1.71	1.57	0.00	0.29	0.072	0.038	28.6%	4.8%	35.7%	11.0%	5.0%	14.3%
Munir El Haddadi	Barcelona	2.00	2.00	0.00	0.00	0.110	0.055	66.7%	0.0%	0.0%	10.0%	8.4%	0.0%
Jordi Alba	Barcelona	1.25	1.25	0.00	0.00	0.086	0.079	62.5%	37.5%	0.0%	7.8%	7.9%	0.0%
Sergio Busquets	Barcelona	1.25	1.00	0.25	0.00	0.093	0.082	62.5%	0.0%	37.5%	6.9%	12.6%	0.0%
Andrés Iniesta	Barcelona	1.00	1.00	0.00	0.00	0.011	0.011	0.0%	40.0%	0.0%	5.9%	1.0%	0.0%

Real Madrid's shooting load remained concentrated on Cristiano Ronaldo at 36.3% shot share and 39.6% goal share. Gareth Bale ranked second at 20.0% shot share and recorded no goals across the entire campaign. Karim Benzema, Isco, and James Rodríguez followed. Sergio Ramos returned the squad's highest rationality at 0.175 from 100% box shots, reflecting his role as a set-piece target rather than an open-play contributor — 0.40 of his 1.40 shots per match were set pieces, the highest proportion in either squad.

Barcelona's distribution is the more balanced. Messi, Neymar and Suárez held shot shares of 36.1%, 20.6% and 18.5%, and goal shares of 38.1%, 11.1% and 38.9%. Suárez converted 0.89 goals per match from 3.33 shots at a rationality of 0.121 and an 85.2% box concentration — the most efficient profile of any player in either squad this season.

The goal-type data records a further divergence. Barcelona scored 19 goals, of which 17 came from open play and 2 from penalties, with no set-piece or free-kick goals. Real Madrid scored 27, comprising 21 from open play, 1 from a set piece, 2 from free kicks and 3 from penalties. Real Madrid retained the broader mechanism profile identified in 2013-2014, though as noted in Module 4.4 this has not proven a reliable predictor of outcome across seasons.

Player Tier Comparison

Table 52 presents the player-level tier breakdown using the same player list as Table 51. Players with no appearances in a tier are shown with zero values.

Table 52 Player Tier Comparison — Shot Analysis 2015-2016 (Shot Share > 5%)

Player	Team	T1 Sh	T1 OP	T1 SP	T1 G	T1 xG	T1 Rat	T1 Box%	T1 Sh%	T2 Sh	T2 OP	T2 SP	T2 G	T2 xG	T2 Rat	T2 Box%	T2 Sh%
Cristiano Ronaldo	Real Madrid	5.50	4.50	0.00	0	0.345	0.056	42.9%	31.8%	9.13	7.25	0.13	2.00	0.821	0.089	67.2%	38.6%
Gareth Bale	Real Madrid	5.33	4.67	0.00	0	0.385	0.081	48.1%	30.2%	2.80	2.40	0.20	0.00	0.120	0.038	55.3%	13.8%
Karim Benzema	Real Madrid	1.50	1.00	0.50	0	0.171	0.090	100.0%	7.9%	4.00	3.83	0.17	0.67	0.382	0.091	78.6%	18.3%

Player	Team	T1 Sh	T1 OP	T1 SP	T1 G	T1 xG	T1 Rat	T1 Box%	T1 Sh%	T2 Sh	T2 OP	T2 SP	T2 G	T2 xG	T2 Rat	T2 Box%	T2 Sh%
Isco	Real Madrid	3.50	3.00	0.50	0	0.204	0.065	90.0%	24.8%	2.40	2.40	0.00	0.00	0.096	0.029	30.0%	9.4%
James Rodríguez	Real Madrid	0.00	0.00	0.00	0	0.000	0.000	0.0%	0.0%	2.67	2.67	0.00	0.33	0.110	0.040	50.0%	9.2%
Sergio Ramos	Real Madrid	1.00	0.33	0.67	1	0.232	0.231	100.0%	6.5%	2.00	2.00	0.00	0.00	0.194	0.090	100.0%	12.1%
Casemiro	Real Madrid	2.00	2.00	0.00	0	0.139	0.087	44.4%	10.8%	1.80	1.80	0.00	0.00	0.066	0.031	26.7%	6.8%
Toni Kroos	Real Madrid	1.20	1.20	0.00	0	0.013	0.008	10.0%	7.9%	2.00	2.00	0.00	0.00	0.023	0.009	8.3%	7.8%
Marcelo	Real Madrid	1.00	0.67	0.33	0	0.007	0.006	0.0%	5.8%	2.40	2.40	0.00	0.00	0.070	0.033	80.0%	8.4%
Lucas Vázquez	Real Madrid	1.33	1.33	0.00	0	0.061	0.048	83.3%	7.3%	1.00	1.00	0.00	0.00	0.003	0.003	0.0%	2.7%
Jesé	Real Madrid	1.33	1.33	0.00	0	0.072	0.058	83.3%	8.5%	1.00	1.00	0.00	0.20	0.044	0.044	60.0%	4.4%
Luka Modrić	Real Madrid	1.00	1.00	0.00	0	0.051	0.051	33.3%	6.1%	1.67	1.67	0.00	0.33	0.053	0.044	44.4%	5.6%
Danilo	Real Madrid	2.00	2.00	0.00	0	0.029	0.015	50.0%	8.0%	1.50	1.50	0.00	0.00	0.049	0.025	25.0%	4.5%
Mateo Kovačić	Real Madrid	0.00	0.00	0.00	0	0.000	0.000	0.0%	0.0%	1.25	1.25	0.00	0.25	0.027	0.022	25.0%	5.5%
Lionel Messi	Barcelona	4.50	3.50	0.00	0	0.151	0.028	25.0%	25.9%	6.00	5.40	0.00	1.20	0.627	0.115	75.0%	40.1%
Neymar	Barcelona	3.50	3.50	0.00	0	0.260	0.045	33.4%	18.2%	4.00	3.43	0.00	0.43	0.430	0.096	76.6%	21.3%
Luis Suárez	Barcelona	2.50	2.50	0.00	1	0.323	0.121	100.0%	14.9%	3.57	3.29	0.14	0.86	0.446	0.121	81.0%	19.6%
Sandro Ramírez	Barcelona	0.00	0.00	0.00	0	0.000	0.000	0.0%	0.0%	1.67	1.67	0.00	0.00	0.109	0.040	33.3%	13.0%
Ivan Rakitić	Barcelona	1.00	1.00	0.00	0	0.004	0.004	0.0%	4.8%	1.83	1.67	0.00	0.33	0.084	0.044	33.3%	12.1%
Munir El Haddadi	Barcelona	0.00	0.00	0.00	0	0.000	0.000	0.0%	0.0%	2.00	2.00	0.00	0.00	0.110	0.055	66.7%	10.0%
Jordi Alba	Barcelona	1.50	1.50	0.00	0	0.075	0.062	75.0%	10.1%	1.00	1.00	0.00	0.00	0.097	0.097	50.0%	5.5%
Sergio Busquets	Barcelona	1.00	1.00	0.00	0	0.097	0.097	100.0%	7.7%	1.33	1.00	0.33	0.00	0.092	0.077	50.0%	6.6%
Andrés Iniesta	Barcelona	1.00	1.00	0.00	0	0.006	0.006	0.0%	6.3%	1.00	1.00	0.00	0.00	0.014	0.014	0.0%	5.6%

The Tier 1 column produces the season's defining finding. Real Madrid's two goals against elite opposition were scored by Sergio Ramos, from a set piece in the final, and by Nacho, from open play against Paris Saint-Germain — a centre-back and a squad defender. No recognised forward or attacking midfielder scored a Tier 1 goal for Real Madrid all season.

The volume was there. Ronaldo took 5.50 shots per Tier 1 match and scored none. Bale took 5.33 and scored none. Benzema took 1.50 and scored none. Isco, at 3.50 shots per Tier 1 match with a 90.0% box concentration, scored none. Between them these four players took approximately 78 shots against elite opposition across five matches and did not produce a goal.

Ronaldo's tier split is the starkest of any player in the study to this point: 5.50 shots and no goals against Tier 1, against 9.13 shots and 2.00 goals per match against Tier 2. His Tier 2 rationality of

0.089 exceeded his Tier 1 figure of 0.056, and his box concentration fell from 67.2% at Tier 2 to 42.9% at Tier 1 — he was shooting less often from inside the penalty area against better opposition.

Barcelona's Tier 1 goals came from Suárez, who scored both, from 2.50 shots per match at 100% box concentration. Messi took 4.50 shots per Tier 1 match without scoring, at a rationality of 0.028 and a box concentration of 25.0% — against 0.115 and 75.0% at Tier 2. Neymar likewise took 3.50 Tier 1 shots without scoring, with box concentration falling from 76.6% to 33.4%.

Both clubs' principal attackers therefore show the same pattern against elite opposition: shot volume maintained, box concentration substantially reduced, goals absent. Whether this reflects opponent defensive structures pushing both attacks away from the penalty area, or a change in shot selection under pressure, cannot be determined from shot data. It is examined in Modules 6 and 8.

Critical Match Analysis

Seven matches are examined for this season: five for Real Madrid and two for Barcelona. They are presented in order of decisiveness. Tables 53 through 59 present the player-level shot data for both teams in each match.

Real Madrid's most consequential match was not a defeat by an elite opponent but the quarter-final first leg away to Wolfsburg, a Tier 2 side, which they lost 2:0 and which came closer than any other result to ending the campaign.

Table 53 Player Shot Data — Wolfsburg vs Real Madrid, Quarter-Final Leg 1, Lost 2:0, 2016-04-06

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Bruno Henrique	Wolfsburg	3	0	0.127	0.042	66.7%	30.0%
André Schürrle	Wolfsburg	2	0	0.071	0.036	100.0%	20.0%
Julian Draxler	Wolfsburg	2	0	0.048	0.024	50.0%	20.0%
Max Kruse	Wolfsburg	1	0	0.069	0.069	100.0%	10.0%
Maximilian Arnold	Wolfsburg	1	1	0.277	0.277	100.0%	10.0%
Ricardo Rodríguez	Wolfsburg	1	1	0.760	0.760	100.0%	10.0%
TEAM TOTAL	Wolfsburg	10	2	1.351	0.135	-	100%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	6	0	0.233	0.039	66.7%	28.6%
Gareth Bale	Real Madrid	5	0	0.155	0.031	60.0%	23.8%
Toni Kroos	Real Madrid	3	0	0.017	0.006	0.0%	14.3%
Casemiro	Real Madrid	2	0	0.131	0.066	50.0%	9.5%
Karim Benzema	Real Madrid	2	0	0.294	0.147	100.0%	9.5%
Marcelo	Real Madrid	2	0	0.061	0.030	100.0%	9.5%
Jesé	Real Madrid	1	0	0.022	0.022	0.0%	4.8%
TEAM TOTAL	Real Madrid	21	0	0.913	0.043	-	100%

Real Madrid took 21 shots to Wolfsburg's 10 and generated 0.913 xG against 1.351. Their accuracy was 14.3%, the lowest single-match figure recorded by either club in the study. Ronaldo took 6 shots for 0.233 xG and Bale 5 for 0.155 — eleven attempts between the two principal attackers producing less than four tenths of an expected goal. Wolfsburg scored twice from ten attempts, including a penalty converted by Ricardo Rodríguez worth 0.760 xG, the highest single-shot value in the match.

A Tier 2 opponent generated 48% more chance quality than Real Madrid from less than half the shot volume. Had the tie ended here, the season's narrative would have been very different.

Table 54 Player Shot Data — Real Madrid vs Wolfsburg, Quarter-Final Leg 2, Won 3:0, 2016-04-12

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	5	3	0.406	0.081	60.0%	31.2%
Karim Benzema	Real Madrid	4	0	0.212	0.053	75.0%	25.0%
Gareth Bale	Real Madrid	3	0	0.153	0.051	66.7%	18.8%
Sergio Ramos	Real Madrid	3	0	0.311	0.104	100.0%	18.8%
Jesé	Real Madrid	1	0	0.031	0.031	100.0%	6.2%
TEAM TOTAL	Real Madrid	16	3	1.113	0.070	-	100%
André Schürrle	Wolfsburg	2	0	0.058	0.029	50.0%	22.2%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Vieirinha	Wolfsburg	2	0	0.013	0.006	0.0%	22.2%
Luiz Gustavo	Wolfsburg	2	0	0.007	0.004	0.0%	22.2%
Bruno Henrique	Wolfsburg	1	0	0.138	0.138	100.0%	11.1%
Josuha Guilavogui	Wolfsburg	1	0	0.003	0.003	0.0%	11.1%
Dante	Wolfsburg	1	0	0.190	0.190	100.0%	11.1%
TEAM TOTAL	Wolfsburg	9	0	0.410	0.046	-	100%

Ronaldo scored three goals from five shots worth 0.406 xG in total — a return far above the distance-based expectation for those attempts. Real Madrid took fewer shots than in the first leg, 16 against 21, and generated more xG, 1.113 against 0.913. Wolfsburg were restricted to 9 shots and 0.410 xG. The tie was recovered in a single half.

Table 55 Player Shot Data — Manchester City vs Real Madrid, Semi-Final Leg 1, Drew 0:0, 2016-04-26

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Kevin De Bruyne	Man City	2	0	0.016	0.008	0.0%	50.0%
Nicolás Otamendi	Man City	1	0	0.110	0.110	100.0%	25.0%
Sergio Agüero	Man City	1	0	0.011	0.011	0.0%	25.0%
TEAM TOTAL	Man City	4	0	0.137	0.034	-	100%
Gareth Bale	Real Madrid	5	0	0.335	0.067	40.0%	38.5%
Casemiro	Real Madrid	1	0	0.176	0.176	100.0%	7.7%
Jesé	Real Madrid	1	0	0.110	0.110	100.0%	7.7%
Karim Benzema	Real Madrid	1	0	0.017	0.017	100.0%	7.7%
Luka Modrić	Real Madrid	1	0	0.004	0.004	0.0%	7.7%
Marcelo	Real Madrid	1	0	0.005	0.005	0.0%	7.7%
Pepe	Real Madrid	1	0	0.414	0.414	100.0%	7.7%
Sergio Ramos	Real Madrid	1	0	0.124	0.124	100.0%	7.7%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Toni Kroos	Real Madrid	1	0	0.004	0.004	0.0%	7.7%
TEAM TOTAL	Real Madrid	13	0	1.188	0.091	-	100%

Table 56 Player Shot Data — Real Madrid vs Manchester City, Semi-Final Leg 2, Won 1:0, 2016-05-04

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	7	0	0.402	0.057	71.4%	46.7%
Gareth Bale	Real Madrid	3	0	0.351	0.117	66.7%	20.0%
Jesé	Real Madrid	1	0	0.023	0.023	100.0%	6.7%
Lucas Vázquez	Real Madrid	1	0	0.026	0.026	100.0%	6.7%
Luka Modrić	Real Madrid	1	0	0.142	0.142	100.0%	6.7%
Pepe	Real Madrid	1	0	0.140	0.140	100.0%	6.7%
Toni Kroos	Real Madrid	1	0	0.003	0.003	0.0%	6.7%
TEAM TOTAL	Real Madrid	15	0	1.088	0.073	-	100%
Kevin De Bruyne	Man City	2	0	0.048	0.024	50.0%	40.0%
Fernandinho	Man City	1	0	0.020	0.020	0.0%	20.0%
Fernando	Man City	1	0	0.127	0.127	100.0%	20.0%
Sergio Agüero	Man City	1	0	0.005	0.005	0.0%	20.0%
TEAM TOTAL	Man City	5	0	0.200	0.040	-	100%

The Manchester City semi-final is the defensive extreme of the study. Across both legs Manchester City took 9 shots and generated 0.337 xG in total — an average of 4.5 shots and 0.169 xG per match. No opponent in any season examined so far has been restricted to comparable figures across a two-legged tie.

Real Madrid did not score from open play in either leg. The player goals column records zero for both matches; the 1:0 second-leg result was decided by an own goal, which this dataset excludes from player totals. Real Madrid generated 2.276 xG across the tie and converted none

of it. They advanced to the final having scored no attributable goals in 180 minutes of semi-final football.

Table 57 Player Shot Data — Real Madrid vs Atletico Madrid, Final, Drew 1:1 AET, Won 5:3 on Penalties, 2016-05-28

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Gareth Bale	Real Madrid	8	0	0.470	0.059	37.5%	32.0%
Cristiano Ronaldo	Real Madrid	6	0	0.339	0.056	33.3%	24.0%
Casemiro	Real Madrid	2	0	0.014	0.007	0.0%	8.0%
Danilo	Real Madrid	2	0	0.029	0.014	50.0%	8.0%
Karim Benzema	Real Madrid	2	0	0.325	0.162	100.0%	8.0%
Lucas Vázquez	Real Madrid	1	0	0.080	0.080	100.0%	4.0%
Luka Modrić	Real Madrid	1	0	0.006	0.006	0.0%	4.0%
Marcelo	Real Madrid	1	0	0.012	0.012	0.0%	4.0%
Sergio Ramos	Real Madrid	1	1	0.456	0.456	100.0%	4.0%
Toni Kroos	Real Madrid	1	0	0.005	0.005	0.0%	4.0%
TEAM TOTAL	Real Madrid	25	1	1.737	0.069	-	100%
Antoine Griezmann	Atletico	5	0	0.858	0.172	40.0%	29.4%
Koke	Atletico	3	0	0.064	0.021	33.3%	17.6%
Saúl Ñíguez	Atletico	2	0	0.082	0.041	50.0%	11.8%
Fernando Torres	Atletico	2	0	0.220	0.110	100.0%	11.8%
Yannick Carrasco	Atletico	2	1	0.403	0.202	100.0%	11.8%
Gabi	Atletico	1	0	0.002	0.002	0.0%	5.9%
Juanfran	Atletico	1	0	0.007	0.007	0.0%	5.9%
Stefan Savić	Atletico	1	0	0.265	0.265	100.0%	5.9%
TEAM TOTAL	Atletico	17	1	1.900	0.112	-	100%

The final was the most evenly contested match of Real Madrid's campaign and the only Tier 1 match in which they conceded. Real Madrid took 25 shots to Atletico's 17, yet Atletico generated the higher xG total at 1.900 against 1.737 and more than half again the rationality at 0.112 against 0.069. Antoine Griezmann alone accounted for 0.858 xG without scoring, the largest individual xG figure in the match.

Real Madrid's goal came from Sergio Ramos, from a single shot worth 0.456 xG at 100% box concentration — a set piece. Bale and Ronaldo took 14 shots between them for a combined 0.809 xG and did not score, with box concentrations of 37.5% and 33.3% respectively. The match was decided on penalties after 120 minutes. Note that this table excludes the penalty shootout, which is not treated as open-match shot data in this study.

Barcelona's elimination is examined in the two quarter-final legs against Atletico Madrid.

Table 58 Player Shot Data — Barcelona vs Atletico Madrid, Quarter-Final Leg 1, Won 2:1, 2016-04-05

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Neymar	Barcelona	6	0	0.515	0.086	66.7%	28.6%
Lionel Messi	Barcelona	6	0	0.263	0.044	50.0%	28.6%
Luis Suárez	Barcelona	3	2	0.492	0.164	100.0%	14.3%
Rafinha	Barcelona	2	0	0.098	0.049	50.0%	9.5%
Jordi Alba	Barcelona	1	0	0.096	0.096	100.0%	4.8%
Javier Mascherano	Barcelona	1	0	0.005	0.005	0.0%	4.8%
Ivan Rakitić	Barcelona	1	0	0.004	0.004	0.0%	4.8%
Andrés Iniesta	Barcelona	1	0	0.005	0.005	0.0%	4.8%
TEAM TOTAL	Barcelona	21	2	1.478	0.070	-	100%
Antoine Griezmann	Atletico	3	0	0.034	0.011	0.0%	42.9%
Augusto Fernández	Atletico	1	0	0.023	0.023	0.0%	14.3%
Fernando Torres	Atletico	1	1	0.048	0.048	100.0%	14.3%
Filipe Luís	Atletico	1	0	0.095	0.095	100.0%	14.3%
Yannick Carrasco	Atletico	1	0	0.006	0.006	0.0%	14.3%

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
TEAM TOTAL	Atletico	7	1	0.207	0.030	-	100%

Barcelona won the first leg 2:1 with a Field Tilt of 75.0%, taking 21 shots to Atletico's 7 and generating 1.478 xG against 0.207. Suárez scored both goals from three shots at 100% box concentration. Atletico scored from a single attempt worth 0.048 xG — a return far above expectation from minimal opportunity. A one-goal lead from a match of this territorial imbalance proved insufficient.

Table 59 Player Shot Data — Atletico Madrid vs Barcelona, Quarter-Final Leg 2, Lost 0:2, 2016-04-13

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Antoine Griezmann	Atletico	4	2	0.945	0.236	100.0%	44.4%
Yannick Carrasco	Atletico	2	0	0.012	0.006	0.0%	22.2%
Gabi	Atletico	1	0	0.036	0.036	100.0%	11.1%
Koke	Atletico	1	0	0.015	0.015	100.0%	11.1%
Saúl Ñíguez	Atletico	1	0	0.063	0.063	100.0%	11.1%
TEAM TOTAL	Atletico	9	2	1.071	0.119	-	100%
Lionel Messi	Barcelona	3	0	0.039	0.013	0.0%	23.1%
Luis Suárez	Barcelona	2	0	0.154	0.077	100.0%	15.4%
Jordi Alba	Barcelona	2	0	0.054	0.027	50.0%	15.4%
Andrés Iniesta	Barcelona	1	0	0.007	0.007	0.0%	7.7%
Arda Turan	Barcelona	1	0	0.010	0.010	0.0%	7.7%
Javier Mascherano	Barcelona	1	0	0.007	0.007	0.0%	7.7%
Dani Alves	Barcelona	1	0	0.002	0.002	0.0%	7.7%
Neymar	Barcelona	1	0	0.004	0.004	0.0%	7.7%
Sergio Busquets	Barcelona	1	0	0.097	0.097	100.0%	7.7%
TEAM TOTAL	Barcelona	13	0	0.374	0.029	-	100%

The second leg is Barcelona's weakest attacking performance in the study to this point. They took 13 shots for a total of 0.374 xG at a rationality of 0.029 — the lowest quality-per-shot figure recorded by either club in any critical match across four seasons. Messi took 3 shots for 0.039 xG with no attempt from inside the box. Neymar took one, worth 0.004. Nine Barcelona players registered a shot and the combined output was less than four tenths of an expected goal.

Atletico took 9 shots for 1.071 xG. Griezmann scored twice from four attempts worth 0.945 xG at 100% box concentration — one player generating more than twice Barcelona's entire team output. Barcelona were eliminated 3:2 on aggregate.

The Atletico Test

Both clubs faced Atletico Madrid in this season, at different stages and seven weeks apart. As with the Juventus comparison in Module 4.4, holding the opponent constant removes the archetype variable that complicates comparisons across different fixtures.

Barcelona met Atletico over two legs and were eliminated. Across the tie they took 34 shots for 1.852 xG and scored 2, while Atletico took 16 shots for 1.278 xG and scored 3. Real Madrid met Atletico once, in the final, taking 25 shots for 1.737 xG and scoring 1, while Atletico took 25 shots for 1.900 xG and scored 1. Real Madrid won on penalties.

Table 60 The Atletico Test — Barcelona Quarter-Final and Real Madrid Final Compared, UCL 2015-2016

Metric	BA Leg 1 (H)	BA Leg 2 (A)	BA Tie Total	RM Final
Result	Won 2:1	Lost 0:2	Out 2:3 agg	Drew 1:1, won 5:3 pens
Shots	21	13	34	25
Goals	2	0	2	1
xG	1.478	0.374	1.852	1.737
Rationality (xG/Shot)	0.070	0.029	0.054	0.069
Box Shot %	57.1%	30.8%	47.1%	40.0%
Accuracy %	28.6%	30.8%	29.4%	28.0%
Field Tilt %	75.0%	59.1%	68.0%	59.5%
Atletico Shots	7	9	16	17
Atletico xG	0.207	1.071	1.278	1.900

Metric	BA Leg 1 (H)	BA Leg 2 (A)	BA Tie Total	RM Final
Atletico Goals	1	2	3	1

The comparison is less clean than the Juventus Test, because the samples differ in length and stage — Barcelona played two legs, Real Madrid one final. Read with that caveat, three observations hold.

First, both Spanish clubs out-shot Atletico and neither out-scored them. Barcelona took 34 shots to Atletico's 16 across the tie and lost it; Real Madrid took 25 to Atletico's 17 and were held. Across all three matches Atletico took 33 shots for 3.178 xG and scored 4, against their opponents' 59 shots for 3.589 xG and 3 goals.

Second, Atletico generated the higher xG total in two of the three matches, and in the third — the Camp Nou first leg, where they were held to 7 shots and 0.207 xG — they still scored, from an attempt worth 0.048 xG. Their per-shot quality against Barcelona away, at 0.119, and against Real Madrid in the final, at 0.112, exceeded both opponents' figures in those matches.

Third, both opponents' box access collapsed in the decisive match. Barcelona's Box Shot percentage fell from 57.1% in the first leg to 30.8% in the second. Real Madrid's stood at 40.0% in the final, against a season Tier 1 average of 55.7%. Neither club maintained penetration when the tie was live.

Set against 2013-2014, where the same two ties produced the same two outcomes, this is now a repeated pattern rather than a single result. The shot data records that Atletico consistently converted low-volume, high-quality opportunities against both Barcelona and Real Madrid across two seasons. It cannot explain how those opportunities were constructed, or why both opponents' attacks were less productive than their shot volumes implied. Those questions belong to Modules 6 and 8.

Key Observations — Season 2015-2016

The 2015-2016 season is the clearest demonstration in the study that the three conditions set out in Module 3 are not substitutable, and that the shot pillar alone cannot determine outcomes.

At season level the two clubs were indistinguishable. Finishing Delta of 0.69 and 0.70, Quality Margin of 0.71 and 0.78, accuracy within a percentage point, goals within 0.12 per match. Both satisfied the first condition and neither held a meaningful advantage.

At tier level they diverged in opposite directions from what the outcomes would suggest. Barcelona scored 1.75 goals per Tier 1 match with a positive Finishing Delta of 0.53. Real Madrid scored 0.40 with a Finishing Delta of -0.74 — two goals in five matches, neither scored by a

forward. On attacking output against elite opposition Barcelona were substantially the better team, and they were eliminated in the quarter-finals.

Real Madrid's compensation was defensive. They conceded 0.62 opponent xG per Tier 1 match and one goal across five matches, restricting Manchester City to 9 shots and 0.337 xG across a two-legged semi-final. The shot pillar can record this suppression through opponent shot volume and quality; it cannot explain it. That Real Madrid won a competition while scoring twice against elite opposition is a finding the shot data can state but not account for.

At critical match level, the most consequential result of Real Madrid's season was a 2:0 defeat away to Wolfsburg, a Tier 2 opponent, in which they recorded 14.3% accuracy from 21 shots and generated 0.913 xG against their opponent's 1.351. The campaign that ended in a European title came closest to ending in the quarter-finals against standard opposition. This is the variance point made in Module 4.4 in reverse: in 2014-2015 a single leg eliminated a team that had performed well across the tier sample; in 2015-2016 a single recovery leg preserved a team that had nearly gone out.

On the diversity question raised in Module 4.4, this season adds a further complication. Real Madrid retained the broader mechanism profile — open play, set pieces, free kicks and penalties — while Barcelona scored 17 of 19 goals from open play. But Real Madrid's mechanism diversity did not manifest against elite opposition in the form previously described: their two Tier 1 goals came from a set piece and an open-play strike by a defender, with no contribution from the attacking players who took the overwhelming majority of the shots. Neither mechanism diversity nor personnel diversity, as defined in Module 4.4, describes what happened here. What Real Madrid possessed against elite opposition was not an alternative route to goal but an alternative route to victory: conceding less than they failed to score.

Three questions pass forward. Why did both clubs' principal attackers lose box concentration against elite opposition while maintaining shot volume — Ronaldo from 67.2% to 42.9%, Messi from 75.0% to 25.0%, Neymar from 76.6% to 33.4%? How were Real Madrid's elite opponents restricted to 0.62 xG per match? And why did Barcelona's shot distribution against Tier 1 opposition shift from the box to the edge of the box, reversing the pattern of every other tier-season in the study? Chance construction is examined in Module 6, defensive structure in Module 8.

4.6 Season 2016-2017

The 2016-2017 UEFA Champions League produced Real Madrid's twelfth European title — the first successful defence of the trophy in the Champions League era — and Barcelona's elimination in the quarter-finals by Juventus. Real Madrid defeated Juventus 4:1 in the Cardiff final. Barcelona, having overturned a 4:0 first-leg deficit against Paris Saint-Germain in the round of 16 with a 6:1 second leg, were then eliminated 3:0 on aggregate by Juventus, failing to score across the two quarter-final legs.

This season inverts the shot-pillar pattern of the two campaigns before it. In 2014-2015 and 2015-2016 Real Madrid's defining feature at elite level was a conversion collapse — a sharply negative Tier 1 Finishing Delta, central chances created and not taken. In 2016-2017 that pattern disappears. Real Madrid scored eighteen goals in seven Tier 1 matches and recorded a positive Tier 1 Finishing Delta for the first time in the study. The counterintuitive result of the prior seasons does not recur; what requires explanation this season is Barcelona's failure, not Real Madrid's success.

It also supplies the study's fourth same-opponent comparison. Both clubs met Juventus — Real Madrid in the final, Barcelona in the quarter-finals — and the pairing is examined as The Juventus Test at the close of the critical-match analysis.

Season Overview

Table 61 presents the season average shot metrics for both teams across all matches in the 2016-2017 UEFA Champions League campaign.

Table 61 Season Overview — Shot Analysis 2016-2017

Metric	Real Madrid	Barcelona
Avg Shots per Match	19.5	13.1
Avg Goals per Match	2.62	2.50
xG per Match	1.35	1.37
Rationality (xG/Shot)	0.073	0.106
Box Shot %	59.8%	65.8%
Accuracy %	37.0%	44.4%
Big Misses per Match	0.31	0.30
Opp Shots per Match	12.1	8.9

Opp xG per Match	0.86	0.65
Finishing Delta (Goals – xG)	+1.26	+1.13
Quality Margin (xG – Opp xG)	+0.50	+0.72
Field Tilt %	61.8%	61.1%

Note: Goals throughout this module are player-attributed — that is, credited to the player recorded as taking the shot. Own goals therefore appear in match scorelines but are excluded from all goal counts, xG comparisons and Finishing Delta figures, for both clubs and their opponents alike. In 2016-2017 this affects four goals: two scored for Real Madrid (against Legia Warszawa and Napoli, both Tier 2), one scored for Barcelona (against Paris Saint-Germain, Tier 1), and one conceded by Real Madrid (against Bayern Munich, Tier 1). Real Madrid's seven Tier 1 matches are unaffected on the attacking side; their eighteen Tier 1 goals are identical on either basis.

At season level the two clubs again resemble one another on the headline outputs and separate on volume. Real Madrid scored 2.62 goals per match to Barcelona's 2.50 and generated near-identical chance quality, 1.35 xG against 1.37. Both finished well above their distance-based expectation — a Finishing Delta of +1.26 for Real Madrid and +1.13 for Barcelona, the two largest positive season figures either club has recorded in the study to this point. As established in 4.1, the xG model is distance-only, so a large positive Finishing Delta records that both clubs scored above the model's expectation without establishing why. [DATA]

Real Madrid took 19.5 shots per match to Barcelona's 13.1, a wider gap than in any prior season, and Barcelona's higher rationality of 0.106 against 0.073 follows from that smaller, more selective shot count. Barcelona were the more accurate and more box-concentrated side, at 44.4% accuracy and 65.8% box shots against 37.0% and 59.8%, while Real Madrid conceded more shots per match, 12.1 against 8.9. The two Field Tilt figures are almost identical at 61.8% and 61.1%.

Quality Margin favoured Barcelona, +0.72 against +0.50, driven by their lower opponent xG conceded of 0.65 against 0.86. On the season aggregates Barcelona are, if anything, marginally the stronger side on chance-quality balance. As in 2015-2016, the season averages satisfy the first of the three conditions set out in Module 3 for both clubs and do not, on their own, predict the divergence in outcome. The separation appears only when the sample is partitioned by opponent quality.

Tier Comparison

Table 62 presents the tier breakdown separating performance against Tier 1 (Elite, Elo > 1850) and Tier 2 (Standard, Elo ≤ 1850) opponents. Real Madrid played seven Tier 1 matches and six Tier 2; Barcelona played six Tier 1 and four Tier 2.

Table 62 Tier Comparison — Shot Analysis 2016-2017

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Avg Shots per Match	19.4	19.5	13.7	12.3
Avg Goals per Match	2.57	2.67	1.67	3.75
xG per Match	1.38	1.33	1.24	1.57
Rationality (xG/Shot)	0.077	0.069	0.089	0.132
Box Shot %	59.7%	59.9%	62.5%	70.7%
Accuracy %	40.4%	33.0%	33.3%	61.1%
Big Misses per Match	0.14	0.50	0.33	0.25
Opp Shots per Match	13.7	10.2	11.8	4.5
Opp xG per Match	1.05	0.63	0.82	0.40
Finishing Delta (Goals – xG)	+1.20	+1.34	+0.43	+2.18
Quality Margin (xG – Opp xG)	+0.33	+0.70	+0.42	+1.17
Field Tilt %	59.9%	64.0%	52.4%	74.2%

The tier split is where 2016-2017 departs from the two seasons before it. Real Madrid scored 2.57 goals per Tier 1 match — eighteen across seven elite matches — against 0.40 in 2015-2016. Their Tier 1 Finishing Delta was +1.20, positive against elite opposition for the first time in the study, where in the two prior seasons it had been sharply negative. The conversion failure that defined Real Madrid's elite-level shooting in 4.4 and 4.5 does not appear in this table. [DATA]

Barcelona's Tier 1 profile is the more paradoxical. On the quality-balance metrics they were the better side: a Tier 1 Quality Margin of +0.42 against Real Madrid's +0.33, and a lower opponent xG conceded of 0.82 against 1.05. Barcelona out-created their elite opponents by a wider margin than Real Madrid did and conceded less chance quality. Yet they scored 1.67 goals per Tier 1 match to Real Madrid's 2.57 and were eliminated two rounds earlier. The separation is on the Finishing Delta line, +1.20 against +0.43: both clubs converted above expectation against elite opposition, and Real Madrid did so by nearly three times the margin. [DATA]

Barcelona's Tier 1 Field Tilt of 52.4% is the recurring signal tracked across this module — the lowest of the four tier-club cells this season and the closest to the 50% threshold, against Real Madrid's 59.9%. Barcelona only narrowly out-shot their elite opponents by territory.

Against Tier 2 opposition both clubs were dominant and Barcelona the more extreme, with a Tier 2 Finishing Delta of +2.18 and Field Tilt of 74.2% from four matches. Real Madrid's Tier 2 figures are strong but flatter — a Finishing Delta of +1.34 and 2.67 goals per match — and

notably close to their Tier 1 figures. Real Madrid's tier gap in goal output, 2.67 to 2.57, is the narrowest recorded by either club in the study; the near eight-to-one ratio of 2015-2016 has closed almost entirely. Whatever suppressed Real Madrid's elite-level scoring in the two prior seasons was not present in this one. The shot data records the change; it cannot establish its cause, which is examined in Modules 6 and 8.

Shot Map Analysis

The shot maps for both clubs are presented in Figures 9 and 10, each preceding its spatial profile. The zone distribution and zone xG data underlying the discussion are presented together in Tables 63 and 64, which follow the comparative insight.

Real Madrid Spatial Profile

Real Madrid reached the box freely against elite opposition and, unlike in the two prior seasons, converted once there. Box shots accounted for 60.3% of their Tier 1 attempts and the central corridor for 83.1%, both marginally above their Tier 2 figures of 61.5% and 77.8%. Access to the penalty area was not restricted by opponent quality — the pattern of the prior seasons — but this season access was matched by conversion.

In Box_Central, the most productive zone in every season of this study, Real Madrid took 69 shots against Tier 1 opponents and scored 17, a goal rate of 24.6%. Against Tier 2 opponents the same zone produced 10 goals from 64 shots, a rate of 15.6%. For the first time in the study Real Madrid's Box_Central goal rate was higher against elite opposition than against standard opposition; in 2015-2016 the equivalent Tier 1 figure was 2.9%. The zone that defined Real Madrid's elite-level failure in the prior two seasons became their most productive against the same tier in this one. [DATA]

Their two Tier 1 goals of 2015-2016 became eighteen. Seventeen came from Box_Central, none from the box flanks, and one from long range. The distribution is narrow and central, and this season it produced goals.

Shot Analysis: Real Madrid (Red) vs Opponents (Blue)
Champions League 2016-2017

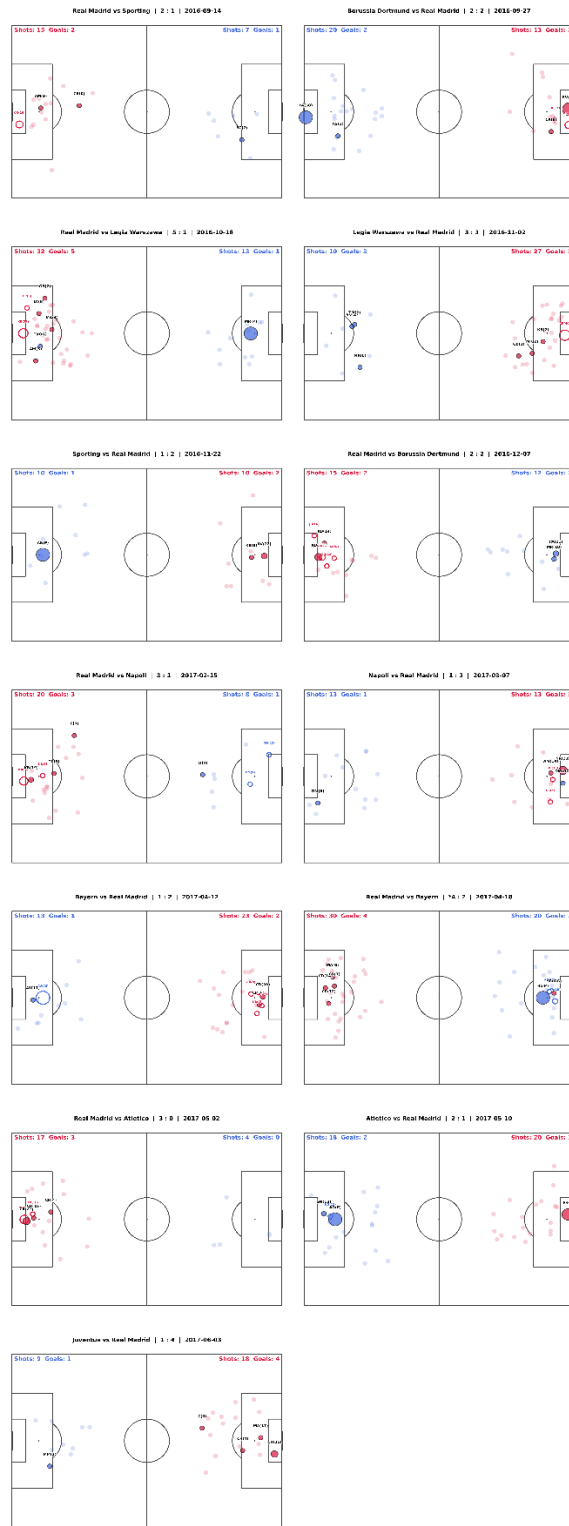


Figure 9 Real Madrid Shot Map — UCL 2016-2017

Barcelona Spatial Profile

Barcelona's Tier 1 spatial profile shows adequate access and thinner conversion. Box shots accounted for 61.0% of their Tier 1 attempts, almost identical to Real Madrid's 60.3%, and the central corridor for 79.3%. On depth and lateral distribution the two clubs' elite-level shot maps are close. The difference is in volume and return: Barcelona took 82 Tier 1 shots to Real Madrid's 136, and 42 Box_Central shots to Real Madrid's 69.

From those 42 Box_Central shots Barcelona scored 8, a goal rate of 19.0%, against Real Madrid's 24.6% from 69. Against Tier 2 opponents Barcelona's Box_Central rate was 39.4% from 33 shots — the tier pattern that Real Madrid's season lacked, with conversion higher against weaker opposition. Barcelona reached the same central zone as Real Madrid, at similar proportions, but less often and to lesser effect against elite opponents. [DATA]

Their edge shooting carried a small return this season, 2 goals from 32 Tier 1 edge shots, where in 2015-2016 the edge produced nothing. Barcelona's Tier 1 attack was neither displaced to the edge as in 2015-2016 nor blocked from the box; it simply generated fewer high-value central attempts than Real Madrid and converted a smaller share of them.

Shot Analysis: Barcelona (Blue) vs Opponents (Red) Champions League 2016-2017

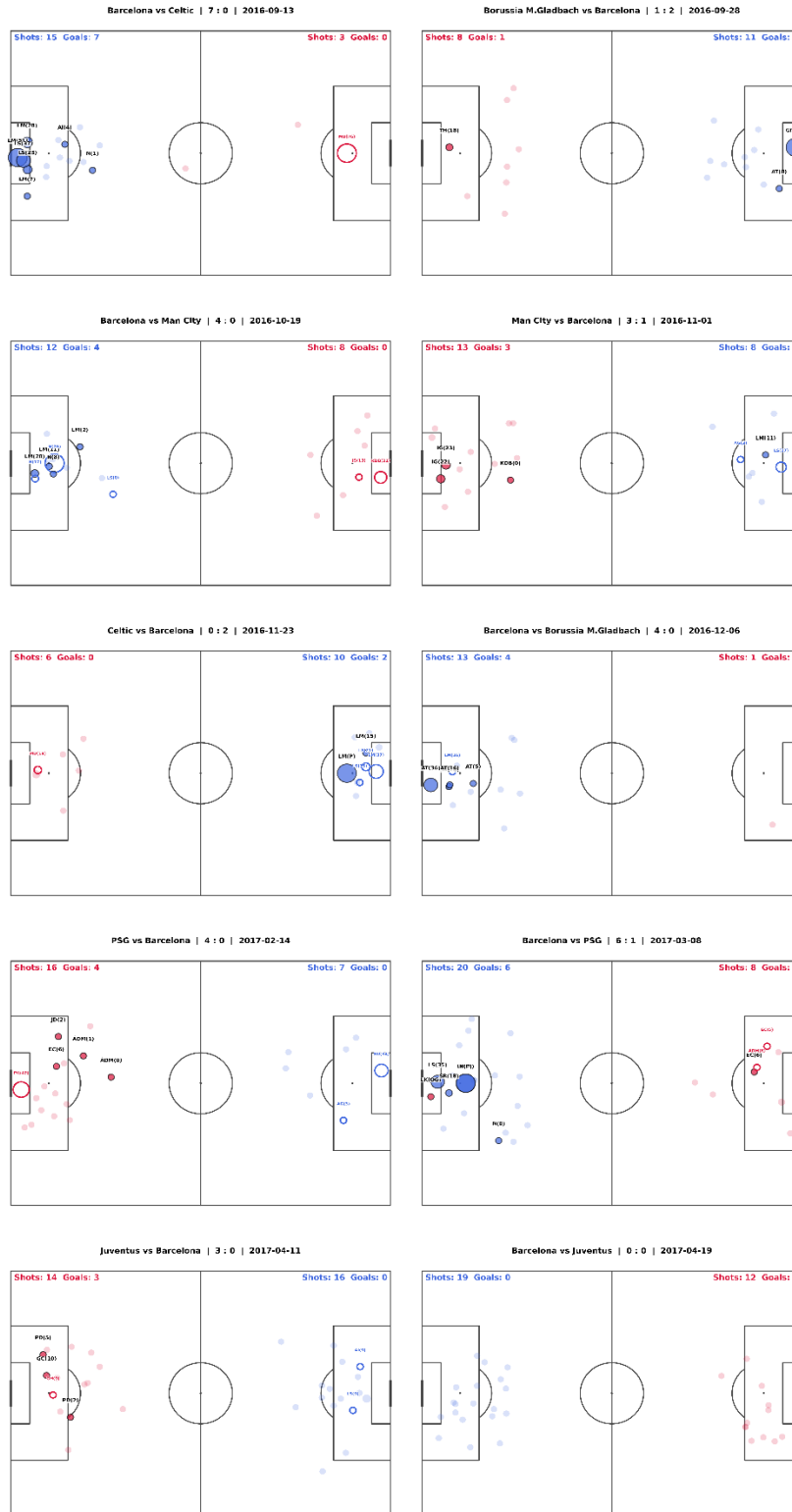


Figure 10 Barcelona Shot Map — UCL 2016-2017

Comparative Spatial Insight

The spatial records of the two clubs against elite opposition are, this season, unusually similar in shape and different in outcome. Both reached the box on roughly 60% of their Tier 1 shots; both funnelled around 80% of attempts through the central corridor. The divergence is in the Box_Central cell: Real Madrid 69 shots and 17 goals, Barcelona 42 shots and 8 goals. Real Madrid had more central access and converted it at the higher rate.

This is the inverse of the 2015-2016 comparison, where Real Madrid had the access and not the conversion and Barcelona the conversion and not the access. In 2016-2017 Real Madrid had both. The shot data can establish that Real Madrid generated more central chances against elite opposition and scored a higher proportion of them; it cannot establish whether the difference lay in how those chances were constructed, in the finishing itself, or in the defensive structures each club faced. Chance construction is examined in Module 6 and defensive compression in Module 8.

Table 63 Zone Shot Distribution — Real Madrid and Barcelona, UCL 2016-2017

Depth Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box	82	60.3%	72	61.5%	50	61.0%	34	69.4%
Edge	50	36.8%	42	35.9%	32	39.0%	15	30.6%
Long Range	4	2.9%	3	2.6%	0	0.0%	0	0.0%
TOTAL	136	100%	117	100%	82	100%	49	100%

Lateral Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Left	11	8.1%	12	10.3%	12	14.6%	2	4.1%
Central	113	83.1%	91	77.8%	65	79.3%	47	95.9%
Right	12	8.8%	14	12.0%	5	6.1%	0	0.0%
TOTAL	136	100%	117	100%	82	100%	49	100%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Left	7	5.1%	4	3.4%	5	6.1%	1	2.0%
Box_Central	69	50.7%	64	54.7%	42	51.2%	33	67.3%
Box_Right	6	4.4%	4	3.4%	3	3.7%	0	0.0%

Edge_Left	4	2.9%	8	6.8%	7	8.5%	1	2.0%
Edge_Central	40	29.4%	24	20.5%	23	28.0%	14	28.6%
Edge_Right	6	4.4%	10	8.5%	2	2.4%	0	0.0%
Long Range_Left	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Long Range_Central	4	2.9%	3	2.6%	0	0.0%	0	0.0%
Long Range_Right	0	0.0%	0	0.0%	0	0.0%	0	0.0%
TOTAL	136	100%	117	100%	82	100%	49	100%

Table 64 Zone xG and Goals by Zone — Real Madrid and Barcelona, UCL 2016-2017

Depth Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box	82	0.1104	17	20.7%	72	0.1031	11	15.3%	50	0.1408	8	16.0%	34	0.1785	14	41.2%
Edge	50	0.0114	0	0.0%	42	0.0126	5	11.9%	32	0.0127	2	6.2%	15	0.0140	1	6.7%
Long Range	4	0.0024	1	25.0%	3	0.0024	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%

Lateral Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Left	11	0.0178	0	0.0%	12	0.0145	0	0.0%	12	0.0220	1	8.3%	2	0.0379	1	50.0%
Central	113	0.0815	18	15.9%	91	0.0826	14	15.4%	65	0.1096	9	13.8%	47	0.1320	14	29.8%
Right	12	0.0188	0	0.0%	14	0.0191	2	14.3%	5	0.0114	0	0.0%	0	0.0000	0	0.0%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box_Left	7	0.0236	0	0.0%	4	0.0246	0	0.0%	5	0.0442	0	0.0%	1	0.0708	1	100%
Box_Central	69	0.1260	17	24.6%	64	0.1119	10	15.6%	42	0.1612	8	19.0%	33	0.1818	13	39.4%
Box_Right	6	0.0328	0	0.0%	4	0.0411	1	25.0%	3	0.0164	0	0.0%	0	0.0000	0	0.0%
Edge_Left	4	0.0077	0	0.0%	8	0.0094	0	0.0%	7	0.0061	1	14.3%	1	0.0050	0	0.0%
Edge_Central	40	0.0127	0	0.0%	24	0.0146	4	16.7%	23	0.0155	1	4.3%	14	0.0146	1	7.1%

Edge_Right	6	0.0048	0	0.0%	10	0.0103	1	10.0%	2	0.0040	0	0.0%	0	0.0000	0	0.0%
Long Range_Left	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%
Long Range_Central	4	0.0024	1	25.0%	3	0.0024	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%
Long Range_Right	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%

Player Highlights

Table 65 presents the season average shot metrics for all players with a shot share above 5% for both clubs.

Table 65 Player Season Average — Shot Analysis 2016-2017 (Shot Share > 5%)

Player	Team	Shots	OP	SP	Goals	xG	Rat.	Box %	Left %	Right %	Shot %	xG %	Goal %
Cristiano Ronaldo	Real Madrid	5.62	4.54	0.08	0.92	0.375	0.062	55.8	20.1	18.3	28.5	26.8	32.7
Karim Benzema	Real Madrid	3.08	3.08	0.00	0.42	0.360	0.107	84.8	12.5	8.3	15.6	25.3	18.1
Gareth Bale	Real Madrid	3.14	2.57	0.14	0.29	0.190	0.063	49.2	17.5	42.1	14.9	11.8	8.3
Raphaël Varane	Real Madrid	1.50	1.33	0.17	0.33	0.281	0.183	100.0	0.0	8.3	10.6	23.4	16.7
Isco	Real Madrid	2.00	2.00	0.00	0.25	0.139	0.050	29.2	45.8	0.0	9.7	10.1	25.0
Luka Modric	Real Madrid	1.71	1.43	0.29	0.00	0.016	0.008	4.8	0.0	61.9	9.4	1.3	0.0
Marcelo	Real Madrid	1.78	1.67	0.11	0.00	0.039	0.023	38.9	68.5	0.0	9.3	3.0	0.0
James Rodríguez	Real Madrid	1.50	1.25	0.25	0.00	0.082	0.046	66.7	75.0	0.0	9.0	5.9	0.0
Casemiro	Real Madrid	1.63	1.50	0.13	0.25	0.066	0.039	39.6	12.5	37.5	8.4	4.9	7.3
Dani Carvajal	Real Madrid	1.50	1.50	0.00	0.00	0.097	0.079	75.0	0.0	37.5	8.4	8.2	0.0
Marco Asensio	Real Madrid	2.20	1.80	0.40	0.60	0.093	0.058	65.0	25.0	26.7	8.1	7.5	15.0
Lucas Vázquez	Real Madrid	1.29	1.29	0.00	0.14	0.074	0.060	78.6	14.3	28.6	8.0	6.2	3.6
Sergio Ramos	Real Madrid	1.38	1.00	0.25	0.13	0.149	0.106	87.5	0.0	12.5	7.9	13.0	6.3
Toni Kroos	Real Madrid	1.67	1.67	0.00	0.17	0.045	0.026	22.2	25.0	8.3	7.8	3.4	5.6
Álvaro Morata	Real Madrid	1.25	1.25	0.00	0.75	0.084	0.075	87.5	37.5	0.0	6.6	7.1	31.3
Lionel Messi	Barcelona	4.22	3.11	0.00	1.22	0.521	0.107	53.8	17.1	5.6	31.7	30.1	40.3
Luis Suárez	Barcelona	2.63	2.50	0.13	0.38	0.328	0.125	82.3	6.3	14.6	20.9	26.4	6.1
Neymar	Barcelona	2.50	1.75	0.13	0.50	0.281	0.089	62.5	43.8	0.0	18.4	16.8	9.9

Arda Turan	Barcelona	2.33	2.33	0.00	1.33	0.247	0.083	75.0	8.3	50.0	18.0	20.4	41.7
André Gomes	Barcelona	1.20	1.00	0.20	0.00	0.084	0.075	70.0	0.0	40.0	12.7	7.1	0.0
Sergi Roberto	Barcelona	1.50	1.00	0.50	0.25	0.090	0.054	62.5	0.0	25.0	10.3	5.7	5.0
Gerard Piqué	Barcelona	1.67	1.33	0.33	0.33	0.321	0.245	100.0	0.0	0.0	9.9	28.8	16.7
Ivan Rakitic	Barcelona	1.14	0.86	0.29	0.00	0.059	0.057	71.4	14.3	14.3	9.9	4.3	0.0
Andrés Iniesta	Barcelona	1.33	1.33	0.00	0.17	0.050	0.034	38.9	22.2	0.0	8.6	4.0	2.4

Real Madrid's shooting load remained concentrated on Cristiano Ronaldo, at 28.5% shot share and 32.7% goal share, with 12 goals — eleven from open play and one free kick. Karim Benzema followed at 15.6% shot share and 5 goals. Beyond the two forwards the distribution widens: Raphaël Varane, a centre-back, ranked fourth at 10.6% shot share with 2 goals at 100% box concentration, of which only 0.17 of his 1.5 shots per match were set pieces — the remainder open-play box attempts, an unusual profile for a defender. [DATA] Álvaro Morata returned the highest goal share relative to shot share in either squad, 6.6% of shots and 31.3% of goals, with three goals from limited minutes, all open play and all against Tier 2. Gareth Bale, third in shot share at 14.9%, scored twice for an 8.3% goal share, continuing the under-conversion noted in 2015-2016. Marco Asensio, at 8.1% shot share, contributed three goals including two against Tier 1 opposition.

Barcelona's load was more concentrated at the top and thinner below. Lionel Messi held a 31.7% shot share and 40.3% goal share with 11 goals — nine open play, two penalties. Luis Suárez, second at 20.9% shot share, returned only 3 goals and a 6.1% goal share, the season's largest under-performance relative to shot volume in either squad. Arda Turan, from an 18.0% shot share, scored four goals for a 41.7% goal share — all four against Tier 2 opposition, a Finishing Delta of +1.09 concentrated entirely in the standard tier. Neymar took 18.4% of shots for 4 goals. Gerard Piqué, a centre-back, held a 9.9% shot share at 100% box concentration as a set-piece target.

Goal-type data records a reversal of the module's recurring mechanism-diversity pattern. Real Madrid scored 34 goals, of which 31 came from open play, 2 from set pieces and 1 from a free kick, with no penalties across the entire campaign — the first season in the study in which Real Madrid scored none. Barcelona scored 25, comprising 19 from open play, 3 penalties, 2 free kicks and 1 set piece. Against Tier 1 opposition the contrast is sharper still: Real Madrid's eighteen elite goals were seventeen open play and one set piece, near-exclusively one mechanism, while Barcelona's ten drew on open play, a penalty, a free kick and a set piece. This season it is Barcelona, not Real Madrid, that carried the broader mechanism profile against elite opposition. As noted in 4.4 and 4.5, mechanism diversity has not proven a reliable predictor of outcome, and it did not save Barcelona here. [DATA]

Player Tier Comparison

Table 66 presents the player-level tier breakdown using the same player list as Table 65. Players with no appearances in a tier are shown with zero values.

Table 66 Player Tier Comparison — Shot Analysis 2016-2017 (Shot Share > 5%)

Player	Team	T1 Sh	T1 OP	T1 SP	T1 G	T1 xG	T1 Rat	T1 Box%	T1 Sh%	T2 Sh	T2 OP	T2 SP	T2 G	T2 xG	T2 Rat	T2 Box%	T2 Sh%
Cristiano Ronaldo	Real Madrid	6.14	5.29	0.00	1.57	0.419	0.068	52.4	32.0	5.00	3.67	0.17	0.17	0.323	0.055	59.7	24.4
Karim Benzema	Real Madrid	2.71	2.71	0.00	0.29	0.345	0.121	90.0	15.5	3.60	3.60	0.00	0.60	0.380	0.087	77.5	15.8
Gareth Bale	Real Madrid	1.67	1.67	0.00	0.00	0.148	0.075	50.0	9.9	4.25	3.25	0.25	0.50	0.221	0.054	48.6	18.6
Raphaël Varane	Real Madrid	1.50	1.50	0.00	0.50	0.417	0.246	100.0	10.7	1.50	1.25	0.25	0.25	0.213	0.152	100.0	10.5
Isco	Real Madrid	2.00	2.00	0.00	0.25	0.139	0.050	29.2	9.7	0.00	0.00	0.00	0.00	0.000	0.000	0.0	0.0
Luka Modric	Real Madrid	1.60	1.40	0.20	0.00	0.014	0.007	6.7	8.2	2.00	1.50	0.50	0.00	0.022	0.011	0.0	12.5
Marcelo	Real Madrid	1.80	1.60	0.20	0.00	0.059	0.034	70.0	9.0	1.75	1.75	0.00	0.00	0.015	0.008	0.0	9.7
James Rodríguez	Real Madrid	2.00	1.50	0.50	0.00	0.123	0.050	83.4	13.9	1.00	1.00	0.00	0.00	0.042	0.041	50.0	4.1
Casemiro	Real Madrid	1.80	1.60	0.20	0.20	0.078	0.035	43.3	8.6	1.33	1.33	0.00	0.33	0.046	0.044	33.3	8.1
Dani Carvajal	Real Madrid	1.50	1.50	0.00	0.00	0.054	0.046	50.0	6.3	1.50	1.50	0.00	0.00	0.140	0.112	100.0	10.5
Marco Asensio	Real Madrid	2.00	1.67	0.33	0.67	0.113	0.077	66.7	8.1	2.50	2.00	0.50	0.50	0.063	0.030	62.5	8.1
Lucas Vázquez	Real Madrid	1.00	1.00	0.00	0.00	0.050	0.050	66.7	5.3	1.50	1.50	0.00	0.25	0.093	0.068	87.5	10.0
Sergio Ramos	Real Madrid	1.50	1.00	0.25	0.00	0.126	0.096	75.0	6.5	1.25	1.00	0.25	0.25	0.172	0.116	100.0	9.3
Toni Kroos	Real Madrid	2.00	2.00	0.00	0.00	0.055	0.027	33.3	8.6	1.00	1.00	0.00	0.50	0.024	0.024	0.0	6.4
Álvaro Morata	Real Madrid	1.00	1.00	0.00	0.00	0.021	0.021	100.0	5.6	1.33	1.33	0.00	1.00	0.104	0.093	83.3	6.9
Lionel Messi	Barcelona	3.67	2.17	0.00	0.83	0.282	0.071	36.3	25.7	5.33	5.00	0.00	2.00	0.998	0.181	88.9	43.6
Luis Suárez	Barcelona	2.60	2.40	0.20	0.20	0.293	0.110	85.0	19.9	2.67	2.67	0.00	0.67	0.386	0.149	77.8	22.4
Neymar	Barcelona	2.83	2.00	0.17	0.50	0.346	0.099	58.3	20.7	1.50	1.00	0.00	0.50	0.085	0.060	75.0	11.2
Arda Turan	Barcelona	1.00	1.00	0.00	0.00	0.051	0.051	100.0	5.0	3.00	3.00	0.00	2.00	0.345	0.098	62.5	24.5
André Gomes	Barcelona	1.50	1.00	0.50	0.00	0.060	0.039	75.0	19.7	1.00	1.00	0.00	0.00	0.099	0.099	66.7	8.1
Sergi Roberto	Barcelona	1.50	1.00	0.50	0.25	0.090	0.054	62.5	10.3	0.00	0.00	0.00	0.00	0.000	0.000	0.0	0.0
Gerard Piqué	Barcelona	2.00	1.50	0.50	0.00	0.229	0.114	100.0	10.3	1.00	1.00	0.00	1.00	0.505	0.505	100.0	9.1
Ivan Rakitic	Barcelona	1.00	0.75	0.25	0.00	0.048	0.048	75.0	8.5	1.33	1.00	0.33	0.00	0.073	0.070	66.7	11.6

Andrés Iniesta	Barcelona	1.50	1.50	0.00	0.00	0.059	0.036	33.3	9.4	1.00	1.00	0.00	0.50	0.031	0.031	50.0	7.2
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The tier split identifies where each squad's carriers held or vanished against elite opposition.

Cristiano Ronaldo is the season's clearest case of Tier 1 consistency: eleven of Real Madrid's eighteen Tier 1 goals, a Tier 1 goal share of 53.6%, from a 32.0% Tier 1 shot share, all eleven from open play, at a Tier 1 Finishing Delta of +1.15 per match. His Tier 1 box concentration of 52.4% held close to his season figure. Ronaldo did not merely maintain output against elite opposition; he raised it. [DATA]

Lionel Messi is the corresponding Barcelona case and the more familiar pattern. He scored five of Barcelona's ten Tier 1 goals for a 32.5% Tier 1 goal share, over-performing at +0.55 Finishing Delta, but his Tier 1 box concentration fell to 36.3% from 88.9% against Tier 2 — the box-displacement pattern recorded for Messi in every season of this study. Against elite opposition Messi took the majority of his shots from outside the box. [DATA]

Below the two principals the squads diverge in depth. Real Madrid drew Tier 1 goals from six players — Ronaldo, Benzema (2), Asensio (2), Casemiro, Isco and Varane — none beyond Ronaldo scoring more than two, but the contributions spread across the tier. Barcelona's Tier 1 goals came from four — Messi (5), Neymar (3), Suárez (1) and Sergi Roberto (1). The Tier 2 over-performers did not carry into the elite tier: all four of Arda Turan's goals and Gerard Piqué's single strike came against Tier 2. Nor did Luis Suárez, whose one Tier 1 goal against two at Tier 2 is a thin open-play return from Barcelona's second-most-frequent shooter. [DATA]

Critical Match Analysis

Nine matches are examined: five for Real Madrid and four for Barcelona, presented in Tables 67 through 75, each showing player-level shot data for both teams. They are ordered by tie within each club.

Real Madrid's five critical matches were all won or survived; the campaign's nearest approach to elimination was the quarter-final second leg against Bayern Munich, won 4:2 after extra time, level on the night before Real Madrid pulled clear in the additional period. Barcelona's decisive failure was the quarter-final first leg away to Juventus, lost 0:3, from which the second-leg goalless draw offered no recovery.

Table 67 Player Shot Data — Bayern vs Real Madrid, Quarter-Final Leg 1, Won 2:1, 2017-04-12

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Arturo Vidal	Bayern	5	1	1.113	0.223	100.0%	38.5%
Thomas Müller	Bayern	3	0	0.304	0.101	66.7%	23.1%

Franck Ribéry	Bayern	2	0	0.040	0.020	0.0%	15.4%
Arjen Robben	Bayern	1	0	0.003	0.003	0.0%	7.7%
David Alaba	Bayern	1	0	0.004	0.004	0.0%	7.7%
Javi Martínez	Bayern	1	0	0.077	0.077	100.0%	7.7%
TEAM TOTAL	Bayern	13	1	1.543	0.119	-	100%
Cristiano Ronaldo	Real Madrid	8	2	0.601	0.075	62.5%	34.8%
Karim Benzema	Real Madrid	4	0	0.359	0.090	100.0%	17.4%
Toni Kroos	Real Madrid	2	0	0.054	0.027	50.0%	8.7%
Gareth Bale	Real Madrid	2	0	0.178	0.089	50.0%	8.7%
Marcelo	Real Madrid	2	0	0.082	0.041	100.0%	8.7%
Marco Asensio	Real Madrid	2	0	0.014	0.007	0.0%	8.7%
Casemiro	Real Madrid	1	0	0.008	0.008	0.0%	4.3%
Luka Modric	Real Madrid	1	0	0.005	0.005	0.0%	4.3%
Sergio Ramos	Real Madrid	1	0	0.006	0.006	0.0%	4.3%
TEAM TOTAL	Real Madrid	23	2	1.306	0.057	-	100%

Real Madrid won away from home while Bayern generated the higher chance quality, 1.543 xG to 1.306, most of it in a single Arturo Vidal attempt worth 1.113. Ronaldo took 8 shots and scored both Real Madrid goals from 0.601 xG. The result gave Real Madrid the advantage carried into the second leg.

Table 68 Player Shot Data — Real Madrid vs Bayern, Quarter-Final Leg 2, Won 4:2 (AET), 2017-04-18

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	9	3	0.536	0.060	77.8%	30.0%
Marco Asensio	Real Madrid	3	1	0.154	0.051	100.0%	10.0%
Toni Kroos	Real Madrid	3	0	0.076	0.025	33.3%	10.0%
Sergio Ramos	Real Madrid	3	0	0.179	0.060	100.0%	10.0%

Isco	Real Madrid	2	0	0.029	0.014	50.0%	6.7%
Dani Carvajal	Real Madrid	2	0	0.034	0.017	0.0%	6.7%
Casemiro	Real Madrid	2	0	0.031	0.016	50.0%	6.7%
Marcelo	Real Madrid	2	0	0.063	0.032	100.0%	6.7%
Luka Modric	Real Madrid	1	0	0.007	0.007	0.0%	3.3%
Lucas Vázquez	Real Madrid	1	0	0.067	0.067	100.0%	3.3%
Karim Benzema	Real Madrid	1	0	0.149	0.149	100.0%	3.3%
Nacho	Real Madrid	1	0	0.115	0.115	100.0%	3.3%
TEAM TOTAL	Real Madrid	30	4	1.440	0.048	-	100%
Arjen Robben	Bayern	4	0	0.263	0.066	75.0%	20.0%
Arturo Vidal	Bayern	4	0	0.357	0.089	50.0%	20.0%
Robert Lewandowski	Bayern	3	1	0.906	0.302	100.0%	15.0%
David Alaba	Bayern	2	0	0.024	0.012	50.0%	10.0%
Douglas Costa	Bayern	2	0	0.062	0.031	50.0%	10.0%
Franck Ribéry	Bayern	2	0	0.052	0.026	50.0%	10.0%
Thiago Alcântara	Bayern	2	0	0.203	0.102	50.0%	10.0%
Mats Hummels	Bayern	1	0	0.103	0.103	100.0%	5.0%
TEAM TOTAL	Bayern	20	1	1.972	0.099	-	100%

Note: Bayern's team total shows one goal against a final score of 4:2. Their second goal was an own goal by Sergio Ramos (77'), which under the player-attribution convention set out at Table 61 is credited to no shooter and therefore does not appear in either team's shot data.

This was the campaign's hinge — level on the night before extra time. Bayern generated 1.972 xG, the highest Real Madrid conceded in any critical match this season, with Lewandowski worth 0.906 from three shots. Ronaldo scored a hat-trick from 0.536 xG, taking his two-leg total against Bayern to five goals, and Real Madrid's finishing, four goals from 1.440 xG, carried the tie once it reached the additional period.

Table 69 Player Shot Data — Real Madrid vs Atletico Madrid, Semi-Final Leg 1, Won 3:0, 2017-05-02

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	5	3	0.495	0.099	60.0%	29.4%
Karim Benzema	Real Madrid	4	0	0.585	0.146	100.0%	23.5%
Luka Modric	Real Madrid	3	0	0.041	0.014	33.3%	17.6%
Casemiro	Real Madrid	1	0	0.054	0.054	100.0%	5.9%
Isco	Real Madrid	1	0	0.008	0.008	0.0%	5.9%
Dani Carvajal	Real Madrid	1	0	0.075	0.075	100.0%	5.9%
Lucas Vázquez	Real Madrid	1	0	0.071	0.071	100.0%	5.9%
Raphaël Varane	Real Madrid	1	0	0.150	0.150	100.0%	5.9%
TEAM TOTAL	Real Madrid	17	3	1.479	0.087	-	100%
Diego Godín	Atletico	2	0	0.197	0.098	50.0%	50.0%
Koke	Atletico	1	0	0.004	0.004	0.0%	25.0%
Yannick Carrasco	Atletico	1	0	0.007	0.007	0.0%	25.0%
TEAM TOTAL	Atletico	4	0	0.208	0.052	-	100%

Real Madrid restricted Atletico to four shots and 0.208 xG, the lowest opponent volume in any critical match in the study, and Ronaldo scored a second consecutive hat-trick. The three-goal cushion effectively decided the tie before the return.

Table 70 Player Shot Data — Atletico Madrid vs Real Madrid, Semi-Final Leg 2, Lost 1:2, 2017-05-10

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Antoine Griezmann	Atletico	5	1	0.858	0.172	60.0%	27.8%
Gabi	Atletico	3	0	0.018	0.006	0.0%	16.7%
Yannick Carrasco	Atletico	3	0	0.066	0.022	33.3%	16.7%
Koke	Atletico	2	0	0.126	0.063	50.0%	11.1%
Saúl Ñíguez	Atletico	2	1	0.184	0.092	50.0%	11.1%

Fernando Torres	Atletico	1	0	0.124	0.124	100.0%	5.6%
Kevin Gameiro	Atletico	1	0	0.119	0.119	100.0%	5.6%
Stefan Savic	Atletico	1	0	0.040	0.040	100.0%	5.6%
TEAM TOTAL	Atletico	18	2	1.535	0.085	-	100%
Cristiano Ronaldo	Real Madrid	6	0	0.215	0.036	16.7%	30.0%
Casemiro	Real Madrid	3	0	0.291	0.097	66.7%	15.0%
Isco	Real Madrid	3	1	0.503	0.168	66.7%	15.0%
Karim Benzema	Real Madrid	2	0	0.110	0.055	50.0%	10.0%
Luka Modric	Real Madrid	2	0	0.012	0.006	0.0%	10.0%
Toni Kroos	Real Madrid	2	0	0.071	0.036	50.0%	10.0%
Marcelo	Real Madrid	1	0	0.043	0.043	100.0%	5.0%
Sergio Ramos	Real Madrid	1	0	0.144	0.144	100.0%	5.0%
TEAM TOTAL	Real Madrid	20	1	1.389	0.069	-	100%

Real Madrid lost the leg 1:2 but advanced 4:2 on aggregate. With the tie already secured, Ronaldo took six shots without scoring at a 16.7% box concentration, and Isco supplied Real Madrid's goal. Griezmann generated 0.858 of Atletico's 1.535 xG. The defeat did not threaten the aggregate the first leg had built.

Table 71 Player Shot Data — Juventus vs Real Madrid, Final, Won 4:1, 2017-06-03

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Gonzalo Higuaín	Juventus	3	0	0.068	0.023	33.3%	33.3%
Miralem Pjanic	Juventus	3	0	0.053	0.018	33.3%	33.3%
Alex Sandro	Juventus	1	0	0.081	0.081	100.0%	11.1%
Mario Mandzukic	Juventus	1	1	0.029	0.029	100.0%	11.1%
Paulo Dybala	Juventus	1	0	0.003	0.003	0.0%	11.1%
TEAM TOTAL	Juventus	9	1	0.234	0.026	-	100%

Cristiano Ronaldo	Real Madrid	6	2	0.608	0.101	66.7%	33.3%
Casemiro	Real Madrid	2	1	0.005	0.002	0.0%	11.1%
Isco	Real Madrid	2	0	0.016	0.008	0.0%	11.1%
Marcelo	Real Madrid	2	0	0.017	0.008	0.0%	11.1%
Karim Benzema	Real Madrid	1	0	0.055	0.055	100.0%	5.6%
Gareth Bale	Real Madrid	1	0	0.004	0.004	0.0%	5.6%
Luka Modric	Real Madrid	1	0	0.005	0.005	0.0%	5.6%
Marco Asensio	Real Madrid	1	1	0.171	0.171	100.0%	5.6%
Toni Kroos	Real Madrid	1	0	0.019	0.019	0.0%	5.6%
Álvaro Morata	Real Madrid	1	0	0.021	0.021	100.0%	5.6%
TEAM TOTAL	Real Madrid	18	4	0.920	0.051	-	100%

The final is the season's most extreme finishing performance. Real Madrid scored four goals from 0.920 xG — a Finishing Delta of +3.08 — including a Casemiro strike worth 0.005 xG. Ronaldo scored twice from 0.608, and Mandzukic supplied Juventus' only goal. The distance-only model rates the four Real Madrid goals as improbable from their locations; the shot pillar records the divergence and cannot account for it. This match forms one half of The Juventus Test below.

Table 72 Player Shot Data — Paris Saint-Germain vs Barcelona, Round of 16 Leg 1, Lost 0:4, 2017-02-14

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Edinson Cavani	PSG	5	1	0.412	0.082	80.0%	31.2%
Ángel Di María	PSG	3	2	0.163	0.054	33.3%	18.8%
Julian Draxler	PSG	3	1	0.117	0.039	66.7%	18.8%
Lucas Moura	PSG	2	0	0.121	0.060	100.0%	12.5%
Adrien Rabiot	PSG	1	0	0.038	0.038	100.0%	6.2%
Blaise Matuidi	PSG	1	0	0.048	0.048	100.0%	6.2%
Presnel Kimpembe	PSG	1	0	0.421	0.421	100.0%	6.2%

TEAM TOTAL	PSG	16	4	1.320	0.083	-	100%
Neymar	Barcelona	2	0	0.048	0.024	50.0%	28.6%
André Gomes	Barcelona	1	0	0.036	0.036	100.0%	14.3%
Ivan Rakitic	Barcelona	1	0	0.041	0.041	100.0%	14.3%
Lionel Messi	Barcelona	1	0	0.004	0.004	0.0%	14.3%
Samuel Umtiti	Barcelona	1	0	0.347	0.347	100.0%	14.3%
Sergi Roberto	Barcelona	1	0	0.013	0.013	0.0%	14.3%
TEAM TOTAL	Barcelona	7	0	0.488	0.070	-	100%

Barcelona's seven shots are their lowest volume in any critical match in the study, and Messi's single attempt was worth 0.004 xG. PSG scored four from 1.320. Taken alone the result read as an elimination; the second leg overturned it.

Table 73 Player Shot Data — Barcelona vs Paris Saint-Germain, Round of 16 Leg 2, Won 6:1, 2017-03-08

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Neymar	Barcelona	6	2	0.834	0.139	33.3%	30.0%
Lionel Messi	Barcelona	4	1	0.825	0.206	50.0%	20.0%
Sergi Roberto	Barcelona	2	1	0.237	0.118	100.0%	10.0%
Luis Suárez	Barcelona	2	1	0.375	0.188	100.0%	10.0%
Gerard Piqué	Barcelona	2	0	0.218	0.109	100.0%	10.0%
Andrés Iniesta	Barcelona	1	0	0.005	0.005	0.0%	5.0%
Arda Turan	Barcelona	1	0	0.051	0.051	100.0%	5.0%
Javier Mascherano	Barcelona	1	0	0.019	0.019	0.0%	5.0%
Rafinha	Barcelona	1	0	0.012	0.012	100.0%	5.0%
TEAM TOTAL	Barcelona	20	5	2.575	0.129	-	100%
Edinson Cavani	PSG	5	1	0.471	0.094	100.0%	62.5%

Julian Draxler	PSG	1	0	0.008	0.008	0.0%	12.5%
Lucas Moura	PSG	1	0	0.004	0.004	0.0%	12.5%
Ángel Di María	PSG	1	0	0.070	0.070	100.0%	12.5%
TEAM TOTAL	PSG	8	1	0.553	0.069	-	100%

Note: Barcelona's team total shows five goals against a final score of 6:1. The second goal was an own goal by Layvin Kurzawa (40'), which under the player-attribution convention set out at Table 61 is credited to no shooter and therefore does not appear in either team's shot data.

The second leg is the strongest single attacking performance in Barcelona's critical-match record for this season: 20 shots for 2.575 xG. The player-attributed goals total five — Neymar 2, Messi 1, Suárez 1, Sergi Roberto 1 — with the sixth goal in the 6:1 an own goal not captured in the shot data; the Match Log records six. The comeback completed a 6:5 aggregate and carried Barcelona into the quarter-finals.

Table 74 Player Shot Data — Juventus vs Barcelona, Quarter-Final Leg 1, Lost 0:3, 2017-04-11

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Gonzalo Higuaín	Juventus	4	0	0.165	0.041	50.0%	28.6%
Paulo Dybala	Juventus	3	2	0.096	0.032	33.3%	21.4%
Sami Khedira	Juventus	3	0	0.076	0.025	33.3%	21.4%
Giorgio Chiellini	Juventus	1	1	0.102	0.102	100.0%	7.1%
Dani Alves	Juventus	1	0	0.065	0.065	100.0%	7.1%
Miralem Pjanic	Juventus	1	0	0.016	0.016	0.0%	7.1%
Mario Mandzukic	Juventus	1	0	0.003	0.003	0.0%	7.1%
TEAM TOTAL	Juventus	14	3	0.523	0.037	-	100%
Lionel Messi	Barcelona	4	0	0.074	0.018	25.0%	25.0%
Luis Suárez	Barcelona	4	0	0.483	0.121	75.0%	25.0%
Andrés Iniesta	Barcelona	3	0	0.139	0.046	33.3%	18.8%
Neymar	Barcelona	2	0	0.037	0.018	0.0%	12.5%

Ivan Rakitic	Barcelona	1	0	0.016	0.016	100.0%	6.2%
Javier Mascherano	Barcelona	1	0	0.162	0.162	100.0%	6.2%
Sergi Roberto	Barcelona	1	0	0.062	0.062	100.0%	6.2%
TEAM TOTAL	Barcelona	16	0	0.974	0.061	-	100%

Barcelona out-created Juventus, 0.974 xG to 0.523, and lost 0:3. Juventus scored three from low-value chances — Dybala twice, Chiellini from a set piece — while Barcelona took 16 shots without scoring; Messi's four attempts, at 25% box concentration, produced 0.074. This is the decisive leg of Barcelona's elimination: the deficit proved unrecoverable.

Table 75 Player Shot Data — Barcelona vs Juventus, Quarter-Final Leg 2, Drew 0:0, 2017-04-19

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Lionel Messi	Barcelona	7	0	0.314	0.045	42.9%	36.8%
Neymar	Barcelona	3	0	0.094	0.031	66.7%	15.8%
Gerard Piqué	Barcelona	2	0	0.240	0.120	100.0%	10.5%
Luis Suárez	Barcelona	2	0	0.182	0.091	100.0%	10.5%
Sergi Roberto	Barcelona	2	0	0.048	0.024	50.0%	10.5%
Jordi Alba	Barcelona	1	0	0.164	0.164	100.0%	5.3%
Andrés Iniesta	Barcelona	1	0	0.019	0.019	0.0%	5.3%
Ivan Rakitic	Barcelona	1	0	0.011	0.011	0.0%	5.3%
TEAM TOTAL	Barcelona	19	0	1.072	0.056	-	100%
Juan Cuadrado	Juventus	4	0	0.200	0.050	100.0%	33.3%
Gonzalo Higuaín	Juventus	3	0	0.171	0.057	100.0%	25.0%
Mario Lemina	Juventus	1	0	0.028	0.028	100.0%	8.3%
Mario Mandzukic	Juventus	1	0	0.131	0.131	100.0%	8.3%
Miralem Pjanic	Juventus	1	0	0.012	0.012	0.0%	8.3%
Paulo Dybala	Juventus	1	0	0.038	0.038	100.0%	8.3%

Sami Khedira	Juventus	1	0	0.065	0.065	100.0%	8.3%
TEAM TOTAL	Juventus	12	0	0.646	0.056	-	100%

Barcelona took 19 shots for 1.072 xG and again failed to score, with Messi's seven attempts producing 0.314. Across the two legs Barcelona took 35 shots for 2.046 xG and scored none against a Juventus side that mustered 26 shots and 1.169 xG. Messi was present in the shot data throughout — eleven attempts across the tie — but silent. The shot pillar records that Barcelona generated more and better chances across the tie and did not convert them; how Juventus' defensive structure produced that outcome is examined in Module 8.

The Juventus Test

Both clubs faced Juventus in 2016-2017, at different stages — Barcelona over two quarter-final legs, Real Madrid once in the final, six weeks apart. As with the Atletico comparison in 4.5 and the Juventus comparison in 4.4, holding the opponent constant removes the archetype variable that complicates comparisons across different fixtures.

Barcelona took 35 shots across the tie for 2.046 xG and scored 0; Juventus took 26 for 1.169 and scored 3. Real Madrid took 18 shots in the final for 0.920 xG and scored 4; Juventus took 9 for 0.234 and scored 1.

Table 76 The Juventus Test — Barcelona Quarter-Final and Real Madrid Final Compared, UCL 2016-2017

Metric	BA Leg 1 (A)	BA Leg 2 (H)	BA Tie Total	RM Final
Result	Lost 0:3	Drew 0:0	Out 0:3 agg	Won 4:1
Shots	16	19	35	18
Goals	0	0	0	4
xG	0.974	1.072	2.046	0.920
Rationality (xG/Shot)	0.061	0.056	0.058	0.051
Box Shot %	50.0%	57.9%	54.3%	38.9%
Accuracy %	25.0%	5.3%	14.3%	27.8%
Field Tilt %	53.3%	61.3%	57.4%	66.7%
Juventus Shots	14	12	26	9
Juventus xG	0.523	0.646	1.169	0.234

Juventus Goals	3	0	3	1
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The comparison is less clean than a two-leg-to-two-leg pairing — Barcelona played 180 minutes, Real Madrid 90 — but read with that caveat, three observations hold.

First, Barcelona generated more than twice Real Madrid's expected goals against Juventus and scored none, while Real Madrid generated less than half Barcelona's per-match xG and scored four. Barcelona's tie Finishing Delta was -2.05; Real Madrid's final Finishing Delta was +3.08. Against the same opponent, the two clubs sat at opposite extremes of the distance model's error. [DATA]

Second, this reverses the 2014-2015 Juventus Test exactly. In that season Real Madrid generated the higher xG against Juventus and were eliminated; Barcelona generated roughly half as much and won the final scoring three. In 2016-2017 the roles are exchanged: Real Madrid are now the club that scored heavily on low xG, Barcelona the club that generated chances and did not score. The same opponent produced the same structural puzzle two seasons apart, with the clubs' positions reversed. [DATA]

Third, box access did not predict conversion in either direction. Real Madrid's final box concentration of 38.9% was low by their season standard, yet they scored four; Barcelona's tie box concentration of 54.3% was higher, and they scored none. Real Madrid scored four from 0.920 xG, including a Casemiro strike worth 0.005; Barcelona scored none from 2.046, including eleven Messi shots.

Set against the 2014-2015 Juventus Test and the 2013-2014 and 2015-2016 Atletico Tests, this is the fourth same-opponent case in the study and the second involving Juventus. The shot data records that against Juventus, in both directions and across two seasons, the club generating more chance quality did not score and the club generating less did. It cannot explain how Juventus' defensive structure produced that outcome, nor how Real Madrid converted four low-value chances in a single final. Those questions belong to Modules 6 and 8.

Key Observations — Season 2016-2017

The 2016-2017 season is the study's clearest case of a team satisfying all three of the conditions set out in Module 3, and of the opponent-held-constant comparison isolating conversion as the residual the shot pillar cannot explain.

At season level the two clubs were again close. Finishing Delta of +1.26 and +1.13, Quality Margin of +0.50 and +0.72, goals within 0.12 per match. Both satisfied the first condition; Barcelona, on chance-quality balance, marginally more so.

At tier level the divergence ran opposite to the outcome on the quality metrics and with the outcome on conversion. Barcelona out-created their elite opponents by the wider margin — a

Tier 1 Quality Margin of +0.42 against +0.33 — and conceded less. But Real Madrid scored 2.57 Tier 1 goals per match to Barcelona's 1.67, a Tier 1 Finishing Delta of +1.20 against +0.43, and converted 24.6% of their Box_Central chances against elite opposition where Barcelona converted 19.0% from fewer attempts. The conversion collapse that defined Real Madrid's prior two elite campaigns did not recur: this season Real Madrid both created and finished against the best opposition, and Cristiano Ronaldo scored eleven of their eighteen Tier 1 goals.

At critical-match level Barcelona failed the variance test and Real Madrid passed it. Barcelona survived one catastrophic leg — a 0:4 defeat at Paris Saint-Germain overturned by a 6:1 second leg — and then could not survive the next, taking 35 shots across the Juventus quarter-final for 2.05 xG and scoring none across 180 minutes. A single unrecoverable tie ended a campaign that the season and tier aggregates did not mark for elimination. This is the 2014-2015 pattern of 4.3 repeated with the clubs reversed: in that season a single leg eliminated Real Madrid after a strong tier sample; in this one two goalless legs eliminated Barcelona after a strong one. Real Madrid, by contrast, won or survived all five of their critical matches, needing extra time only against Bayern Munich, and defeated the same Juventus 4:1 in the final.

On the diversity question this season inverts the recurring pattern. Barcelona, not Real Madrid, carried the broader mechanism profile against elite opposition — open play, penalty, free kick and set piece across their ten Tier 1 goals, against Real Madrid's near-exclusive open play — and Real Madrid scored no penalties all season. Neither the broader mechanism profile nor Barcelona's superior Tier 1 quality balance prevented elimination. What separated the two clubs at elite level was conversion, and conversion is the one thing a distance-only shot model cannot account for.

Three questions pass forward. Why did Real Madrid's Box_Central conversion against elite opposition rise from 2.9% in 2015-2016 to 24.6% in 2016-2017 without a corresponding rise in chance quality? Why did Barcelona generate 2.05 xG across two legs against Juventus and score nothing, while Real Madrid scored four from 0.92 xG against the same opponent? And why did Barcelona's superior Tier 1 chance-quality balance and broader goal-type profile fail to produce goals when a single tie turned unrecoverable? Chance construction is examined in Module 6, defensive structure in Module 8. The full six-season comparison of these patterns is assembled in Module 9.

4.7 Season 2017-2018

The 2017-2018 UEFA Champions League produced Real Madrid's thirteenth European title and a third consecutive triumph — the first club to retain the trophy twice in the Champions League era — secured with a 3:1 win over Liverpool in Kyiv. Barcelona were eliminated in the quarter-finals by Roma, going out on away goals after a 4:4 aggregate, having led the tie 4:1 after the first leg.

This season completes the six-season sample and closes the pattern the module has tracked throughout. It is the third consecutive campaign in which Barcelona exit at the quarter-final stage — Atletico in 2015-2016, Juventus in 2016-2017, Roma in 2017-2018 — and the third consecutive Real Madrid title. The quarter-final exit is now the most repeated single result in the study.

It also produces the sharpest contrast in the sample. Both clubs reached a quarter-final against a Serie A opponent, both took a commanding first-leg lead, and both conceded three goals in the second leg. The two ties were played one day apart. Real Madrid survived; Barcelona did not. That pairing is examined as The Serie A Test at the close of the critical-match analysis, and it is where the season's central finding sits.

Season Overview

Table 77 presents the season average shot metrics for both teams across all matches in the 2017-2018 UEFA Champions League campaign.

Table 77 Season Overview — Shot Analysis 2017-2018

Metric	Real Madrid	Barcelona
Avg Shots per Match	17.3	12.9
Avg Goals per Match	2.54	1.20
xG per Match	1.51	0.73
Rationality (xG/Shot)	0.086	0.054
Box Shot %	67.0%	49.8%
Accuracy %	44.8%	44.3%
Big Misses per Match	0.46	0.10
Opp Shots per Match	12.0	10.2
Opp xG per Match	1.00	0.59

Finishing Delta (Goals – xG)	+1.03	+0.47
Quality Margin (xG – Opp xG)	+0.51	+0.14
Field Tilt %	58.5%	54.6%

Note: Goals are player-attributed — credited to the player recorded as taking the shot — and own goals are therefore excluded throughout, for both clubs and their opponents alike. Own goals are errors produced by the conceding side rather than shooting output by the attacking side, so excluding them measures each club's own attacking production. Real Madrid recorded no own goals in either direction this season; their 33 goals are identical on either basis. Barcelona benefited from five own goals across four matches — against Sporting (away and home), Olympiacos, and Roma in the quarter-final first leg, which carried two — so their scoreline total of 17 corresponds to 12 player-attributed goals. All Barcelona figures in this section are the player-attributed figures. The gap is discussed where it bears on interpretation, most directly at Table 92.

The two clubs are further apart at season level than in any prior campaign in the study. Real Madrid scored 2.54 goals per match to Barcelona's 1.20 and generated more than twice the chance quality, 1.51 xG against 0.73. Real Madrid took 17.3 shots per match to 12.9 and reached the box on 67.0% of them against Barcelona's 49.8% — the lowest box concentration recorded by either club in six seasons. [DATA]

Both clubs finished above the model's expectation, Real Madrid at +1.03 and Barcelona at +0.47, and both held a positive Quality Margin. But where in 2016-2017 the season aggregates were near-identical and the divergence appeared only under tier partition, here the separation is visible at the top level. Barcelona's 0.73 xG per match is the lowest season figure by either club in the study; Real Madrid's 1.51 is among the highest.

Barcelona's Rationality of 0.054 against Real Madrid's 0.086 follows from the same source: a shot profile weighted away from the box. Their accuracy of 44.3% is essentially level with Real Madrid's 44.8%, and their Big Misses figure of 0.10 per match is the lowest recorded by either club. Barcelona were accurate and rarely wasteful; they were shooting from positions the model rates as low-value, and not many of them. For the first time in the study, the first of the three conditions set out in Module 3 is not satisfied by both clubs. [DATA]

Tier Comparison

Table 78 presents the tier breakdown separating performance against Tier 1 (Elite, Elo > 1850) and Tier 2 (Standard, Elo ≤ 1850) opponents. Real Madrid played nine Tier 1 matches and four Tier 2; Barcelona played four Tier 1 and six Tier 2.

Table 78 Tier Comparison — Shot Analysis 2017-2018

Metric	RM Tier 1	RM Tier 2	BA Tier 1	BA Tier 2
Matches	9	4	4	6
Avg Shots per Match	15.4	21.5	8.3	16.0

Avg Goals per Match	2.44	2.75	1.75	0.83
xG per Match	1.25	2.10	0.49	0.88
Rationality (xG/Shot)	0.081	0.096	0.060	0.049
Box Shot %	69.3%	61.7%	56.0%	45.7%
Accuracy %	46.5%	41.0%	48.5%	41.6%
Big Misses per Match	0.22	1.00	0.25	0.00
Opp Shots per Match	13.2	9.3	12.0	9.0
Opp xG per Match	1.21	0.54	0.48	0.66
Finishing Delta (Goals – xG)	+1.19	+0.65	+1.26	–0.05
Quality Margin (xG – Opp xG)	+0.04	+0.70	+0.01	+0.22
Field Tilt %	53.5%	69.9%	40.9%	63.8%

Note on tier assignment: tiers follow the Elo threshold applied throughout this module (Tier 1 above 1850). For 2017-2018 this places Chelsea in Tier 1 and both Tottenham and Roma in Tier 2. The Roma classification warrants comment, because Roma eliminated Barcelona and went on to the semi-finals. [RESEARCH] Roma's rating at the time of these fixtures lagged their form in this particular campaign: they topped a group containing Chelsea and Atletico Madrid — beating Chelsea 3:0 in Rome and eliminating Atletico altogether — then removed Shakhtar Donetsk in the round of 16 and reached the semi-finals, their deepest European run since the 1984 final. The tier label is correctly applied by the stated rule; it should be read as a rating that trailed a rising side, not as a measure of the opposition Barcelona actually faced.

Real Madrid's tier profile is the more conventional. They faced elite opposition in nine of thirteen matches and scored 2.44 goals per Tier 1 match against 2.75 in Tier 2 — a narrow gap, continuing the closure first noted in 2016-2017. Their Tier 1 Finishing Delta of +1.19 is positive for a second consecutive season, and their Tier 1 Quality Margin of +0.04 is close to level: against the best opposition Real Madrid created barely more than they conceded, and won on conversion rather than on chance-quality dominance. [DATA]

Barcelona's tier split runs in the opposite direction to every prior season in the study, and to expectation. They scored more goals per match against Tier 1 opponents than Tier 2 — 1.75 against 0.83 — and recorded a Tier 1 Finishing Delta of +1.26 against a Tier 2 figure of –0.05. Against standard opposition Barcelona finished below the model's expectation, the only negative tier cell recorded by either club in six seasons. [DATA]

The explanation is not that Barcelona were better against elite opposition. Their Tier 1 sample of four matches comprises two group-stage fixtures against Juventus and the round-of-16 tie against Chelsea, in which they took 8.3 shots per match at a Field Tilt of 40.9% — the lowest tier-club cell in the study by a wide margin, and the only figure below 45% recorded by either

club. Barcelona were territorially dominated in their Tier 1 matches and converted a small number of chances at a high rate. The Finishing Delta records conversion above expectation on a very small base; it does not record control. [DATA]

Real Madrid's Tier 2 sample is likewise small and carries its own anomaly. Four matches, two against APOEL Nicosia and two against Tottenham, and they lost one of them — 3:1 away to a Tottenham side that topped the group ahead of them. Their Tier 2 Field Tilt of 69.9% and xG of 2.10 per match reflect the APOEL fixtures more than the Tottenham ones. Neither club's Tier 2 cell this season carries the interpretive weight it does in earlier seasons.

Shot Map Analysis

The shot maps for both clubs are presented in Figures 11 and 12, each preceding its spatial profile. The zone distribution and zone xG data underlying the discussion are presented together in Tables 79 and 80, which follow the comparative insight.

Real Madrid Spatial Profile

Real Madrid's spatial profile against elite opposition is the strongest recorded by either club in the study. Box shots accounted for 69.1% of their Tier 1 attempts — above their Tier 2 figure of 61.6%, the second consecutive season in which their box access was higher against better opposition. The central corridor took 73.4% of Tier 1 attempts, a slightly wider distribution than the 79.1% of Tier 2, with the flanks carrying more of the load against elite sides.

In Box_Central, Real Madrid took 77 Tier 1 shots and scored 17, a goal rate of 22.1%, against 43 shots and 9 goals for 20.9% in Tier 2. The two rates are nearly level, which is itself the finding: for a second consecutive season Real Madrid's central conversion did not deteriorate against better opposition. In 2015-2016 the equivalent Tier 1 figure was 2.9%, in 2016-2017 it was 24.6%, and here it is 22.1%. The collapse pattern of the earlier seasons has not returned. [DATA]

Their edge shooting produced four Tier 1 goals from 42 attempts, a 9.5% rate that is the highest edge return by either club in the study, and one that the distance-only model rates at 0.0121 average xG per attempt. The shot pillar records that Real Madrid scored from outside the box against elite opposition at a rate the model does not anticipate; it cannot establish why. [CANNOT CONFIRM]

Shot Analysis: Real Madrid (Red) vs Opponents (Blue)
Champions League 2017-2018

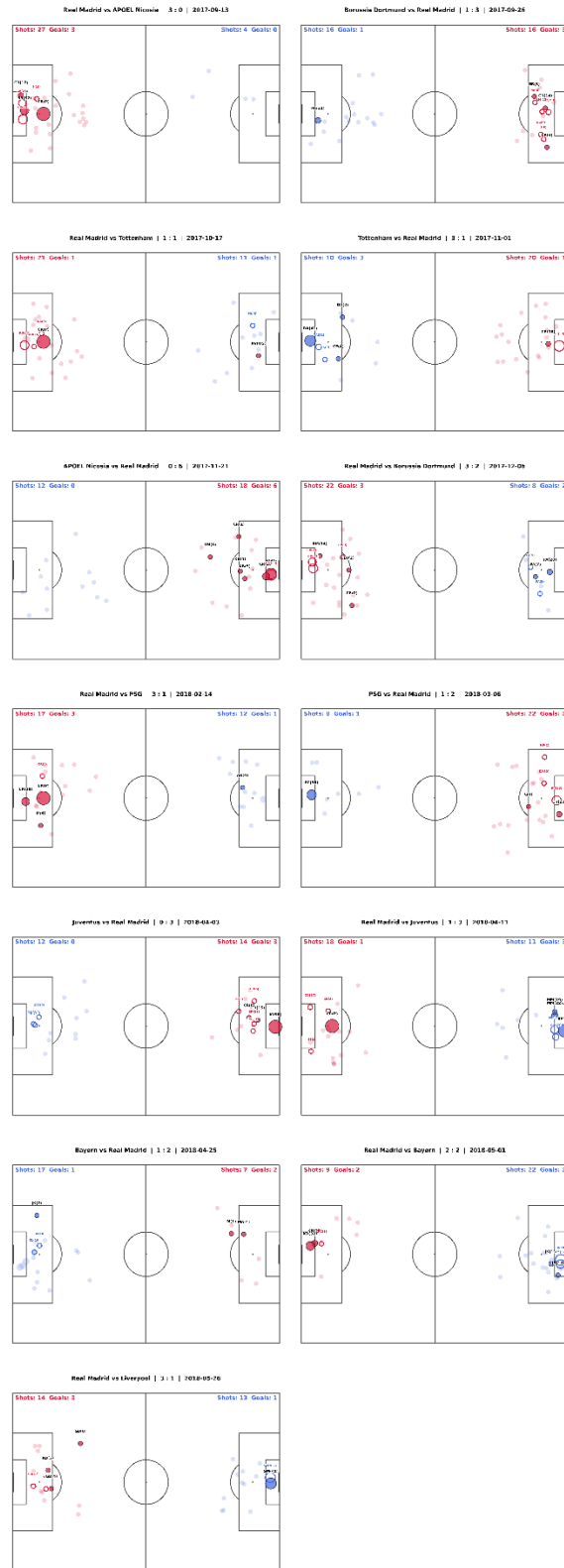


Figure 11 Real Madrid Shot Map — UCL 2017-2018

Barcelona Spatial Profile

Barcelona's spatial profile is the weakest in the study on every access measure. Box shots accounted for 54.5% of their Tier 1 attempts and 50.0% of Tier 2 — against Real Madrid's 69.1% and 61.6% — with the edge taking 45.5% and 49.0% respectively. Close to half of Barcelona's shooting in this campaign came from outside the penalty area, in both tiers. [DATA]

In Box_Central they took 12 Tier 1 shots across four matches, scoring 3 for a 25.0% rate, and 40 Tier 2 shots scoring 4 for 10.0%. The Tier 1 figure rests on twelve attempts and should not be read as a rate. The Tier 2 figure is the substantive one: 40 central box shots for four goals, at an average xG of 0.113, against opposition the tier system rates as standard. Barcelona reached the most productive zone on the pitch forty times against Sporting, Olympiacos and Roma, and scored four times. [DATA]

Their edge volume produced almost nothing: 15 Tier 1 edge shots for one goal, 47 Tier 2 edge shots for one. Across the campaign Barcelona took 62 shots from outside the box and scored twice. This is the displacement pattern first recorded in 2015-2016, but at greater volume and across both tiers rather than against elite opposition alone.

**Shot Analysis: Barcelona (Blue) vs Opponents (Red)
Champions League 2017-2018**

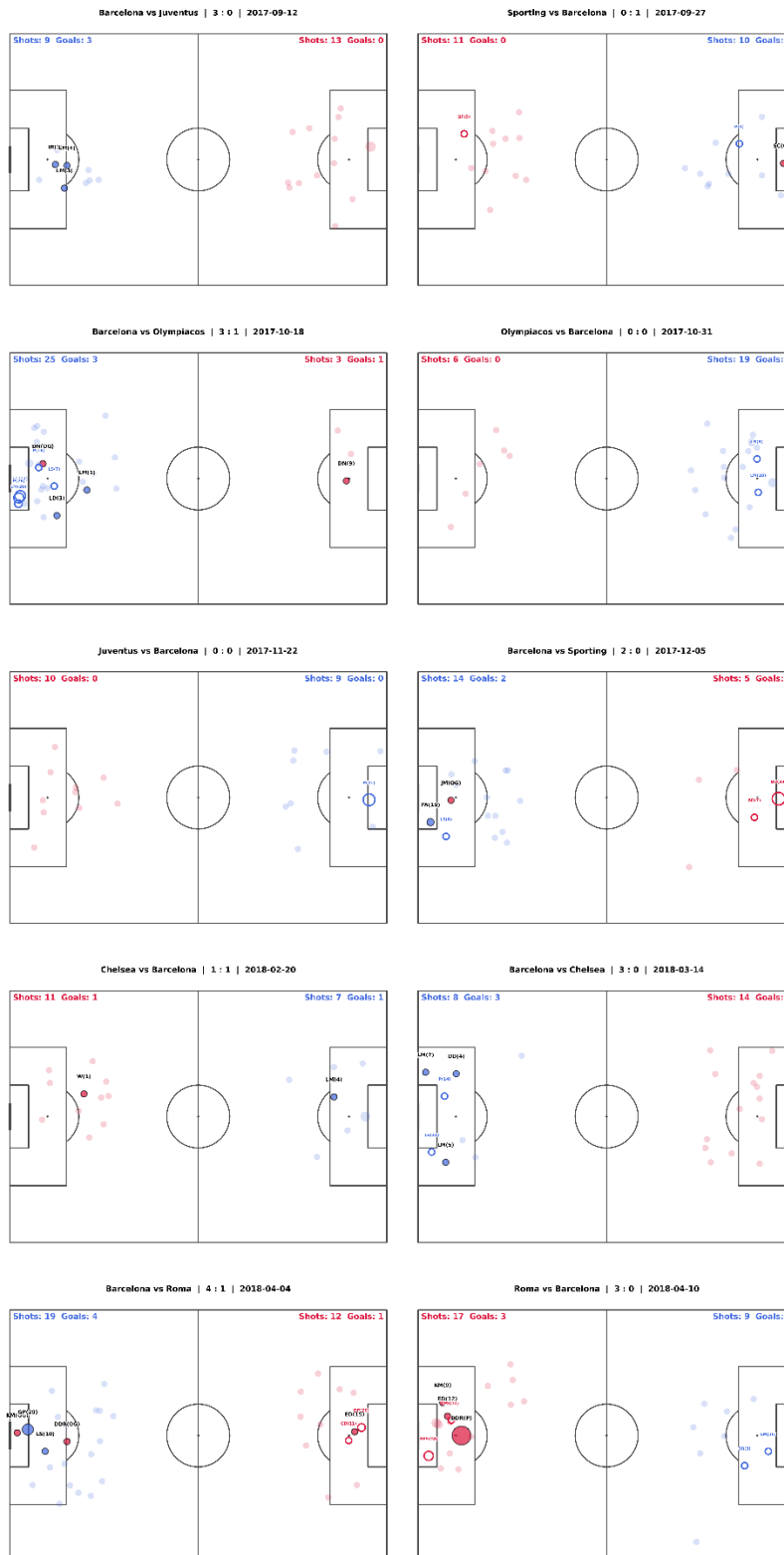


Figure 12 Barcelona Shot Map — UCL 2017-2018

Comparative Spatial Insight

The two spatial records diverge more sharply than in any prior season, and they diverge on access rather than on conversion. Real Madrid reached the box on 69.1% of Tier 1 shots; Barcelona on 54.5%. Real Madrid took 77 Tier 1 Box_Central shots; Barcelona took 12. Even allowing for Barcelona's smaller Tier 1 sample, the per-match rates are 8.6 central box shots for Real Madrid against 3.0 for Barcelona.

In 2015-2016 Real Madrid had access without conversion and Barcelona conversion without access. In 2016-2017 Real Madrid had both. In 2017-2018 Barcelona have neither in sufficient quantity to matter: their conversion rates are respectable and computed on samples too small to carry a campaign. The shot pillar can establish that Barcelona did not arrive in the penalty area often enough to generate chance volume, and that this held against standard opposition as well as elite. Why the supply failed — whether in build-up, in final-third entry, or in the shape of the block they faced — is a Module 6 and Module 8 question. [DATA]

Table 79 Zone Shot Distribution — Real Madrid and Barcelona, UCL 2017-2018

Depth Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box	96	69.1%	53	61.6%	18	54.5%	48	50.0%
Edge	42	30.2%	31	36.0%	15	45.5%	47	49.0%
Long Range	1	0.7%	2	2.3%	0	0.0%	1	1.0%
TOTAL	139	100%	86	100%	33	100%	96	100%

Lateral Zone

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Left	20	14.4%	10	11.6%	6	18.2%	10	10.4%
Central	102	73.4%	68	79.1%	23	69.7%	77	80.2%
Right	17	12.2%	8	9.3%	4	12.1%	9	9.4%
TOTAL	139	100%	86	100%	33	100%	96	100%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 %	RM T2 Shots	RM T2 %	BA T1 Shots	BA T1 %	BA T2 Shots	BA T2 %
Box_Left	11	7.9%	7	8.1%	4	12.1%	4	4.2%
Box_Central	77	55.4%	43	50.0%	12	36.4%	40	41.7%
Box_Right	8	5.8%	3	3.5%	2	6.1%	4	4.2%

Edge_Left	9	6.5%	3	3.5%	2	6.1%	6	6.2%
Edge_Central	24	17.3%	23	26.7%	11	33.3%	36	37.5%
Edge_Right	9	6.5%	5	5.8%	2	6.1%	5	5.2%
Long Range_Left	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Long Range_Central	1	0.7%	2	2.3%	0	0.0%	1	1.0%
Long Range_Right	0	0.0%	0	0.0%	0	0.0%	0	0.0%
TOTAL	139	100%	86	100%	33	100%	96	100%

Table 80 Zone xG and Goals by Zone — Real Madrid and Barcelona, UCL 2017-2018

Depth Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box	96	0.1121	18	18.8%	53	0.1529	9	17.0%	18	0.0992	6	33.3%	48	0.1004	4	8.3%
Edge	42	0.0121	4	9.5%	31	0.0095	2	6.5%	15	0.0112	1	6.7%	47	0.0102	1	2.1%
Long Range	1	0.0016	0	0.0%	2	0.0021	0	0.0%	0	0.0000	0	0.0%	1	0.0028	0	0.0%

Lateral Zone

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Left	20	0.0185	1	5.0%	10	0.0272	1	10.0%	6	0.0340	1	16.7%	10	0.0184	0	0.0%
Central	102	0.1035	19	18.6%	68	0.1178	10	14.7%	23	0.0705	4	17.4%	77	0.0641	5	6.5%
Right	17	0.0202	2	11.8%	8	0.0152	0	0.0%	4	0.0319	2	50.0%	9	0.0208	0	0.0%

9-Zone Grid (Depth x Lateral)

Zone	RM T1 Shots	RM T1 Avg xG	RM T1 Goals	RM T1 Rate%	RM T2 Shots	RM T2 Avg xG	RM T2 Goals	RM T2 Rate%	BA T1 Shots	BA T1 Avg xG	BA T1 Goals	BA T1 Rate%	BA T2 Shots	BA T2 Avg xG	BA T2 Goals	BA T2 Rate%
Box_Left	11	0.0281	0	0.0%	7	0.0330	0	0.0%	4	0.0459	1	25.0%	4	0.0359	0	0.0%
Box_Central	77	0.1322	17	22.1%	43	0.1815	9	20.9%	12	0.1234	3	25.0%	40	0.1130	4	10.0%
Box_Right	8	0.0344	1	12.5%	3	0.0229	0	0.0%	2	0.0608	2	100%	4	0.0395	0	0.0%
Edge_Left	9	0.0069	1	11.1%	3	0.0137	1	33.3%	2	0.0102	0	0.0%	6	0.0067	0	0.0%
Edge_Central	24	0.0156	2	8.3%	23	0.0087	1	4.3%	11	0.0128	1	9.1%	36	0.0114	1	2.8%

Edge_Right	9	0.0077	1	11.1%	5	0.0106	0	0.0%	2	0.0029	0	0.0%	5	0.0059	0	0.0%
Long Range_Left	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%
Long Range_Central	1	0.0016	0	0.0%	2	0.0021	0	0.0%	0	0.0000	0	0.0%	1	0.0028	0	0.0%
Long Range_Right	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%	0	0.0000	0	0.0%

Player Highlights

Table 81 presents the season average shot metrics for all players with a shot share above 5% for both clubs.

Table 81 Player Season Average — Shot Analysis 2017-2018 (Shot Share > 5%)

Player	Team	Shots	OP	SP	Goals	xG	Rat.	Box %	Left %	Right %	Shot %	Goal %
Cristiano Ronaldo	Real Madrid	6.31	5.62	0.15	1.15	0.728	0.108	71.2	20.3	18.0	37.3	52.6
Isco	Real Madrid	2.75	2.63	0.13	0.00	0.102	0.035	38.3	30.2	24.8	15.3	0.0
Karim Benzema	Real Madrid	2.25	2.25	0.00	0.63	0.213	0.091	74.0	13.5	12.5	15.0	20.8
Gareth Bale	Real Madrid	2.20	2.20	0.00	0.60	0.161	0.061	70.0	10.0	30.0	11.4	20.0
Sergio Ramos	Real Madrid	2.00	1.88	0.13	0.13	0.266	0.129	96.9	18.8	3.1	10.7	4.2
Lucas Vázquez	Real Madrid	2.00	2.00	0.00	0.33	0.091	0.041	50.0	16.7	11.1	9.9	11.1
Casemiro	Real Madrid	1.75	1.75	0.00	0.25	0.066	0.038	62.5	25.0	12.5	7.8	12.5
Raphaël Varane	Real Madrid	1.40	1.20	0.20	0.00	0.186	0.137	80.0	0.0	0.0	8.0	0.0
Marco Asensio	Real Madrid	1.29	1.14	0.00	0.14	0.048	0.040	64.3	28.6	21.4	8.1	7.1
Luka Modric	Real Madrid	1.29	1.29	0.00	0.14	0.039	0.037	42.9	0.0	28.6	7.6	2.4
Marcelo	Real Madrid	1.25	1.13	0.13	0.38	0.098	0.093	56.3	56.3	0.0	7.9	14.6
Dani Carvajal	Real Madrid	1.25	1.25	0.00	0.00	0.029	0.027	62.5	0.0	75.0	9.0	0.0
Toni Kroos	Real Madrid	1.14	1.14	0.00	0.00	0.013	0.012	14.3	28.6	14.3	5.8	0.0
Nacho	Real Madrid	1.00	0.80	0.20	0.20	0.177	0.177	100.0	0.0	40.0	6.0	3.3
Lionel Messi	Barcelona	4.40	3.20	0.00	0.60	0.183	0.037	40.4	7.6	23.0	34.7	28.3
Luis Suárez	Barcelona	3.20	2.90	0.10	0.10	0.197	0.054	61.2	48.9	8.4	24.4	5.0
Paulinho	Barcelona	2.40	2.20	0.20	0.00	0.238	0.097	68.7	0.0	0.0	20.0	0.0
Ivan Rakitic	Barcelona	1.67	1.67	0.00	0.33	0.036	0.030	44.4	11.1	0.0	11.3	11.1

Andrés Iniesta	Barcelona	1.33	1.33	0.00	0.00	0.027	0.023	25.0	83.3	0.0	13.0	0.0
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Real Madrid's shooting load was more concentrated on Cristiano Ronaldo than in any prior season: a 37.3% shot share and a 52.6% goal share, with 15 of the club's 33 goals. Eleven came from open play and four from penalties — a reversal of 2016-2017, when Real Madrid scored no penalties at all. His 71.2% box concentration is the highest he records in the study. Behind him the distribution is wide and thin: Benzema 5 goals, Bale and Marcelo 3 each, and single goals from Casemiro, Lucas Vázquez, Modric, Ramos, Nacho, Asensio and Borja Mayoral. Eleven different players scored. [DATA]

Two profiles are worth noting. Isco ranked second in shot share at 15.3% and scored none, from a box concentration of 38.3% — the lowest sustained shooting profile in either squad. Sergio Ramos, a centre-back, ranked fifth at 10.7% with a 96.9% box concentration, almost entirely from open play rather than set pieces.

Barcelona's table is the shorter one, and that is itself the finding: only five players cleared a 5% shot share, against fourteen for Real Madrid. Lionel Messi held a 34.7% shot share and a 28.3% goal share with 6 goals, five from open play and one free kick. His box concentration of 40.4% is the lowest he records in the study — the displacement pattern noted in every prior season, here at its most pronounced. Luis Suárez, second at 24.4% shot share, scored once all campaign for a 5.0% goal share, and Paulinho took a 20.0% share without scoring. [DATA]

Barcelona's twelve player-attributed goals came from seven scorers — Messi 6, and one each from Piqué, Dembélé, Alcácer, Rakitic, Digne and Suárez. Set against Real Madrid's eleven scorers across 33 goals, the contrast is in volume rather than distribution. On goal type, Real Madrid scored 29 from open play and 4 penalties; Barcelona 11 from open play and 1 free kick, with no penalties and no set-piece goals. Neither club drew on a broad mechanism profile this season, and the mechanism-diversity question that ran through 4.3 to 4.6 does not separate them here. [DATA]

Player Tier Comparison

Table 82 presents the player-level tier breakdown using the same player list as Table 81. Real Madrid's Tier 1 sample is nine matches and Barcelona's four; the per-match figures should be read with that difference in mind.

Table 82 Player Tier Comparison — Shot Analysis 2017-2018 (Shot Share > 5%)

Player	Team	T1 Sh	T1 G	T1 xG	T1 Rat	T1 Box%	T1 Sh%	T1 Goal%	T2 Sh	T2 G	T2 xG	T2 Rat	T2 Box%	T2 Sh%	T2 Goal%
Cristiano Ronaldo	Real Madrid	6.44	1.00	0.699	0.109	71.6	41.9	40.9	6.00	1.50	0.792	0.132	70.2	27.9	54.5
Isco	Real Madrid	2.86	0.00	0.117	0.041	40.6	18.5	0.0	2.50	0.00	0.070	0.028	33.3	11.6	0.0

Karim Benzema	Real Madrid	2.17	0.50	0.191	0.088	76.4	14.1	13.6	2.50	1.00	0.278	0.111	67.0	11.6	18.2
Gareth Bale	Real Madrid	2.00	1.00	0.164	0.082	75.0	13.0	13.6	2.50	0.00	0.156	0.062	61.7	11.6	0.0
Sergio Ramos	Real Madrid	2.20	0.00	0.281	0.128	95.5	14.3	0.0	1.67	0.33	0.241	0.144	100.0	7.8	9.1
Lucas Vázquez	Real Madrid	1.80	0.20	0.083	0.046	44.4	11.7	4.5	2.33	0.50	0.104	0.045	58.3	10.8	9.1
Casemiro	Real Madrid	1.83	0.33	0.077	0.042	63.6	11.9	4.5	1.50	0.00	0.033	0.022	58.3	7.0	0.0
Raphaël Varane	Real Madrid	1.50	0.00	0.204	0.136	83.3	9.7	0.0	1.00	0.00	0.114	0.114	66.7	4.7	0.0
Marco Asensio	Real Madrid	1.20	0.20	0.043	0.036	60.0	7.8	4.5	1.50	0.00	0.061	0.041	75.0	7.0	0.0
Luka Modric	Real Madrid	1.29	0.00	0.041	0.032	44.4	8.4	0.0	1.33	0.33	0.033	0.025	33.3	6.2	9.1
Marcelo	Real Madrid	1.29	0.43	0.115	0.089	55.6	8.4	19.5	1.00	0.00	0.031	0.031	66.7	4.7	0.0
Dani Carvajal	Real Madrid	1.20	0.00	0.020	0.017	50.0	7.8	0.0	1.50	0.00	0.064	0.043	100.0	7.0	0.0
Toni Kroos	Real Madrid	1.13	0.00	0.014	0.012	11.1	7.3	0.0	1.00	0.00	0.008	0.008	33.3	4.7	0.0
Nacho	Real Madrid	1.00	0.00	0.036	0.036	100.0	6.5	0.0	1.00	0.33	0.318	0.318	100.0	4.7	9.1
Lionel Messi	Barcelona	2.50	1.25	0.072	0.029	46.7	29.6	71.4	5.67	0.17	0.256	0.042	36.2	38.1	20.0
Luis Suárez	Barcelona	2.00	0.00	0.094	0.047	50.0	24.5	0.0	4.00	0.17	0.266	0.058	68.7	24.3	20.0
Paulinho	Barcelona	1.67	0.00	0.187	0.115	77.8	20.0	0.0	3.50	0.00	0.316	0.070	55.0	20.0	0.0
Ivan Rakitic	Barcelona	1.00	1.00	0.075	0.075	100.0	11.1	14.3	2.00	0.00	0.017	0.007	16.7	11.5	0.0
Andrés Iniesta	Barcelona	1.33	0.00	0.039	0.035	50.0	17.4	0.0	1.33	0.00	0.015	0.011	0.0	8.5	0.0

Cristiano Ronaldo scored 9 of Real Madrid's 22 Tier 1 goals, a 40.9% share, at almost exactly his Tier 2 shot volume and box concentration — 6.44 Tier 1 shots per match against 6.00, and 71.6% box against 70.2%. His profile is effectively unchanged by opponent quality, the second consecutive season in which that holds. Seven of the nine were open play and two penalties. [DATA]

Beneath him Real Madrid drew Tier 1 goals from seven players: Ronaldo 9, Benzema 3, Bale 3, Marcelo 3, and one each from Casemiro, Lucas Vázquez, Asensio and Borja Mayoral. Marcelo's three, all in Tier 1 and all open play from a full-back position, is the most unusual line in the table. Isco, Ramos, Varane, Modric, Carvajal and Kroos scored none against elite opposition between them, from a combined Tier 1 shot share of 55.8%. [DATA]

Lionel Messi scored 5 of Barcelona's 7 Tier 1 goals — a 71.4% share, the highest single-player dependency recorded in the study — from 2.5 shots per Tier 1 match. His Tier 1 box concentration of 46.7% is higher than his Tier 2 figure of 36.2%, inverting the displacement pattern of earlier seasons on a four-match sample. Ivan Rakitic supplied the other Tier 1 goal. Luis Suárez and Paulinho, between them holding a 44.5% Tier 1 shot share, scored none. [DATA]

Barcelona's Tier 2 record is the more troubling line. Across six matches against Sporting, Olympiacos and Roma, five player-attributed goals: Messi one free kick, and one each from Piqué, Alcácer, Digne and Suárez. Messi took 5.67 shots per Tier 2 match at a 36.2% box concentration and scored once from open play in six matches. [DATA]

Critical Match Analysis

Eleven matches are examined: seven for Real Madrid and four for Barcelona, presented in Tables 83 through 93, each showing player-level shot data for both teams. They are ordered by tie within each club. Real Madrid's full knockout campaign is included because all four ties were competitive; Barcelona's comprises both knockout ties they played.

Real Madrid won or survived every tie, twice advancing on a 4:3 aggregate after losing or drawing the second leg. Barcelona controlled the Chelsea tie and then lost a 4:1 quarter-final lead in a single match.

Table 83 Player Shot Data — Real Madrid vs PSG, Round of 16 Leg 1, Won 3:1, 2018-02-14

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	10	2	1.313	0.131	70.0%	58.8%
Sergio Ramos	Real Madrid	2	0	0.260	0.130	100.0%	11.8%
Karim Benzema	Real Madrid	1	0	0.021	0.021	0.0%	5.9%
Marcelo	Real Madrid	1	1	0.047	0.047	100.0%	5.9%
Nacho	Real Madrid	1	0	0.039	0.039	100.0%	5.9%
Raphaël Varane	Real Madrid	1	0	0.002	0.002	0.0%	5.9%
Toni Kroos	Real Madrid	1	0	0.011	0.011	0.0%	5.9%
TEAM TOTAL	Real Madrid	17	3	1.692	0.100	-	100%
Adrien Rabiot	PSG	2	1	0.093	0.046	100.0%	16.7%
Giovani Lo Celso	PSG	2	0	0.059	0.030	50.0%	16.7%
Neymar	PSG	2	0	0.139	0.070	50.0%	16.7%
Kylian Mbappé	PSG	2	0	0.090	0.045	100.0%	16.7%
Julian Draxler	PSG	1	0	0.011	0.011	0.0%	8.3%

Edinson Cavani	PSG	1	0	0.064	0.064	100.0%	8.3%
Presnel Kimpembe	PSG	1	0	0.241	0.241	100.0%	8.3%
Yuri Berchiche	PSG	1	0	0.022	0.022	100.0%	8.3%
TEAM TOTAL	PSG	12	1	0.720	0.060	-	100%

Ronaldo took ten of Real Madrid's seventeen shots and scored twice from 1.313 xG, the highest single-player chance quality in any critical match this season. PSG, with Neymar, Mbappé and Cavani on the pitch, generated 0.720 across twelve attempts. The two-goal margin gave Real Madrid the tie.

Table 84 Player Shot Data — PSG vs Real Madrid, Round of 16 Leg 2, Won 2:1, 2018-03-06

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Edinson Cavani	PSG	2	1	0.418	0.209	100.0%	25.0%
Ángel Di María	PSG	2	0	0.050	0.025	50.0%	25.0%
Javier Pastore	PSG	1	0	0.126	0.126	100.0%	12.5%
Kylian Mbappé	PSG	1	0	0.055	0.055	100.0%	12.5%
Thiago Motta	PSG	1	0	0.012	0.012	0.0%	12.5%
Thiago Silva	PSG	1	0	0.104	0.104	100.0%	12.5%
TEAM TOTAL	PSG	8	1	0.765	0.096	-	100%
Cristiano Ronaldo	Real Madrid	4	1	0.357	0.089	75.0%	18.2%
Sergio Ramos	Real Madrid	4	0	0.213	0.053	75.0%	18.2%
Lucas Vázquez	Real Madrid	3	0	0.195	0.065	100.0%	13.6%
Karim Benzema	Real Madrid	3	0	0.102	0.034	66.7%	13.6%
Marco Asensio	Real Madrid	2	0	0.084	0.042	50.0%	9.1%
Dani Carvajal	Real Madrid	1	0	0.003	0.003	0.0%	4.5%
Casemiro	Real Madrid	1	1	0.043	0.043	100.0%	4.5%
Gareth Bale	Real Madrid	1	0	0.004	0.004	0.0%	4.5%

Isco	Real Madrid	1	0	0.007	0.007	0.0%	4.5%
Raphaël Varane	Real Madrid	1	0	0.336	0.336	100.0%	4.5%
Toni Kroos	Real Madrid	1	0	0.008	0.008	0.0%	4.5%
TEAM TOTAL	Real Madrid	22	2	1.351	0.061	-	100%

Real Madrid won away as well, out-shooting PSG 22 to 8 and taking the tie 5:2 on aggregate. PSG, at home and needing three goals, managed eight shots for 0.765 xG. This is the least competitive of Real Madrid's four knockout ties on the shot data.

Table 85 Player Shot Data — Juventus vs Real Madrid, Quarter-Final Leg 1, Won 3:0, 2018-04-03

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Paulo Dybala	Juventus	4	0	0.101	0.025	50.0%	33.3%
Rodrigo Bentancur	Juventus	3	0	0.060	0.020	33.3%	25.0%
Gonzalo Higuaín	Juventus	2	0	0.184	0.092	50.0%	16.7%
Giorgio Chiellini	Juventus	1	0	0.106	0.106	100.0%	8.3%
Mattia De Sciglio	Juventus	1	0	0.002	0.002	0.0%	8.3%
Juan Cuadrado	Juventus	1	0	0.155	0.155	100.0%	8.3%
TEAM TOTAL	Juventus	12	0	0.609	0.051	-	100%
Cristiano Ronaldo	Real Madrid	7	2	0.548	0.078	100.0%	50.0%
Lucas Vázquez	Real Madrid	1	0	0.035	0.035	0.0%	7.1%
Luka Modric	Real Madrid	1	0	0.038	0.038	0.0%	7.1%
Marcelo	Real Madrid	1	1	0.582	0.582	100.0%	7.1%
Marco Asensio	Real Madrid	1	0	0.071	0.071	100.0%	7.1%
Mateo Kovacic	Real Madrid	1	0	0.049	0.049	100.0%	7.1%
Raphaël Varane	Real Madrid	1	0	0.113	0.113	100.0%	7.1%
Toni Kroos	Real Madrid	1	0	0.006	0.006	0.0%	7.1%
TEAM TOTAL	Real Madrid	14	3	1.441	0.103	-	100%

Real Madrid won 3:0 in Turin on 1.441 xG against 0.609, with Ronaldo scoring twice from seven shots all taken inside the box and Marcelo converting the highest single-shot xG of the tie at 0.582. This is the first leg the Serie A Test compares directly with Barcelona's: a three-goal away lead built on a clear chance-quality advantage.

Table 86 Player Shot Data — Real Madrid vs Juventus, Quarter-Final Leg 2, Lost 1:3, 2018-04-11

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Cristiano Ronaldo	Real Madrid	6	1	1.097	0.183	100.0%	33.3%
Isco	Real Madrid	4	0	0.176	0.044	75.0%	22.2%
Raphaël Varane	Real Madrid	3	0	0.365	0.122	100.0%	16.7%
Gareth Bale	Real Madrid	2	0	0.260	0.130	100.0%	11.1%
Marcelo	Real Madrid	1	0	0.024	0.024	100.0%	5.6%
Marco Asensio	Real Madrid	1	0	0.005	0.005	0.0%	5.6%
Toni Kroos	Real Madrid	1	0	0.024	0.024	0.0%	5.6%
TEAM TOTAL	Real Madrid	18	1	1.951	0.108	-	100%
Mario Mandzukic	Juventus	3	2	0.709	0.236	100.0%	27.3%
Blaise Matuidi	Juventus	2	1	0.533	0.266	50.0%	18.2%
Sami Khedira	Juventus	2	0	0.073	0.036	50.0%	18.2%
Gonzalo Higuaín	Juventus	2	0	0.260	0.130	50.0%	18.2%
Alex Sandro	Juventus	1	0	0.004	0.004	0.0%	9.1%
Douglas Costa	Juventus	1	0	0.005	0.005	0.0%	9.1%
TEAM TOTAL	Juventus	11	3	1.583	0.144	-	100%

Real Madrid lost 1:3 at home and advanced 4:3, the tie settled in stoppage time. They generated 1.951 xG to Juventus' 1.583 and out-shot them 18 to 11 — losing the leg while creating more, and creating more than in the first leg they had won 3:0. Ronaldo alone produced 1.097 xG from six shots all inside the box and scored once. The shot data does not record a collapse in chance generation; it records a leg in which Real Madrid created well and conceded three from eleven attempts. [DATA]

Table 87 Player Shot Data — Bayern vs Real Madrid, Semi-Final Leg 1, Won 2:1, 2018-04-25

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Franck Ribéry	Bayern	4	0	0.201	0.050	100.0%	23.5%
Robert Lewandowski	Bayern	4	0	0.563	0.141	100.0%	23.5%
Rafinha	Bayern	2	0	0.010	0.005	0.0%	11.8%
Mats Hummels	Bayern	2	0	0.433	0.216	100.0%	11.8%
Thomas Müller	Bayern	2	0	0.330	0.165	100.0%	11.8%
Javi Martínez	Bayern	1	0	0.263	0.263	100.0%	5.9%
Joshua Kimmich	Bayern	1	1	0.030	0.030	100.0%	5.9%
Thiago Alcântara	Bayern	1	0	0.007	0.007	0.0%	5.9%
TEAM TOTAL	Bayern	17	1	1.837	0.108	-	100%
Cristiano Ronaldo	Real Madrid	2	0	0.174	0.087	50.0%	28.6%
Dani Carvajal	Real Madrid	1	0	0.028	0.028	100.0%	14.3%
Isco	Real Madrid	1	0	0.014	0.014	0.0%	14.3%
Karim Benzema	Real Madrid	1	0	0.052	0.052	100.0%	14.3%
Marcelo	Real Madrid	1	1	0.018	0.018	0.0%	14.3%
Marco Asensio	Real Madrid	1	1	0.041	0.041	100.0%	14.3%
TEAM TOTAL	Real Madrid	7	2	0.328	0.047	-	100%

Real Madrid took seven shots, the fewest by either club in any match in the study, and won 2:1 away from 0.328 xG. Bayern generated 1.837 from seventeen attempts. Marcelo and Asensio scored from shots worth 0.018 and 0.041. This is the single most extreme conversion performance in the six-season sample — a Finishing Delta of +1.67 in one match — and the distance-only model rates both goals as improbable from their locations. [CANNOT CONFIRM]

Table 88 Player Shot Data — Real Madrid vs Bayern, Semi-Final Leg 2, Drew 2:2, 2018-05-01

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
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Cristiano Ronaldo	Real Madrid	5	0	0.342	0.068	60.0%	55.6%
Karim Benzema	Real Madrid	2	2	0.555	0.278	100.0%	22.2%
Luka Modric	Real Madrid	1	0	0.011	0.011	0.0%	11.1%
Sergio Ramos	Real Madrid	1	0	0.090	0.090	100.0%	11.1%
TEAM TOTAL	Real Madrid	9	2	0.998	0.111	-	100%
James Rodríguez	Bayern	5	1	0.910	0.182	100.0%	22.7%
Mats Hummels	Bayern	4	0	0.510	0.128	100.0%	18.2%
Thomas Müller	Bayern	4	0	0.591	0.148	100.0%	18.2%
Corentin Tolisso	Bayern	3	0	0.132	0.044	33.3%	13.6%
Robert Lewandowski	Bayern	3	0	0.436	0.145	100.0%	13.6%
David Alaba	Bayern	1	0	0.009	0.009	0.0%	4.5%
Joshua Kimmich	Bayern	1	1	0.192	0.192	100.0%	4.5%
Niklas Süle	Bayern	1	0	0.037	0.037	0.0%	4.5%
TEAM TOTAL	Bayern	22	2	2.818	0.128	-	100%

Bayern generated 2.818 xG across the two legs' highest single-match figure and scored twice; Real Madrid took nine shots for 0.998 and also scored twice, both through Benzema. Across the tie Bayern out-shot Real Madrid 39 to 16 and out-created them 4.655 xG to 1.326, and lost 4:3. This is the widest chance-quality deficit overcome by either club in the study. [DATA]

Table 89 Player Shot Data — Real Madrid vs Liverpool, Final, Won 3:1, 2018-05-26

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Karim Benzema	Real Madrid	4	1	0.210	0.052	75.0%	28.6%
Cristiano Ronaldo	Real Madrid	3	0	0.292	0.097	100.0%	21.4%
Gareth Bale	Real Madrid	2	2	0.049	0.024	50.0%	14.3%
Isco	Real Madrid	2	0	0.211	0.106	100.0%	14.3%
Luka Modric	Real Madrid	1	0	0.031	0.031	100.0%	7.1%

Marcelo	Real Madrid	1	0	0.004	0.004	0.0%	7.1%
Nacho	Real Madrid	1	0	0.043	0.043	100.0%	7.1%
TEAM TOTAL	Real Madrid	14	3	0.841	0.060	-	100%
Sadio Mané	Liverpool	3	1	0.482	0.161	66.7%	23.1%
Roberto Firmino	Liverpool	2	0	0.174	0.087	100.0%	15.4%
Mohamed Salah	Liverpool	2	0	0.062	0.031	50.0%	15.4%
Adam Lallana	Liverpool	1	0	0.036	0.036	0.0%	7.7%
Dejan Lovren	Liverpool	1	0	0.140	0.140	100.0%	7.7%
James Milner	Liverpool	1	0	0.019	0.019	0.0%	7.7%
Georginio Wijnaldum	Liverpool	1	0	0.008	0.008	0.0%	7.7%
Trent Alexander-Arnold	Liverpool	1	0	0.040	0.040	100.0%	7.7%
Virgil van Dijk	Liverpool	1	0	0.375	0.375	100.0%	7.7%
TEAM TOTAL	Liverpool	13	1	1.334	0.103	-	100%

Real Madrid won the final scoring three from 0.841 xG while Liverpool generated 1.334 — a Finishing Delta of +2.16 against an opponent who out-created them. Gareth Bale scored twice from two shots worth 0.049 combined. For a second consecutive season Real Madrid won the final on a chance-quality deficit and an extreme positive Finishing Delta; in 2016-2017 the figures were 4 goals from 0.920 xG. [DATA]

Table 90 Player Shot Data — Chelsea vs Barcelona, Round of 16 Leg 1, Drew 1:1, 2018-02-20

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Willian	Chelsea	4	1	0.087	0.022	25.0%	36.4%
Eden Hazard	Chelsea	2	0	0.028	0.014	0.0%	18.2%
Victor Moses	Chelsea	2	0	0.040	0.020	50.0%	18.2%
Antonio Rüdiger	Chelsea	1	0	0.148	0.148	100.0%	9.1%
Cesc Fàbregas	Chelsea	1	0	0.005	0.005	0.0%	9.1%

Marcos Alonso	Chelsea	1	0	0.007	0.007	0.0%	9.1%
TEAM TOTAL	Chelsea	11	1	0.315	0.029	-	100%
Andrés Iniesta	Barcelona	2	0	0.023	0.012	50.0%	28.6%
Luis Suárez	Barcelona	2	0	0.053	0.026	50.0%	28.6%
Gerard Piqué	Barcelona	1	0	0.261	0.261	100.0%	14.3%
Lionel Messi	Barcelona	1	1	0.044	0.044	100.0%	14.3%
Paulinho	Barcelona	1	0	0.094	0.094	100.0%	14.3%
TEAM TOTAL	Barcelona	7	1	0.476	0.068	-	100%

Barcelona took seven shots away at Chelsea and drew, Messi scoring from a 0.044 xG attempt — his only shot of the match. Both sides generated under 0.5 xG. The away goal proved decisive for the tie.

Table 91 Player Shot Data — Barcelona vs Chelsea, Round of 16 Leg 2, Won 3:0, 2018-03-14

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Lionel Messi	Barcelona	3	2	0.133	0.044	66.7%	37.5%
Luis Suárez	Barcelona	2	0	0.173	0.086	100.0%	25.0%
Andrés Iniesta	Barcelona	1	0	0.022	0.022	0.0%	12.5%
Ousmane Dembélé	Barcelona	1	1	0.042	0.042	100.0%	12.5%
Paulinho	Barcelona	1	0	0.143	0.143	100.0%	12.5%
TEAM TOTAL	Barcelona	8	3	0.512	0.064	-	100%
Marcos Alonso	Chelsea	4	0	0.163	0.041	75.0%	28.6%
Eden Hazard	Chelsea	3	0	0.045	0.015	33.3%	21.4%
Antonio Rüdiger	Chelsea	2	0	0.247	0.124	100.0%	14.3%
Willian	Chelsea	2	0	0.014	0.007	0.0%	14.3%
Davide Zappacosta	Chelsea	1	0	0.006	0.006	0.0%	7.1%

N'Golo Kanté	Chelsea	1	0	0.052	0.052	100.0%	7.1%
Olivier Giroud	Chelsea	1	0	0.045	0.045	100.0%	7.1%
TEAM TOTAL	Chelsea	14	0	0.571	0.041	-	100%

Barcelona won 3:0 from eight shots and 0.512 xG while being out-shot 14 to 8 — a Finishing Delta of +2.49, their highest single-match figure in the study. Messi scored twice from three shots worth 0.133 combined. Barcelona advanced 4:1 on aggregate having generated 0.988 xG across the tie against Chelsea's 0.886. [DATA]

Table 92 Player Shot Data — Barcelona vs Roma, Quarter-Final Leg 1, Won 4:1, 2018-04-04

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Lionel Messi	Barcelona	7	0	0.155	0.022	42.9%	36.8%
Luis Suárez	Barcelona	5	1	0.251	0.050	80.0%	26.3%
Ivan Rakitic	Barcelona	3	0	0.030	0.010	33.3%	15.8%
Andrés Iniesta	Barcelona	2	0	0.019	0.010	0.0%	10.5%
Gerard Piqué	Barcelona	1	1	0.295	0.295	100.0%	5.3%
Jordi Alba	Barcelona	1	0	0.008	0.008	0.0%	5.3%
TEAM TOTAL	Barcelona	19	2	0.758	0.040	-	100%
Aleksandar Kolarov	Roma	4	0	0.068	0.017	25.0%	33.3%
Diego Perotti	Roma	2	0	0.209	0.104	50.0%	16.7%
Edin Dzeko	Roma	2	1	0.178	0.089	100.0%	16.7%
Alessandro Florenzi	Roma	1	0	0.010	0.010	0.0%	8.3%
Grégoire Defrel	Roma	1	0	0.110	0.110	100.0%	8.3%
Kevin Strootman	Roma	1	0	0.050	0.050	100.0%	8.3%
Lorenzo Pellegrini	Roma	1	0	0.025	0.025	0.0%	8.3%
TEAM TOTAL	Roma	12	1	0.651	0.054	-	100%

Note: Barcelona's team total shows two goals against a final score of 4:1. The other two were own goals by Daniele De Rossi (38') and Kostas Manolas (55'), which under the player-attribution convention set out at Table 77 are credited to no shooter and

appear in neither team's shot data. The distinction carries unusual weight here: half of the four-goal scoreline was produced by the conceding side, and the two goals Barcelona actually generated came from 0.758 xG.

This is the most misleading scoreline in the six-season sample. Barcelona won by three goals while generating 0.758 xG from nineteen shots — a Rationality of 0.040, the lowest of any critical match in the study — against Roma's 0.651 from twelve. Two of the four goals were Roma's own. Messi took seven shots for 0.155 xG at a 42.9% box concentration and did not score. On the shot data this was a marginal match that the scoreline recorded as a rout. [DATA]

Table 93 Player Shot Data — Roma vs Barcelona, Quarter-Final Leg 2, Lost 0:3, 2018-04-10

Player	Team	Shots	Goals	xG	Rationality	Box %	Shot Share %
Edin Dzeko	Roma	4	1	0.360	0.090	75.0%	23.5%
Radja Nainggolan	Roma	3	0	0.057	0.019	33.3%	17.6%
Patrik Schick	Roma	3	0	0.508	0.169	100.0%	17.6%
Daniele De Rossi	Roma	2	1	1.000	0.500	100.0%	11.8%
Aleksandar Kolarov	Roma	1	0	0.053	0.053	100.0%	5.9%
Kevin Strootman	Roma	1	0	0.003	0.003	0.0%	5.9%
Cengiz Ünder	Roma	1	0	0.010	0.010	0.0%	5.9%
Konstantinos Manolas	Roma	1	1	0.096	0.096	100.0%	5.9%
Stephan El Shaarawy	Roma	1	0	0.243	0.243	100.0%	5.9%
TEAM TOTAL	Roma	17	3	2.330	0.137	-	100%
Lionel Messi	Barcelona	5	0	0.211	0.042	20.0%	55.6%
Andrés Iniesta	Barcelona	1	0	0.006	0.006	0.0%	11.1%
Luis Suárez	Barcelona	1	0	0.047	0.047	100.0%	11.1%
Ousmane Dembélé	Barcelona	1	0	0.000	0.000	0.0%	11.1%
Sergi Roberto	Barcelona	1	0	0.039	0.039	100.0%	11.1%
TEAM TOTAL	Barcelona	9	0	0.303	0.034	-	100%

Barcelona were eliminated taking nine shots for 0.303 xG at a Field Tilt of 34.6% — the lowest territorial share recorded by either club in any match in the study. Roma took seventeen shots

for 2.330. Messi took five of Barcelona's nine shots at a 20.0% box concentration, generating 0.211. Across the tie Barcelona were out-shot 29 to 28 and out-created 2.981 xG to 1.061. [DATA] [RESEARCH] Contemporary tactical accounts record that Roma switched to a 3-4-3 and pressed high while Barcelona retained the narrow 4-4-2 they had used throughout the campaign; the shot pillar can confirm the territorial and chance-quality outcome but not the mechanism. [CANNOT CONFIRM]

The Serie A Test

Both clubs met a Serie A opponent in the quarter-final, both won the first leg by a commanding margin, and both conceded three goals in the second. The first legs were played on consecutive days, 3 and 4 April, and the second legs on 10 and 11 April. Real Madrid advanced 4:3; Barcelona went out 4:4 on away goals.

This is not a same-opponent comparison of the kind used in 4.3, 4.4, 4.5 and 4.6 — Juventus and Roma are different sides, and the tier system rates them differently, Juventus in Tier 1 and Roma in Tier 2. What the pairing holds constant is the round, the season, the competitive situation and the broad opponent archetype. [HYPOTHESIS] It should be read as a weaker control than the same-opponent format: it narrows the opponent confound without removing it.

Table 94 The Serie A Test — Real Madrid vs Juventus and Barcelona vs Roma Compared, UCL 2017-2018

Metric	RM vs Juventus	BA vs Roma	Difference
Leg 1	W 3:0 away	W 4:1 home	—
Leg 2	L 1:3 home	L 0:3 away	—
Outcome	Advanced 4:3	Out 4:4 (a.g.)	—
Tie Shots	32	28	+4
Tie Goals	4	2	+2
Tie xG	3.392	1.061	+2.331
Rationality (xG/Shot)	0.106	0.038	+0.068
Box Shot %	81.2%	42.9%	+38.3
Accuracy %	43.8%	46.4%	-2.6
Field Tilt %	58.2%	49.1%	+9.1

Opponent Shots	23	29	-6
Opponent xG	2.192	2.981	-0.789
Opponent Goals	3	4	-1
Finishing Delta	+0.608	+0.939	-0.331
Quality Margin	+1.200	-1.920	+3.120

Note: Barcelona's tie goals are the player-attributed figure of two; the aggregate scoreline of four includes the two Roma own goals recorded at Table 92. On the scoreline basis Barcelona's tie Finishing Delta would read +2.939, which would describe conversion Barcelona did not perform.

Three findings follow, and they run against the pattern of the four preceding same-opponent comparisons.

First, Barcelona did not lose this tie to variance. In 2016-2017 they generated 2.046 xG against Juventus and scored none — chances created and not converted. Here they generated 1.061 xG across 180 minutes and were out-created by Roma 2.981 to 1.061, a Quality Margin of -1.920. Real Madrid's corresponding figure was +1.200. Barcelona were the inferior side on chance quality across their own quarter-final, and by a wide margin. This is the first elimination in the study that the shot data describes as deserved rather than anomalous. [DATA]

Second, the two first legs were not equivalent performances despite similar scorelines. Real Madrid won 3:0 away on 1.441 xG against 0.609 — a two-to-one chance-quality advantage. Barcelona won 4:1 at home on 0.758 xG against 0.651, with two of their four goals scored by Roma players. Real Madrid's lead was earned; Barcelona's was not. When both sides then conceded three in the second leg, only one of the two leads had the substance to survive it. [DATA]

Third, box access separates them where conversion does not. Real Madrid took 81.2% of their tie shots inside the box against Barcelona's 42.9% — a 38-point gap, the widest in any comparison in this module. Accuracy was near-level at 43.8% and 46.4%, and Barcelona's tie Finishing Delta of +0.939 was actually higher than Real Madrid's +0.608. Barcelona shot accurately and converted above expectation. They did so from outside the penalty area, nine or nineteen times a match, against a side that conceded the box to nobody. [DATA]

The tier classification is worth restating here rather than treating as an anomaly. Roma were rated Tier 2 and eliminated Barcelona; they had already topped a group containing Chelsea and Atletico Madrid and went on to a semi-final. The rating lagged the side. But the shot data does not require the rating to be wrong in order to explain the result: across 180 minutes Roma took more shots, generated three times the chance quality, and conceded a penalty area that

Barcelona entered on fewer than half their attempts. Whatever tier label is attached, Barcelona were outplayed. [DATA]

Key Observations — Season 2017-2018

The 2017-2018 season is the only campaign in the study in which Barcelona fail the first of the three conditions set out in Module 3, and the only one in which their elimination is not a variance result.

At season level the clubs separate at the top for the first time. Real Madrid generated 1.51 xG per match to Barcelona's 0.73 and reached the box on 67.0% of shots against 49.8%. In every prior season the season aggregates were close enough that the divergence in outcome had to be located in the tier split or the critical matches. Here it is visible immediately: Barcelona's 0.73 xG per match and 49.8% box concentration are the lowest figures either club records in six seasons. [DATA]

At tier level Barcelona's split inverts. They scored 1.75 goals per Tier 1 match against 0.83 in Tier 2 and recorded a negative Tier 2 Finishing Delta of -0.05 , the only negative tier cell in the study. This is not elite-level strength; their Tier 1 Field Tilt of 40.9% is the lowest tier-club figure recorded by either club, and their four Tier 1 matches produced 8.3 shots per match. Barcelona were territorially dominated against elite opposition and unproductive against standard opposition. Real Madrid's Tier 1 Finishing Delta of $+1.19$ was positive for a second consecutive season on a nine-match sample, with a Quality Margin of only $+0.04$ — they won at elite level on conversion, not on chance-quality control.

At critical-match level the contrast is sharpest. Real Madrid won or survived all seven of their examined matches, twice advancing 4:3 after failing to win the second leg, and won the final scoring three from 0.841 xG against an opponent generating 1.334. Across the Bayern semi-final they were out-created 4.655 xG to 1.326 and advanced. Barcelona controlled the Chelsea tie on a similar basis — 3:0 from 0.512 xG while being out-shot — and then met a side that did not permit it. [DATA]

The Serie A Test carries the season. Both clubs took a commanding first-leg lead over a Serie A opponent and conceded three in the second leg; only Real Madrid's lead was built on chance quality, and only Real Madrid survived. Barcelona generated 1.061 xG across their quarter-final against Roma's 2.981, a Quality Margin of -1.920 , and took 42.9% of their shots inside the box against Real Madrid's 81.2%. Barcelona's 4:1 first leg, half of it scored by Roma players from 0.758 xG, is the most misleading scoreline in the sample.

Three questions pass forward. Why did Barcelona's box access fall to 49.8% for the season and 42.9% across the Roma tie, having sustained above 60% in every prior campaign? Why did Real Madrid win a final and a semi-final while being out-created, converting at rates the distance-

only model rates as improbable in three separate critical matches? And why, across six seasons, has the club generating more chance quality in a decisive tie so often failed to advance? Chance construction is examined in Module 6, defensive structure in Module 8, and the six-season synthesis that these questions now converge on is assembled in Module 9.

4.8 Cross-Season Synthesis — Shot Analysis, 2012–2018

Across the six seasons in this sample, Real Madrid won four European titles — 2013-14, 2015-16, 2016-17, and 2017-18 — and Barcelona won one, in 2014-15. This section closes Module 4 by asking what the shot data alone reveals about that four-to-one asymmetry between two squads of comparable quality, and where it necessarily falls silent.

The shot pillar cannot explain why the asymmetry exists; it can locate it. The framework treats sustained success as requiring three conditions in sequence — a competent season baseline, robustness against elite opposition, and survival of the single unrecoverable match — and the pillar’s task is to walk both clubs down that hierarchy and identify the level at which they diverge. Where a gap appears that the shot data cannot account for, it is named and handed to the modules equipped to explain it; where a mechanism would be asserted, the pillar has left its scope.

Four conventions govern the figures. All metrics are per game, because the clubs did not play equal samples — Real Madrid contested 40 Tier-1 matches to Barcelona’s 29 — so totals would reward mere survival. Goals are player-attributed, with own goals excluded for both clubs and their opponents, keeping the basis symmetric across all six seasons. The expected-goals model is distance-only and blind to pressure, so Finishing Delta (Goals – xG) and Quality Margin (xG – opponent xG) are read as complementary indicators, and a large gap between xG and goals is treated as a flag that the model is missing something, not evidence of what. Finally, two axes run through every table: the between-team comparison (Real Madrid relative to Barcelona) and the within-team trajectory (each club against its own record). A note on the defensive figures introduced below: opponent xG and opponent Finishing Delta describe the defensive picture only as it appears from the shot ledger — chance quality conceded and whether opponents over- or under-converted it. They are a shadow of defending, not a measurement of it; the mechanism belongs to Module 8.

The Season Baseline (Level 1)

The first condition is competence on average. Tables 95 and 96 report the season-average attacking profile across all matches in each campaign; defensive detail is deferred to the tier level, because blending elite and standard opponents into one figure hides the distinction that matters.

Table 95 Volume and Territory — Season Averages, 2012–2018

Season	RM Shots	BA Shots	RM Tilt %	BA Tilt %	RM Box %	BA Box %
12-13	19.3	14.8	58.7	56.8	51.6	50.0

Season	RM Shots	BA Shots	RM Tilt %	BA Tilt %	RM Box %	BA Box %
13-14	16.2	14.8	53.7	64.5	55.8	56.7
14-15	18.7	14.8	62.3	60.0	52.7	60.2
15-16	21.2	16.8	69.8	63.3	58.9	59.5
16-17	19.5	13.1	61.8	61.1	59.8	65.8
17-18	17.3	12.9	58.5	54.6	67.0	49.8

Real Madrid out-shoots Barcelona in every season, by a per-game margin of 1.4 to 6.4 attempts, and holds a positive Field Tilt throughout. Barcelona's territorial control is more volatile and its share of shots from inside the box overtakes Real Madrid's in three seasons. On raw volume and territory both clubs clear the baseline; Real Madrid does so more consistently.

Table 96 Quality and Finishing — Season Averages, 2012–2018

Season	RM Goals	BA Goals	RM xG	BA xG	RM QM	BA QM	RM FinΔ	BA FinΔ
12-13	2.17	1.50	1.25	0.84	+0.38	+0.12	+0.91	+0.66
13-14	3.15	2.10	1.28	1.11	+0.38	+0.58	+1.87	+0.99
14-15	1.92	2.38	1.42	1.00	+0.89	+0.35	+0.50	+1.39
15-16	2.08	2.20	1.39	1.50	+0.71	+0.78	+0.69	+0.70
16-17	2.62	2.50	1.35	1.37	+0.50	+0.72	+1.26	+1.13
17-18	2.54	1.20	1.51	0.73	+0.51	+0.14	+1.03	+0.47

Goals per game, player-attributed. QM = Quality Margin (xG – opponent xG); FinΔ = Finishing Delta (Goals – xG).

At season-average resolution the two clubs are close, and in two seasons Barcelona is ahead: both post a positive Quality Margin every year, and the between-team gap reverses in 2015-16 and 2016-17, when Barcelona out-creates Real Madrid over the year. A reader handed only these two tables could not tell which club won four titles and which won one. The championship gap is invisible at Level 1, which — by the logic of necessary conditions — means it lies further down.

Robustness Against Elite Opposition (Level 2)

The second condition is that the baseline survives elite opposition. Tables 97 and 98 restrict the sample to Tier-1 opponents (opponent Elo above 1850) and, for the first time, separate the attacking and defensive views.

Table 97 Tier-1 Attacking Robustness — Elite Opponents Only

Season	RM Goals	BA Goals	RM xG	BA xG	RM FinΔ	BA FinΔ	RM Tilt %	BA Tilt %
12-13	1.62	0.75	1.32	0.76	+0.30	-0.01	60.4	49.2
13-14	2.29	1.25	1.09	0.85	+1.20	+0.40	48.3	58.1
14-15	0.75	2.33	1.29	0.97	-0.54	+1.37	69.0	53.3
15-16	0.40	1.75	1.14	1.22	-0.74	+0.53	63.9	61.3
16-17	2.57	1.67	1.38	1.24	+1.20	+0.43	59.9	52.4
17-18	2.00	1.50	1.38	0.48	+0.62	+1.02	54.5	44.1

Tier-1 matches (RM / BA): 12-13 8/4; 13-14 7/4; 14-15 4/9; 15-16 5/4; 16-17 7/6; 17-18 9/2. Small samples in the tail limit inference.

Table 98 Tier-1 Defensive Robustness (shot-side) — Elite Opponents Only

Season	RM OppxG	BA OppxG	RM OppG	BA OppG	RM OppFinΔ	BA OppFinΔ	RM QM	BA QM
12-13	1.05	1.15	1.38	2.25	+0.33	+1.10	+0.28	-0.39
13-14	1.05	0.61	0.86	0.75	-0.19	+0.14	+0.04	+0.24
14-15	0.75	0.83	0.75	1.00	+0.00	+0.17	+0.54	+0.13
15-16	0.62	0.82	0.20	1.00	-0.42	+0.18	+0.52	+0.39
16-17	1.05	0.82	1.29	1.83	+0.24	+1.01	+0.33	+0.42
17-18	1.25	0.51	1.33	0.00	+0.08	-0.51	+0.13	-0.03

OppxG = opponent xG conceded; OppFinΔ = opponent Goals – opponent xG (positive = opponents over-converted against this club). QM = Quality Margin.

On creation, Real Madrid sustains a positive Tier-1 Quality Margin in all six seasons; Barcelona's turns negative in 2012-13 and 2017-18, the same two years its Tier-1 Field Tilt falls below the 50% threshold (49.2% and 44.1%). But the diagnostic finding is in conversion, at both ends. Real Madrid's Tier-1 Finishing Delta is negative in exactly two seasons — 2014-15 (–0.54) and 2015-16 (–0.74) — and 2015-16 is the pivotal case: its own finishing was –0.74 and it should, on that number, have gone out, but its opponents finished even worse (Opp FinΔ –0.42), and penalties completed the rescue. Real Madrid won the competition that year while net-negative on Tier-1 finishing. The defensive column makes the survival legible: where a club advances on poor finishing, the opponent's finishing was poorer still. The distinction between out-creating and out-scoring an elite opponent is handed to Module 6; the defensive mechanism behind opponents' conversion is handed to Module 8.

Squad Character: Load and Mechanism

The preceding levels compare performance; the next two compare construction — how each squad is built to shoot. These are descriptions of persistent structural difference, not claims of cause.

Table 99 Squad Dependency — Primary Carrier and Top-Three Concentration

Season	RM Carrier	RM Share %	RM Top-3 %	BA Carrier	BA Share %	BA Top-3 %
12-13	Ronaldo	37.0	68.4	Messi	29.6	73.9
13-14	Ronaldo	41.3	75.6	Messi	32.9	75.7
14-15	Ronaldo	33.6	68.1	Messi	30.0	82.9
15-16	Ronaldo	36.3	72.0	Messi	36.1	75.1
16-17	Ronaldo	28.5	58.9	Messi	31.7	70.9
17-18	Ronaldo	37.3	67.6	Messi	34.7	79.1

Both clubs lean on a single carrier — Ronaldo 28.5–41.3%, Messi 29.6–36.1% — but Barcelona’s top-three share exceeds Real Madrid’s in every season (70.9–82.9% against 58.9–75.6%), peaking in the 2014-15 MSN year at 82.9%. The concentration behind the carrier is Barcelona’s, not the carrier dependence itself.

Table 100 Mechanism Diversity — Goal-Type Split (player goals)

Season	Team	Open Play	Set Piece	Free Kick	Penalty	Total
12-13	RM	25	1	0	0	26
12-13	BA	17	0	0	1	18
13-14	RM	37	2	0	2	41
13-14	BA	17	1	0	3	21
14-15	RM	20	0	0	3	23
14-15	BA	30	1	0	0	31
15-16	RM	22	1	2	2	27
15-16	BA	19	1	0	2	22
16-17	RM	31	2	1	0	34
16-17	BA	19	1	2	3	25
17-18	RM	29	0	0	4	33
17-18	BA	11	0	1	0	12

Real Madrid draws on a wider set of mechanisms — set-piece goals in five of six seasons, penalties in four — so its goal base is spread across open play, dead balls, and the spot. Barcelona’s narrows toward open play in its decisive years: 30 of 31 player goals from open play in the 2014-15 title season, 11 of 12 in the 2017-18 collapse. A diversified base for one club, an open-play-dependent one for the other, most exclusive when it mattered most.

The shot pillar cannot promote these to causes. One hypothesis is carried forward for the later pillars: that Real Madrid’s system opens space through personnel complexity, horizontal width, and vertical depth at once, and that this underlies its more durable finishing [**HYPOTHESIS**]. It is stated to be tested in Modules 5, 6, and 8, not concluded here.

Critical Matches (Level 3)

The third condition is survival of the tie whose result is unrecoverable. The critical matches were designated within each season’s section; consolidated, they comprise 26 knockout ties — 16 for Real Madrid, whose runs reached deeper, and 10 for Barcelona. Table 101 gives the full record with both attacking and defensive shot metrics, two-legged ties aggregated.

Table 101 Critical-Match Record — All Designated Ties, 2012–2018

Season	Team	Tie	Result	G	xG	FinΔ	OppG	OppxG	OppFinΔ
12-13	RM	SF Dortmund	OUT	3	2.58	+0.42	4	2.38	+1.62
12-13	BA	SF Bayern	OUT	0	1.29	-1.29	6	2.57	+3.43
13-14	RM	QF Dortmund	adv	3	1.96	+1.04	2	1.56	+0.44
13-14	RM	SF Bayern	adv	5	1.58	+3.42	0	1.69	-1.69
13-14	RM	Final Atletico	WON	4	2.06	+1.94	1	0.65	+0.35
13-14	BA	QF Atletico	OUT	1	1.77	-0.77	2	0.95	+1.05
14-15	RM	QF Atletico	adv	1	1.76	-0.76	0	1.09	-1.09
14-15	RM	SF Juventus	OUT	2	3.40	-1.40	3	1.92	+1.08
14-15	BA	QF PSG	adv	5	1.09	+3.91	0	0.82	-0.82
14-15	BA	SF Bayern	adv	5	1.73	+3.27	3	1.73	+1.27
14-15	BA	Final Juventus	WON	3	1.00	+2.00	1	0.75	+0.25

Season	Team	Tie	Result	G	xG	FinΔ	OppG	OppxG	OppFinΔ
15-16	RM	QF Wolfsburg	adv	3	2.03	+0.97	2	1.76	+0.24
15-16	RM	SF Man City	adv	0	2.28	-2.28	0	0.34	-0.34
15-16	RM	Final Atletico	WON	1	1.74	-0.74	1	1.90	-0.90
15-16	BA	QF Atletico	OUT	2	1.85	+0.15	3	1.28	+1.72
16-17	RM	QF Bayern	adv	6	2.75	+3.25	2	3.51	-1.51
16-17	RM	SF Atletico	adv	4	2.87	+1.13	2	1.74	+0.26
16-17	RM	Final Juventus	WON	4	0.92	+3.08	1	0.23	+0.77
16-17	BA	R16 PSG	adv	5	3.06	+1.94	5	1.87	+3.13
16-17	BA	QF Juventus	OUT	0	2.05	-2.05	3	1.17	+1.83
17-18	RM	R16 PSG	adv	5	3.04	+1.96	2	1.48	+0.52
17-18	RM	QF Juventus	adv	4	3.39	+0.61	3	2.19	+0.81
17-18	RM	SF Bayern	adv	4	1.33	+2.67	3	4.66	-1.66
17-18	RM	Final Liverpool	WON	3	0.84	+2.16	1	1.33	-0.33
17-18	BA	R16 Chelsea	adv	4	0.99	+3.01	1	0.89	+0.11
17-18	BA	QF Roma	OUT	2	1.06	+0.94	4	2.98	+1.02

Result: WON = won the Final; adv = advanced; OUT = eliminated at this tie. Goals player-attributed; two-legged ties aggregated.

Table 102 distils the record into a per-season variance test, holding the benchmark constant within each season. Four seasons offer a same-opponent comparison; 2017-18 and 2012-13, which have no shared opponent, are read through a same-league test instead — the Serie A Test defined in section 4.7, and a parallel Bundesliga Test for the opening season, where both clubs were eliminated in the semifinal by a German side.

Table 102 Per-Season Variance Test — Finishing at Both Ends

Season	Benchmark	Club — Tie (result)	xG	FinΔ	OppFinΔ	Net
12-13	Bundesliga	RM — SF Dortmund (OUT)	2.58	+0.42	+1.62	-1.20
12-13	Bundesliga	BA — SF Bayern (OUT)	1.29	-1.29	+3.43	-4.72

Season	Benchmark	Club — Tie (result)	xG	FinΔ	OppFinΔ	Net
13-14	Atlético	RM — Final Atlético (WON)	2.06	+1.94	+0.35	+1.59
13-14	Atlético	BA — QF Atlético (OUT)	1.77	-0.77	+1.05	-1.82
14-15	Juventus	RM — SF Juventus (OUT)	3.40	-1.40	+1.08	-2.48
14-15	Juventus	BA — Final Juventus (WON)	1.00	+2.00	+0.25	+1.75
15-16	Atlético	RM — Final Atlético (WON)	1.74	-0.74	-0.90	+0.16
15-16	Atlético	BA — QF Atlético (OUT)	1.85	+0.15	+1.72	-1.57
16-17	Juventus	RM — Final Juventus (WON)	0.92	+3.08	+0.77	+2.31
16-17	Juventus	BA — QF Juventus (OUT)	2.05	-2.05	+1.83	-3.88
17-18	Serie A	RM — QF Juventus (adv)	3.39	+0.61	+0.81	-0.20
17-18	Serie A	BA — QF Roma (OUT)	1.06	+0.94	+1.02	-0.08

Net = own Finishing Delta – opponent Finishing Delta (positive = this club won the finishing exchange). Serie A / Bundesliga rows are same-league, not same-opponent, comparisons.

In three of the four same-opponent seasons, the club that advanced was the one that finished better relative to its chances — the higher Finishing Delta — regardless of which side created more. The 2016-17 Juventus pairing is the extreme: Barcelona took 35 shots worth 2.05 xG across two legs and scored none (Net –3.88, eliminated), while Real Madrid beat the same defence in the Final from 0.92 xG and scored four (Net +2.31). The 2014-15 Juventus pairing runs the same way reversed — Real Madrid generated 3.40 xG for two goals and went out, Barcelona turned 1.00 xG into three and the title — and the 2013-14 Atlético pairing repeats the finishing gap. The 2015-16 Atlético pairing is the honest exception: Real Madrid advanced on penalties despite the poorer finishing (Net +0.16 only after the shoot-out), a variance outcome rather than a finishing edge. The two same-league tests complete the picture. The Serie A Test marks 2017-18 as the one season Barcelona's exit is not variance: across the tie they were out-created by Roma, 1.06 xG to 2.98, and lost the finishing exchange as well (Net –0.08), the season they fail the baseline condition itself. The Bundesliga Test shows that even in their shared 2012-13 semifinal exits the clubs differed in kind — Real Madrid competitive against Dortmund (Net –1.20), Barcelona

overwhelmed by Bayern (Net -4.72). Level 3 is where the shot-pillar evidence separates the clubs: not at the baseline, where they are close, nor decisively at tier creation, where both largely hold, but in the single ties, where the chance-to-goal relationship breaks — repeatedly against Barcelona, and, when it breaks against Real Madrid, more often survived through variance.

4.8.7 Evolution and Adaptability

The tables above are cross-sectional. This closing subsection asks a longitudinal question the analyst has long suspected from watching the clubs: did Real Madrid grow more adaptable across the six seasons, and did Barcelona fail to? The claim is interpretive and the evidence is thinner than the descriptive tables, so it is presented as a pattern with hypotheses attached, not a proven law. Figure 13 places each club, each season, in the space that decides its fate: territorial dominance on the horizontal axis, and net finishing — own Finishing Delta minus the opponent's — on the vertical, aggregated over that season's critical matches. The path runs in time order; stars mark title seasons; marker size scales with the number of critical matches behind each point.

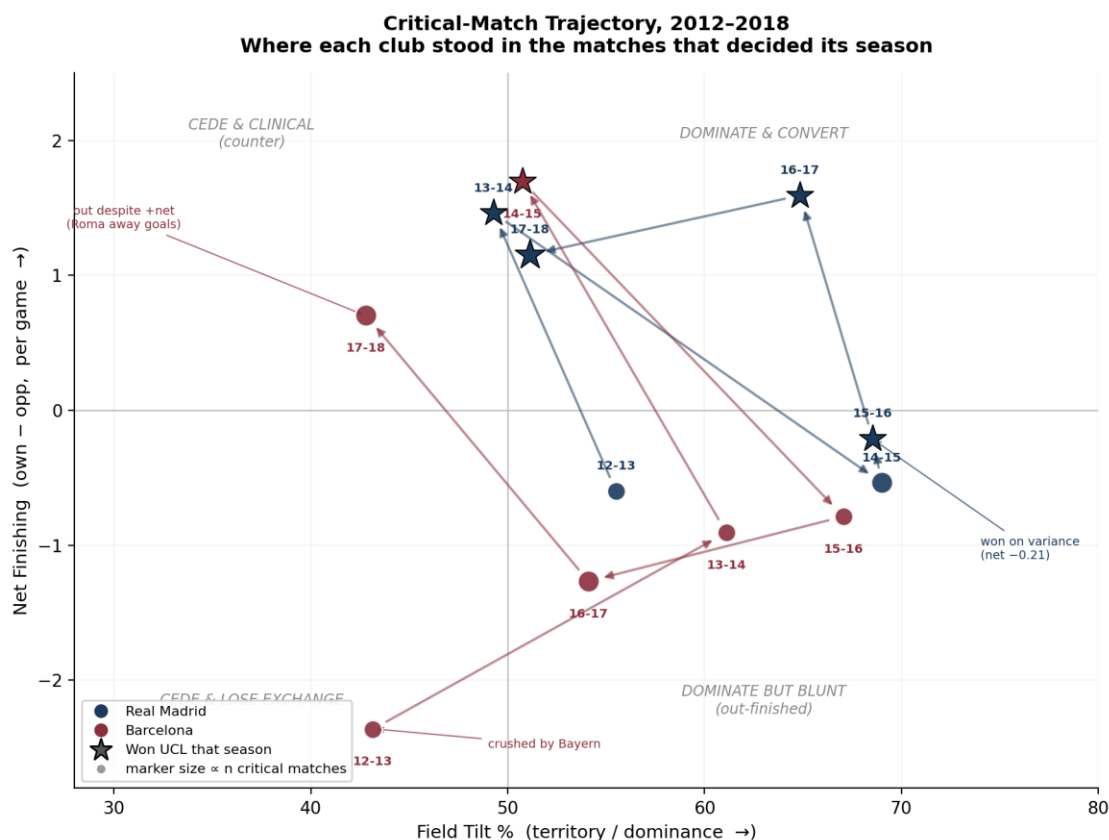


Figure 13 Critical-match trajectory, 2012–2018. Horizontal: Field Tilt (territory). Vertical: net finishing (own – opponent Finishing Delta, per game) over the season's critical matches. Stars = seasons the club won the competition; marker size $\propto$ number of critical matches.

Two things are visible. Real Madrid stays in the upper half and its floor holds: every season sits at or above net -0.60 , and its four title stars land in three different regions of the map — cede-and-clinical (2013-14, 2017-18), dominate-and-convert (2016-17), and the variance survival (2015-16, sitting just below the line). Barcelona swings across the whole vertical range: one clean title in the upper band (2014-15) but a floor of -2.36 , where it was overwhelmed by Bayern, and repeated descents into the lower half, including 2017-18, where it finished net-positive yet still went out. Real Madrid occupies the top from several positions; Barcelona spikes once and cannot hold altitude.

Three match-level measures put numbers to the impression. Opponent-responsiveness — the correlation between opponent strength and a club's own Field Tilt, where a more negative value means ceding territory to stronger sides — grew for Real Madrid from -0.12 in the first three seasons to -0.50 in the last three; it was developed, not innate. In games where Real Madrid ceded territory, its mean net finishing rose from $+0.56$ early to $+1.16$ late: it did not merely cede more thoughtfully, it punished more ruthlessly. Barcelona, by contrast, was already the more opponent-responsive club early (-0.56) but did not build on it, and its ceding-game finishing improved only to $+0.30$ — real, but never approaching Real Madrid's level. The honest framing is therefore not adaptable versus static but compounding versus plateauing: Real Madrid added a second gear — the counter-attacking, cede-and-finish mode — and refined it season over season, while Barcelona kept a single possession-finishing mode that, once the MSN finishing peak faded, had no alternative to fall back on.

Three limits are stated plainly. The late-era ceding sample is only four matches, so the $+1.16$ rests on few games; several trajectory points aggregate just two critical matches; and because the xG model is distance-only, net finishing in low-tilt games partly reflects the quality of transition shots the model cannot see — which supports the counter-attacking reading but keeps the mechanism a hypothesis for Module 6, not a claim the shot pillar can close.

The Three-Level Verdict and Handoff

Real Madrid satisfies all three conditions: a strong baseline every year, a positive Tier-1 Quality Margin in all six seasons, and repeated survival of the critical tie — with the caveat that it under-finishes at Tier-1 in two seasons and is rescued by variance, which cut in its favour more often than not. Barcelona satisfies the baseline every year and largely holds Tier-1 creation, but fails the critical-match condition repeatedly; its problem is not that it is worse on average or worse at creating against elite opponents — in several seasons it is better on both — but that in the decisive ties its conversion collapses relative to the chances it makes. Five eliminations against one title is the outcome. Stated at the pillar's resolution, the championship gap is a critical-match conversion phenomenon, sharpened by Real Madrid's compounding second mode — a location and a trajectory, not an explanation.

Four questions leave this module named. Why does conversion collapse relative to chance quality in the decisive ties — a matter of how the chances were built, or of what they were built against? This divides between Module 6, which separates chances constructed through circulation from those created in transition, and Module 8, which measures the compression of the opposing block. Does Barcelona's higher top-three concentration and narrower goal base make its output more fragile to a single elite defensive performance? For Modules 5 and 6. Is Real Madrid's distributed load and spatial profile — the Space Creation Thesis, and the counter-attacking second gear identified in Figure 13 — the mechanism behind its more resilient finishing? Tested across Modules 5, 6, and 8. And the integration of these shot-side findings with the passing, danger-pass, and defending pillars into a single account of the four-to-one gap belongs to Module 9, for which this section supplies the shooting evidence and, deliberately, none of the explanation.